3 The Molecules of Life

Why Macromolecules Matter

If you drink coffee, odds are good that you spooned beets into your cup today—in the form of table sugar.



A long-distance runner who carbo-loads the night before a race is banking glycogen to be used the next day.





The structure of your DNA is nearly indistinguishable from the DNA of a mosquito or an elephant: The differences in animal species result from the way nucleotides are arranged.



Chocolate melts in your mouth (and in your hands) because of the structure of cocoa butter, which is high in saturated fats.

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Lactose Intolerance

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Lactose Intolerance BIOLOGY AND SOCIETY

Got Lactose?

You've probably seen ads that associate a milk mustache with good health. Indeed, milk is a very healthy food: It's rich in protein, minerals, and vitamins, and can be low in fat. But for the majority of adults in the world, a glass of milk delivers a heavy dose of digestive discomfort that can include bloating, gas, and abdominal pain. These are symptoms of lactose intolerance, the inability to digest lactose, the main sugar found in milk.

For people with lactose intolerance, the problem starts when lactose enters the small intestine. To absorb this sugar, digestive cells in the small intestine must produce a molecule called lactase. Lactase is an enzyme, a protein that helps drive chemical reactions—in this case, the breakdown of lactose into smaller sugars. Most people are born with the ability to digest lactose. But after about age 2, lactase levels in most people decline significantly. Lactose that is not broken down in the small intestine passes into the large intestine, where bacteria feed on it and belch out gaseous by-products. An accumulation of gas produces uncomfortable symptoms. Sufficient lactase can thus mean the difference between delight and discomfort when someone drinks a tall glass of chocolate milk.



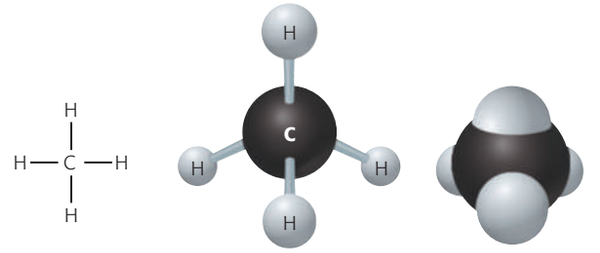
Lactose-rich chocolate milk. Lactose digestion results from interactions between several classes of the body's molecules.

There is no treatment for the underlying cause of lactose intolerance: the decreased production of lactase. So what are the options for people who do not produce enough of this enzyme? The first option is avoiding lactose-containing foods; abstinence is always the best form of protection! Alternatively, substitutes are available, such as milk made from soy or almonds, or cow's milk that has been pretreated with lactase. If you are lactose intolerant and pizza is your staple food, you can buy lactose-free cheese (made from soy) or you can make your own mozzarella from lactose-free milk. Also, lactase in pill form can be taken along with food to ease digestion by artificially providing the enzyme that the body naturally lacks.

Lactose intolerance illustrates one way the interplay of biological molecules can affect your health. Such molecular interactions, repeated in countless variations, drive all biological processes. In this chapter, we'll explore the structure and function of large molecules that are essential to life. We'll start with an overview of carbon and then examine four classes of molecules: carbohydrates, lipids, proteins, and nucleic acids. Along the way, we'll look at where these molecules occur in your diet and the important roles they play in your body.

Organic Compounds

A cell is mostly made up of water, but the rest is mainly carbon-based molecules. When it comes to the chemistry of life, carbon plays the leading role. This is because carbon is unparalleled in its ability to form the skeletons of large, complex, diverse molecules that are necessary for life's functions. The study of carbon-based molecules, which are called **organic compounds**, lies at the heart of any study of life.



Structural formula **Ball-and-stick model** **Space-filling model**

▲ **Figure 3.2** **Methane, a simple organic compound.**

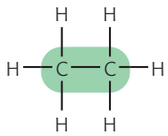
Carbon Chemistry

Carbon is a versatile molecule ingredient because an atom of carbon can share electrons with other atoms in four covalent bonds that can branch off in four directions. Because carbon can use one or more of its bonds to attach to other carbon atoms, it is possible to construct an endless diversity of carbon skeletons varying in size and branching pattern **(Figure 3.1)**. Thus, molecules with multiple carbon "intersections" can form very elaborate shapes. The carbon atoms of organic compounds can also bond with other elements, most commonly hydrogen, oxygen, and nitrogen.

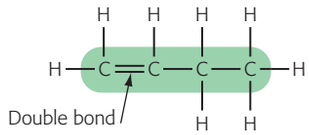
One of the simplest organic compounds is methane, CH_4, with a single carbon atom bonded to four hydrogen atoms **(Figure 3.2)**. Methane is abundant in natural gas and is also produced by prokaryotes that live in swamps (in the form of swamp gas) and in the digestive tracts of grazing animals, such as cows. Larger organic compounds (such as octane, with eight carbons) are the main molecules in the gasoline we burn in cars and other machines. Organic compounds are also important fuels in your body; the energy-rich parts of fat molecules have a structure similar to gasoline **(Figure 3.3)**.

The unique properties of an organic compound depend not only on its carbon skeleton but also on the atoms attached to the skeleton. In an organic compound, the groups of atoms directly involved in chemical reactions are called **functional groups**. Each functional group plays a particular role during chemical reactions. Two examples of functional groups are the hydroxyl group (—OH, found in alcohols such as isopropyl rubbing alcohol) and the carboxyl group (—COOH, found in all proteins). Many biological molecules have two or more functional groups. Keeping in mind this basic scheme—carbon skeletons with attached functional groups—we are now ready to explore how our cells make large molecules out of smaller ones.

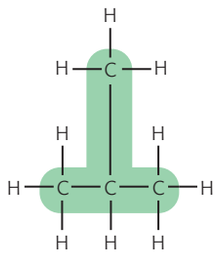
▼ **Figure 3.1** **Variations in carbon skeletons.** All of these examples are organic compounds that consist only of carbon and hydrogen. Notice that each carbon atom forms four bonds and each hydrogen atom forms one bond. One line represents a single bond (sharing of one pair of electrons) and two lines represent a double bond (sharing of two pairs of electrons).



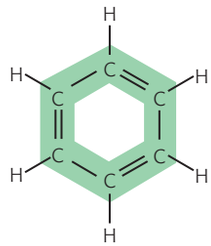
Carbon skeletons vary in length



Carbon skeletons may have double bonds, which can vary in location



Carbon skeletons may be unbranched or branched



Carbon skeletons may be arranged in rings

▼ **Figure 3.3** **Hydrocarbons as fuel.** Energy-rich organic compounds in gasoline provide fuel for machines, and energy-rich molecules in fats provide fuel for cells.



Octane Dietary fat

Giant Molecules from Smaller Building Blocks

On a molecular scale, the members of three categories of biological molecules—carbohydrates (such as those found in starchy foods like French fries and bagels), proteins (such as enzymes and the components of your hair), and nucleic acids (such as DNA)—are gigantic; in fact, biologists call them **macromolecules** (*macro* means "big"). Despite the size of macromolecules, their structures can be easily understood because they are **polymers**, large molecules made by stringing together many smaller molecules called **monomers**. A polymer is like a necklace made by joining together many monomer "beads" or a train made from a chain of box cars. Although, at first, the structure of a long train may seem very complex, it can be easily understood by asking two questions: What kind of cars make up the train? How are they joined together? Similarly, even large and complex biological macromolecules can be understood by considering which monomers make them up and how those monomers are attached together.



Cells link monomers together to form a polymer through a **dehydration reaction**. As the name implies, this chemical reaction involves removing a molecule of water **(Figure 3.4a)**. For each monomer added to a chain, a water molecule (H_2O) is formed by the release of two hydrogen atoms and one oxygen atom from the reactants. This same type of dehydration reaction occurs regardless of the specific monomers involved and the type of polymer the cell is producing. Therefore, you'll see this type of reaction repeated many times throughout this chapter.

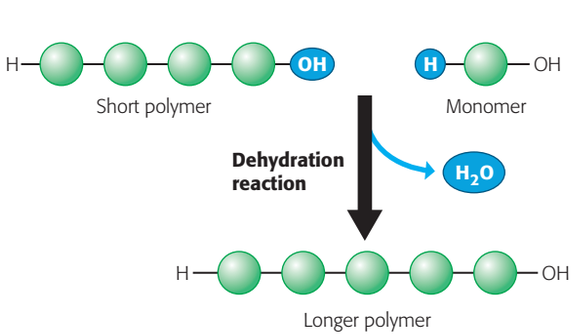
Organisms not only make macromolecules but also break them down. For example, you must digest macromolecules in food to make their monomers available to your cells, which can then rebuild the monomers into the macromolecules that make up your body. Converting macromolecules is like taking apart a car (food) made of interlocking toy blocks and then using these blocks to assemble a new car of your own design (your own body's molecules). The breakdown of polymers occurs by a process called **hydrolysis (Figure 3.4b)**. Hydrolysis means to break (*lyse*) with water (*hydro*). Cells break bonds between monomers by adding water to them, a process that is essentially the reverse of a dehydration reaction. Imagine disassembling a long train by throwing water at the junctions between cars. Each time you add water, one more car or set of cars is broken free. You learned in the Biology and Society section about one real-world example of a hydrolysis reaction: the breakdown of lactose into its monomers by the enzyme lactase. ✓

✓ CHECKPOINT

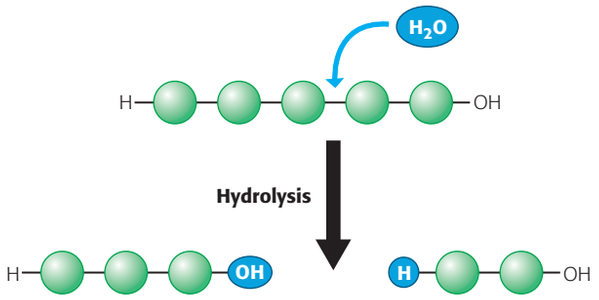
1. Chemical reactions that build polymers from _____ are called _____ reactions because they release a molecule of water.
2. The reverse type of reaction, one that breaks larger molecules down into smaller molecules, is called _____.

Answers: **1.** *monomers; dehydration* **2.** *hydrolysis*

▼ Figure 3.4 Synthesis and breakdown of polymers. For simplicity, the only atoms shown in these diagrams are hydrogens and hydroxyl groups (—OH) in strategic locations.



(a) Building a polymer chain. A polymer grows in length when an incoming monomer and the monomer at the end of the polymer each contribute atoms to form a water molecule. The monomers replace the lost covalent bonds with a bond between each other.



(b) Breaking a polymer chain. Hydrolysis reverses the process by adding a water molecule, which breaks the bond between two monomers, creating two smaller molecules from one bigger one.

Large Biological Molecules

There are four categories of large, important biological molecules found in all living creatures: carbohydrates, lipids, proteins, and nucleic acids. For each category, we'll explore the structure and function of these large molecules by first learning about the smaller molecules used to build them.

Carbohydrates

Carbohydrates, or "carbs," are a class of molecules that includes sugars and polymers of sugars. Some examples are the small sugar molecules dissolved in soft drinks and the long starch molecules in spaghetti and bread. In animals, carbohydrates are a primary source of dietary energy and raw material for manufacturing other kinds of organic compounds; in plants, they serve as a building material for much of the plant body.

Monosaccharides

Simple sugars, or **monosaccharides** (from the Greek *mono*, single, and *sacchar*, sugar), are the monomers of carbohydrates; they cannot be broken down into smaller sugars. Common examples are glucose, found in soft drinks, and fructose, found in fruit. Both of these simple sugars are also in honey **(Figure 3.5)**. The molecular formula for glucose is $C_6H_{12}O_6$. Fructose has the same formula, but its atoms are arranged differently. Glucose and fructose are examples of **isomers**, molecules that have the same molecular formula but different structures. Isomers are like anagrams—words that contain the same letters in a different order, such as *heart* and *earth*. Because molecular shape is so important, seemingly minor differences in the arrangement of atoms give isomers different properties, such as how they react with other molecules. In this case, the rearrangement of functional groups makes fructose taste much sweeter than glucose.

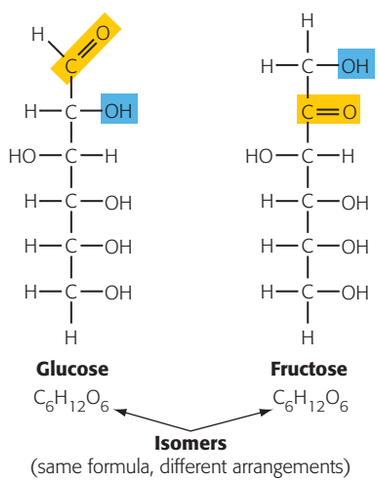
It is convenient to draw sugars as if their carbon skeletons were linear. When dissolved in water, however, many monosaccharides form rings when one end of the molecule forms a bond with another part of the molecule **(Figure 3.6)**. You'll notice this ring shape in the depiction of the structure of many carbohydrates in this chapter.

Monosaccharides, particularly glucose, are the main fuel molecules for cellular work. Like an automobile engine consuming gasoline, your cells break down glucose molecules and extract their stored energy, giving off carbon dioxide as "exhaust." The rapid conversion of glucose to cellular energy is why a solution of glucose dissolved in water (often called dextrose) is given as an IV to sick or injured patients; the glucose provides an immediate energy source to tissues in need of repair. ✓

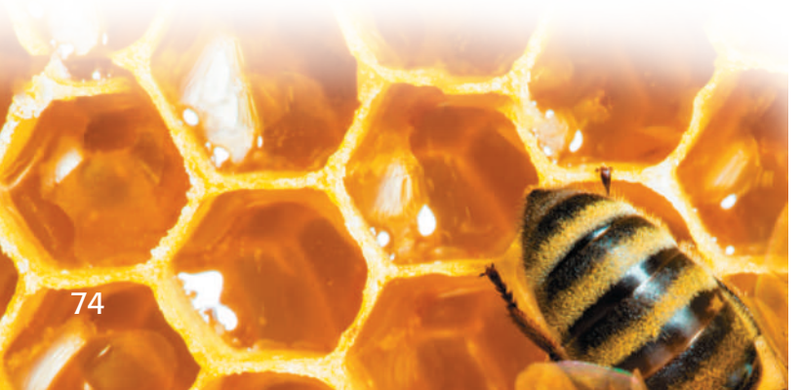
✓ CHECKPOINT

1. All carbohydrates consist of one or more _____, also called simple sugars. Name two examples of simple sugars.
2. Explain why glucose and fructose have the same formula ($C_6H_{12}O_6$) but different properties.

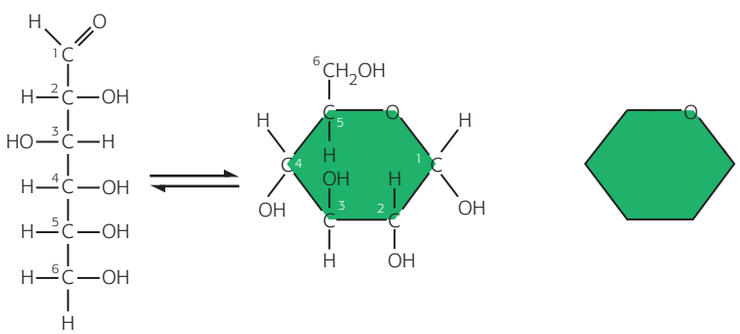
Answers: 1. monosaccharides; 2. Different arrangements of atoms affect molecular shapes and properties.



► Figure 3.5 Monosaccharides (simple sugars). Glucose and fructose, both of which are found in honey, are isomers, molecules with the same atoms arranged differently.



▼ Figure 3.6 The ring structure of glucose.



(a) Linear and ring structures. The carbon atoms are numbered so you can relate the linear and ring versions of the molecule. As the double arrows indicate, ring formation is a reversible process, but at any instant in an aqueous solution, most glucose molecules are rings.

(b) Abbreviated ring structure. In this book, we'll use this abbreviated ring symbol for glucose. Each unmarked corner represents a carbon and its attached atoms.

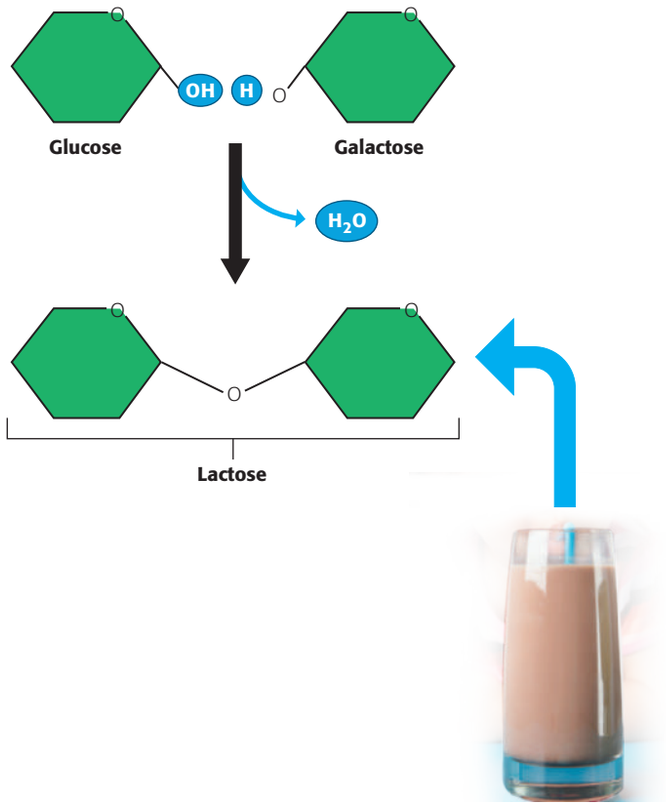
Disaccharides

A **disaccharide**, or double sugar, is constructed from two monosaccharides by a dehydration reaction. The disaccharide lactose, sometimes called "milk sugar," is made from the monosaccharides glucose and galactose **(Figure 3.7)**. Another common disaccharide is maltose, naturally found in germinating seeds. It is used in making beer, malt whiskey and liquor, malted milk shakes, and malted milk ball candy. A molecule of maltose consists of two glucose monomers joined together.

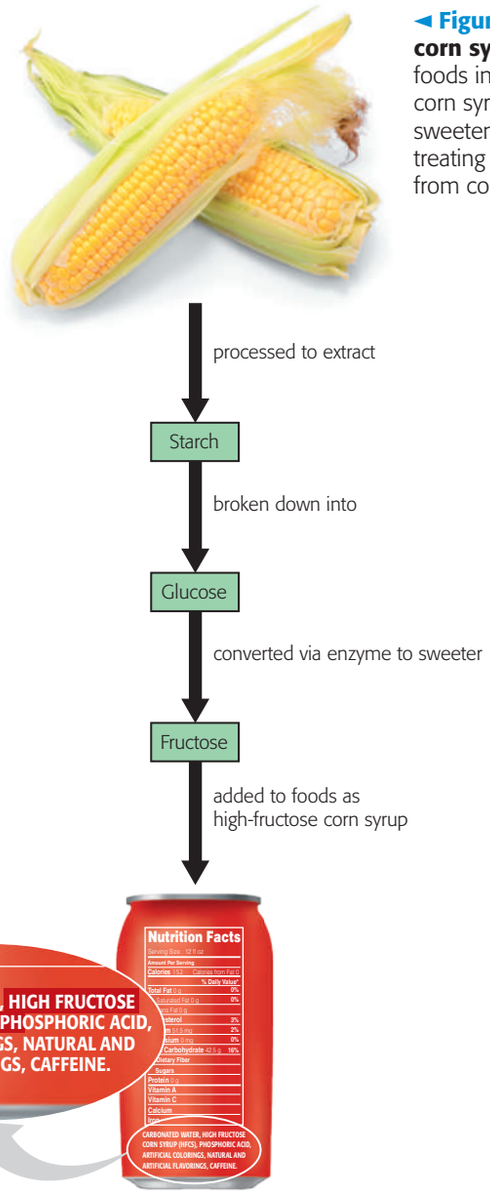
The most common disaccharide is sucrose (table sugar), which consists of a glucose monomer linked to a fructose monomer. Sucrose is the main carbohydrate in plant sap, and it nourishes all the parts of the plant. Sugar manufacturers extract sucrose from the stems of sugarcane or (much more often in the United States) the roots of sugar beets. Another common sweetener is high-fructose corn syrup (HFCS), made through a commercial process that uses an enzyme to convert natural glucose in corn syrup to the much sweeter fructose. HFCS is a clear, goopy liquid containing about 55% fructose; it is much cheaper than sucrose and easier to mix into drinks and processed foods. If you read the label on a soft drink, you're likely to find that high-fructose corn syrup is one of the first ingredients listed **(Figure 3.8)**.

The average American consumes about 45 kilograms (kg)—a whopping 100 pounds—of sweeteners per year, mainly in the form of sucrose and HFCS. This national "sweet tooth" persists in spite of our growing awareness about how sugar can negatively affect our health. Sugar is a major cause of tooth decay, and overconsumption increases the risk of developing diabetes and heart disease. Moreover, high sugar consumption tends to replace eating more varied and nutritious foods. The description of sugars as "empty calories" is accurate in the sense that most sweeteners contain only negligible

If you drink coffee, odds are good that you spooned beets into your cup today—in the form of table sugar.



▼ **Figure 3.7 Disaccharide (double sugar) formation.** To form a disaccharide, two simple sugars are joined by a dehydration reaction, in this case forming a bond between monomers of glucose and galactose to make the double sugar lactose.



◄ **Figure 3.8 High-fructose corn syrup.** Many processed foods include high-fructose corn syrup, an artificial sweetener made by chemically treating sugars extracted from corn.

✓ CHECKPOINT

How and why do manufacturers produce HFCS?

Answer: They convert glucose to the sweeter fructose. HFCS is cheaper and more easily blended with processed foods.

amounts of nutrients other than carbohydrates. For good health, we also require proteins, fats, vitamins, and minerals. And we need to include substantial amounts of complex carbohydrates—that is, polysaccharides—in our diet. Let's examine these macromolecules next. ✓

Polysaccharides

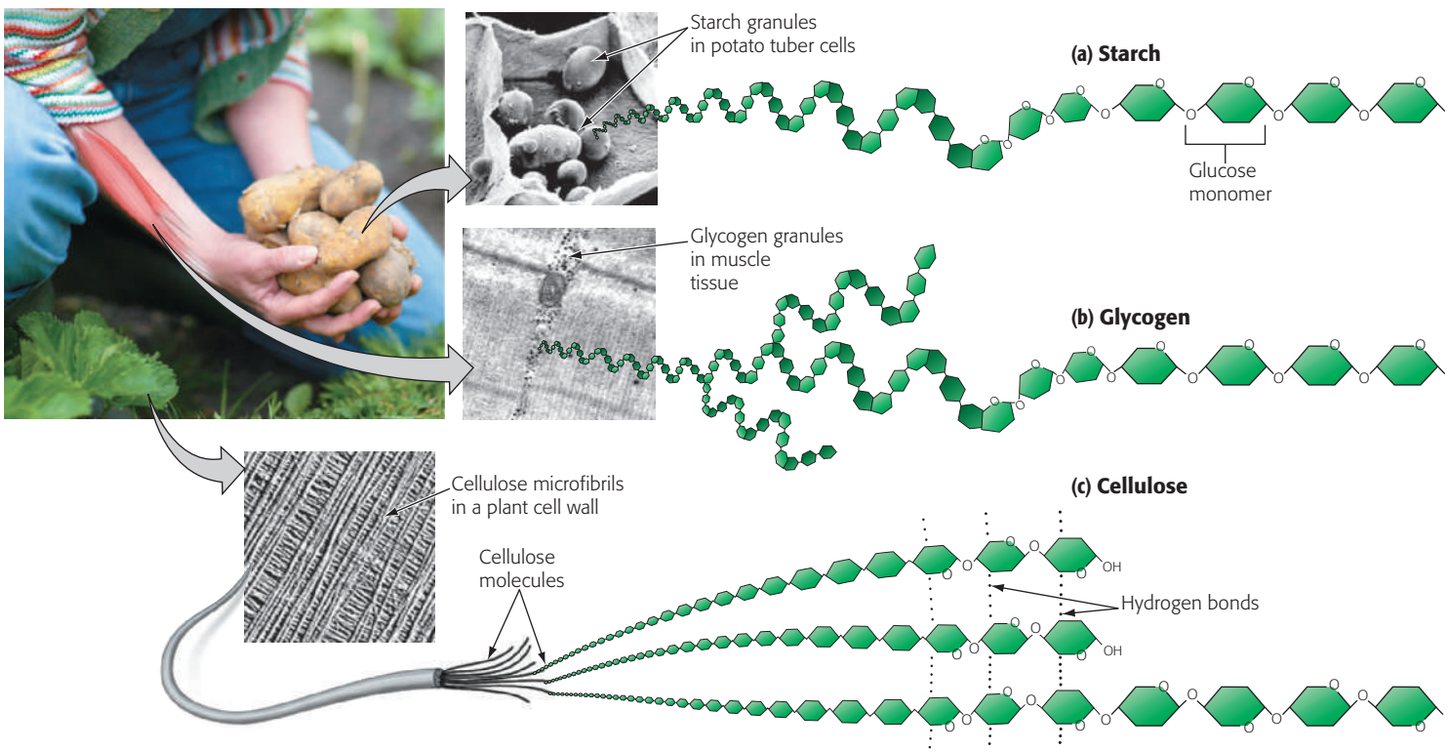
Complex carbohydrates, or **polysaccharides**, are long chains of sugars—polymers of monosaccharides. One familiar example is starch, a storage polysaccharide found in plants. **Starch** consists of long strings of glucose monomers **(Figure 3.9a)**. Plant cells store starch, providing a sugar stockpile that can be tapped when needed. Potatoes and grains, such as wheat, corn, and rice, are the major sources of starch in our diet. Animals can digest starch because enzymes within their digestive systems break the bonds between glucose monomers through hydrolysis reactions.

Animals store excess glucose in the form of a polysaccharide called **glycogen**. Like starch, glycogen is a polymer of glucose monomers, but glycogen is more extensively branched **(Figure 3.9b)**. Most of your glycogen is stored in liver and muscle cells, which break down the glycogen to release glucose when you need energy. This is why some athletes "carbo-load," consuming large amounts of starchy foods the night before an athletic event. The starch is converted to glycogen, which is then available for rapid use during physical activity the next day.

A long-distance runner who carbo-loads the night before a race is banking glycogen to be used the next day.

Cellulose, the most abundant organic compound on Earth, forms cable-like fibrils in the tough walls that enclose plant cells and is a major component of wood and other structural components of plants **(Figure 3.9c)**. We take advantage of that structural strength when we use lumber as a building material. Cellulose is also a polymer of glucose, but its glucose monomers are linked together in a unique way. Unlike the glucose linkages in starch and glycogen, those in cellulose cannot be broken by any enzyme produced by animals. Grazing animals and wood-eating insects such as termites are able to derive nutrition from cellulose because microorganisms inhabiting their digestive tracts break it down. The cellulose in plant foods that you eat, commonly known as dietary fiber (your grandma might call it "roughage") passes through your digestive tract unchanged. Because it remains undigested, fiber does not provide nutrients, but it does help keep your digestive system healthy. The passage of cellulose stimulates cells lining the digestive tract to secrete mucus, which allows food to pass smoothly. The health benefits of dietary fiber include lowering the risk of heart disease, diabetes, and gastrointestinal disease. However, most Americans do not get the recommended levels of fiber in their diet. Foods rich in fiber include fruits and vegetables, whole grains, bran, and beans.

▼ Figure 3.9 Three common polysaccharides.



Lipids

Almost all carbohydrates are **hydrophilic** ("water-loving") molecules that dissolve readily in water. In contrast, **lipids** are **hydrophobic** ("water-fearing"); they do not mix with water. You've probably seen this chemical behavior when you combine oil and vinegar: The oil, which is a type of lipid, separates from the vinegar, which is mostly water **(Figure 3.10)**. If you shake vigorously, you can force a temporary mixture long enough to douse your salad, but what remains in the bottle will quickly separate. Lipids also differ from carbohydrates, proteins, and nucleic acids in that they are neither huge macromolecules nor are they necessarily polymers built from repeating monomers. Lipids are a diverse group of molecules made from different molecular building blocks. In this section, we'll look at two types of lipids: fats and steroids.



◄ Figure 3.10 The separation of hydrophobic (oil) and hydrophilic (vinegar) components in salad dressing.

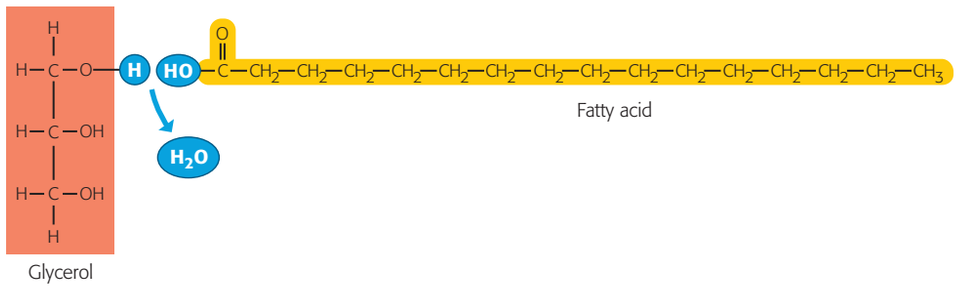
Fats

A typical **fat** consists of a glycerol molecule joined with three fatty acid molecules by dehydration reactions **(Figure 3.11a)**. The resulting fat molecule is called a **triglyceride** **(Figure 3.11b)**, a term you may hear in the results of a blood test. A fatty acid is a long molecule that stores a lot of energy. A pound of fat packs more than twice as much energy as a pound of carbohydrate. The downside to this energy efficiency in fats is that it is very difficult for a person trying to lose weight to "burn off" excess fat. We stock these long-term food stores in specialized reservoirs called adipose cells, which swell and shrink when we deposit and withdraw fat from them. This adipose tissue, or body fat, not only stores energy but also cushions vital organs and insulates us, helping maintain a constant, warm body temperature.

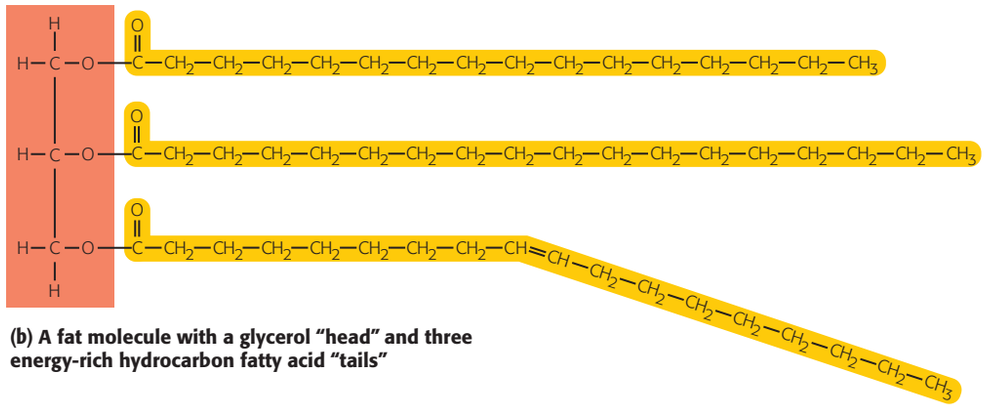
Notice in Figure 3.11b that the bottom fatty acid bends where there is a double bond in the carbon skeleton. That fatty acid is **unsaturated** because it has fewer than the maximum number of hydrogens at the double bond. The other two fatty acids in the fat molecule lack double bonds in their tails. Those fatty acids are **saturated**, meaning that they contain the maximum number of hydrogen atoms, giving them a straight shape. A saturated fat is one with all three of its fatty acid tails saturated. If one or more of the fatty acids is unsaturated, then it's an unsaturated fat, like the one in Figure 3.11b. A polyunsaturated fat has several double bonds within its fatty acids.

Most animal fats, such as lard and butter, have a relatively high proportion of saturated fatty acids. The linear shape of saturated fatty acids allows these molecules to stack easily (like bricks in a wall), so saturated

▼ Figure 3.11 The synthesis and structure of a triglyceride molecule.



(a) A dehydration reaction linking a fatty acid to glycerol



(b) A fat molecule with a glycerol "head" and three energy-rich hydrocarbon fatty acid "tails"

fats tend to be solid at room temperature **(Figure 3.12)**. Diets rich in saturated fats may contribute to cardiovascular disease by promoting atherosclerosis. In this condition, lipid-containing deposits called plaque build up along the inside walls of blood vessels, reducing blood flow and increasing risk of heart attacks and strokes.

Plant and fish fats are relatively high in unsaturated fatty acids. The bent shape of unsaturated fatty acids makes them less likely to form solids (imagine trying to build a wall using bent bricks!). Most unsaturated fats are liquid at room temperature. Fats that are primarily unsaturated include vegetable oils (such as corn and canola oil) and fish oils (such as cod liver oil).

Chocolate melts in your mouth (and in your hands) because of the structure of cocoa butter, which is high in saturated fats.

Although plant oils tend to be low in saturated fat, tropical plant fats are an exception. Cocoa butter, a main ingredient in chocolate, contains a mix of saturated and unsaturated fat that gives chocolate a melting point near body temperature. Thus, chocolate stays solid at room temperature but melts in your mouth, creating a pleasing "mouth feel" that is one of the reasons chocolate is so appealing.

Sometimes a food manufacturer wants to use a vegetable oil but needs the food product to be solid, as with margarine or peanut butter. To achieve the desired texture, the manufacturer can convert unsaturated fats to saturated fats by adding hydrogen, a process called **hydrogenation**. Unfortunately, hydrogenation can also create **trans fats**, a type of unsaturated fat that is particularly bad for your health. Since 2006, the U.S. Food and Drug Administration has required that trans fats be listed on nutrition labels. But even if a label on a hydrogenated product says 0 grams (g) of trans fats per serving, the product might actually have up to 0.5 g per serving (which legally can be rounded down to 0). Furthermore, trans fats are often found in fast foods, such as French fries and battered-and-fried foods, that may not have labels. Due to the increased awareness of their unhealthy nature, trans fats are becoming less common as food manufacturers substitute other forms of fat. Indeed, trans fats may soon be a dietary thing of the past. New York City banned restaurants from serving trans fats in 2006, and California followed suit in 2010. The countries of Iceland, Switzerland, and Denmark have effectively removed trans fats from the food supply. In the United States, the Food and Drug Administration has ruled that trans fats are not "generally regarded as safe," a designation that will likely result in a phasing out of trans fats from the American food supply.

Although trans fats should generally be avoided and saturated fats limited, it is not true that *all* fats are unhealthy. In fact, some fats perform important functions within the body and are beneficial and even essential to a healthy diet. For example, fats containing omega-3 fatty acids have been shown to reduce the risk of heart disease and relieve the symptoms of arthritis and inflammatory bowel disease. Some sources of these beneficial fats are nuts and oily fish such as salmon. ☑

☑ CHECKPOINT

What are unsaturated fats? What kinds of unsaturated fat are particularly unhealthy? What kinds of unsaturated fat are most healthful?

Answer: fats with less than the maximum number of hydrogens because of double bonds between some carbons; trans fats; fats containing omega-3 fatty acids

▼ **Figure 3.12** **Types of fats.**

TYPES OF FATS

Saturated Fats
(unhealthy fats found primarily in meat and full-fat dairy products; solid at room temperature)



Unsaturated Fats
(fats found primarily in fish and plants; usually liquid at room temperature)



Plant oils
(unhydrogenated; usually liquid at room temperature)



Margarine



Trans fats
(unhealthy fats in hydrogenated processed foods; solid at room temperature)



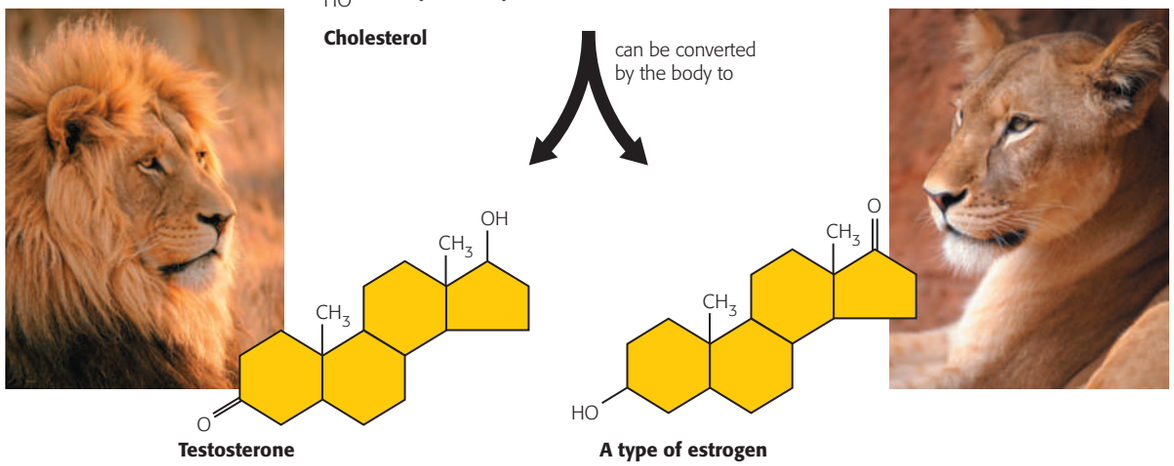
Omega-3 fats
(beneficial fats in some fish and plant oils; liquid at room temperature)

Steroids

Steroids are lipids that are very different from fats in structure and function. All steroids have a carbon skeleton with four fused rings. Different steroids vary in the functional groups attached to this set of rings, and these chemical variations affect their function. One common steroid is cholesterol, which has a bad reputation because of its association with cardiovascular disease. However, cholesterol is a key component of the membranes that surround your cells. It is also the "base steroid" from which your body produces other steroids, such as the hormones estrogen and testosterone **(Figure 3.13)**, which are responsible for the development of female and male sex characteristics, respectively.

Anabolic steroids are synthetic variants of testosterone. In human males, testosterone causes buildup of muscle and bone mass during puberty and maintains masculine traits throughout life. Because anabolic steroids resemble testosterone, they mimic some of its effects. Anabolic steroids are prescribed to treat diseases that cause muscle wasting, such as cancer and AIDS. However, individuals sometimes abuse anabolic steroids to build up their muscles quickly. In recent years, many famous athletes have admitted using chemically modified ("designer") performance-enhancing anabolic steroids **(Figure 3.14)**. Such revelations have raised questions about the validity of home run records and other athletic accomplishments.

Using anabolic steroids is indeed a fast way to increase body size beyond what hard work can produce. But at what cost? Steroid abuse may cause violent mood swings ("roid rage"), depression, liver damage, high cholesterol, shrunken testicles, reduced sex drive, and infertility. Symptoms related to sexuality occur because artificial anabolic steroids often cause the body to reduce its output of natural sex hormones. Most athletic organizations ban the use of anabolic steroids because of their many potential health hazards coupled with the unfairness of an artificial advantage. ☑



▲ **Figure 3.13 Examples of steroids.** The molecular structures of the steroids shown here are abbreviated by omitting all the atoms that make up the rings. The subtle difference between testosterone and estrogen influences the development of the anatomical and physiological differences between male and female mammals, including lions and humans. This example illustrates the importance of molecular structure to function.



Alex Rodriguez



Mark McGwire



Floyd Landis



Ben Johnson

► **Figure 3.14 Steroids and the modern athlete.** Each of the athletes shown here—baseball players Alex Rodriguez and Mark McGwire, Tour de France cyclist Floyd Landis, and Olympic sprinter Ben Johnson—has admitted to steroid use.

☑ CHECKPOINT

What steroid acts as the molecular building block of the human steroid hormones?

Answer: *cholesterol*

Proteins

A **protein** is a polymer of amino acid monomers. Proteins account for more than 50% of the dry weight of most cells, and they are instrumental in almost everything cells do **(Figure 3.15)**. Proteins are the "worker bees" of your body: Chances are, if something is getting done, there is a protein doing it. Your body has tens of thousands of different kinds of proteins, each with a unique three-dimensional shape corresponding to a specific function. In fact, proteins are the most structurally sophisticated molecules in your body.

The Monomers of Proteins: Amino Acids

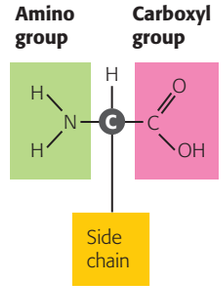
All proteins are made by stringing together a common set of 20 kinds of amino acids. Every **amino acid** consists of a central carbon atom bonded to four covalent partners. Three of those attachments are common to all 20 amino acids: a carboxyl group (—COOH), an amino group ($—NH_2$), and a hydrogen atom. The variable component of amino acids is called the side chain (or R group, for radical group); it is attached to the fourth bond of the central carbon **(Figure 3.16a)**. Each type of amino acid has a unique side chain which gives that amino acid its special chemical properties **(Figure 3.16b)**. Some amino acids have very simple side chains; the amino acid glycine, for example, has a single hydrogen as its side chain. Other amino acids have more complex side chains, some with branches or rings within them. ☑

☑ CHECKPOINT

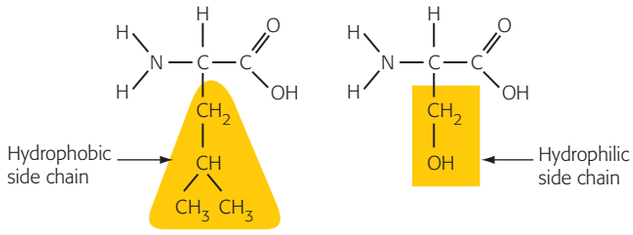
1. Which of the following is *not* made of protein: hair, muscle, cellulose, or enzymes?
2. What are the monomers of all proteins? What is the one part of an amino acid that varies?

Answers: **1.** *Cellulose is a carbohydrate.* **2.** *amino acids; the side chain*

▼ Figure 3.16 Amino acids. All amino acids share common functional groups but vary in their side chains.



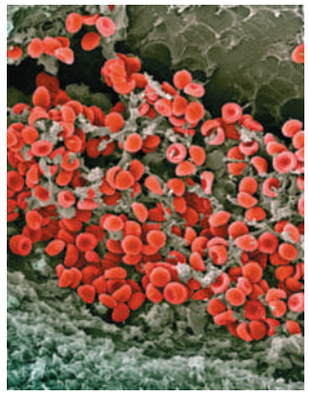
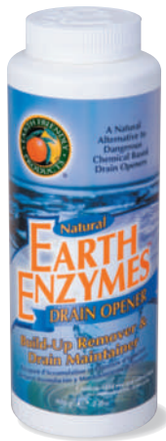
(a) The general structure of an amino acid.



(b) Examples of amino acids with hydrophobic and hydrophilic side chains. The side chain of the amino acid leucine is hydrophobic. In contrast, the side chain of the amino acid serine has a hydroxyl group (–OH), which is hydrophilic.

▼ Figure 3.15 Some of the varied roles played by proteins.

MAJOR TYPES OF PROTEINS

Structural Proteins (provide support)	**Storage Proteins** (provide amino acids for growth)	**Contractile Proteins** (help movement)	**Transport Proteins** (help transport substances)	**Enzymes** (help chemical reactions)
 Structural proteins give hair ligaments and horns	 Seeds and eggs are rich in storage proteins.	 Contractile proteins enable muscles to contract.	 The protein hemoglobin within red blood cells transports oxygen.	 Some cleaning products use enzymes to help break down molecules.

Structure/ Function

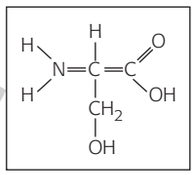
Protein Shape

Cells link amino acid monomers together by—can you guess?—dehydration reactions. The bond that joins adjacent amino acids is called a **peptide bond** (Figure 3.17). The resulting long chain of amino acids is called a **polypeptide**. A functional protein is one or more polypeptide chains precisely twisted, folded, and coiled into a molecule of unique shape. The difference between a polypeptide and a protein can be likened to the relationship between a long strand of yarn and a sweater. To be functional, the long fiber (the yarn) must be precisely knit into a specific shape (the sweater).

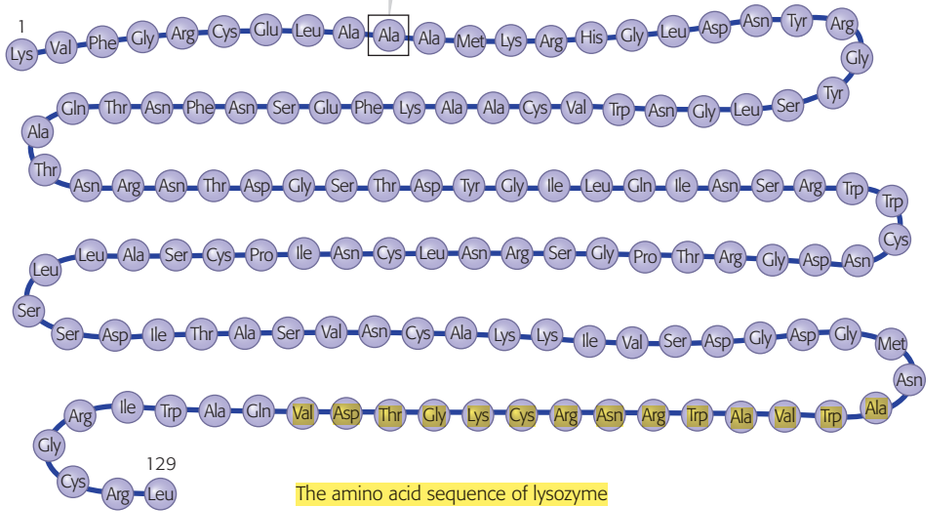
How is it possible to make the huge variety of proteins found in your body from just 20 kinds of amino acids? The answer is arrangement. You know that you can make many different English words by varying the sequence of just 26 letters. Though the protein alphabet is slightly smaller (just 20 "letters"), the "words" are much longer, with a typical polypeptide being hundreds or thousands of amino acids in length. Just as each word is constructed from a unique succession of letters, each protein has a unique linear sequence of amino acids.

The amino acid sequence of each polypeptide determines the three-dimensional structure of the protein. And it is a protein's three-dimensional structure that enables the molecule to carry out its specific function. Nearly all proteins work by recognizing and binding to some other molecule. For example, the specific shape of lactase enables it to recognize and attach to lactose, its molecular target. For all proteins, structure and function are interrelated: What a protein does is a consequence of its shape. The twists and turns of the protein in Figure 3.18 may appear haphazard, but they represent this protein's specific three-dimensional shape, and without that exact shape, the protein could not do its job.

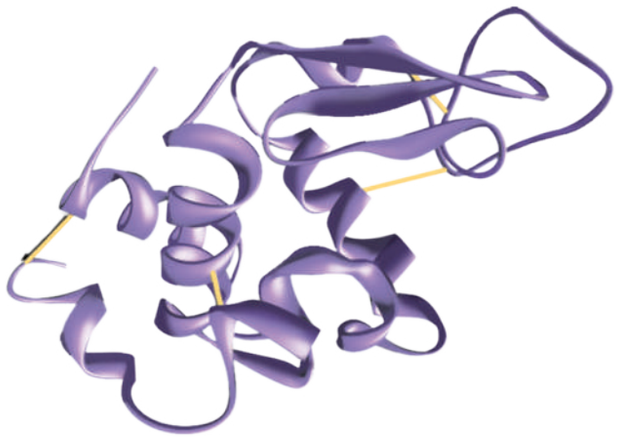
▼ **Figure 3.18 The structure of a protein.** The chain below, drawn in serpentine fashion so that it fits on the page, shows the amino acid sequence of a polypeptide found in lysozyme, an enzyme in your tears and sweat that helps prevent bacterial infections. The names of the amino acids are given as their three-letter abbreviations; for example, the amino acid alanine is abbreviated "ala." This amino acid sequence folds into a protein of specific shape, shown at bottom in two computer-generated representations. Without this specific shape, the protein could not perform its function.



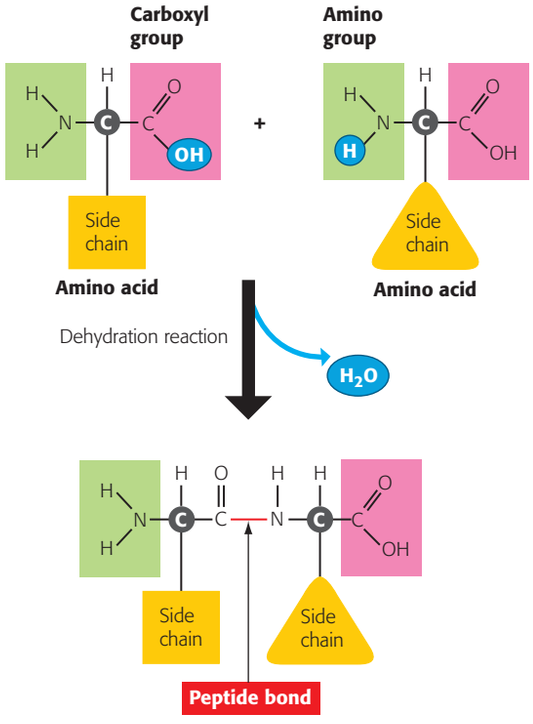
One amino acid (alanine)



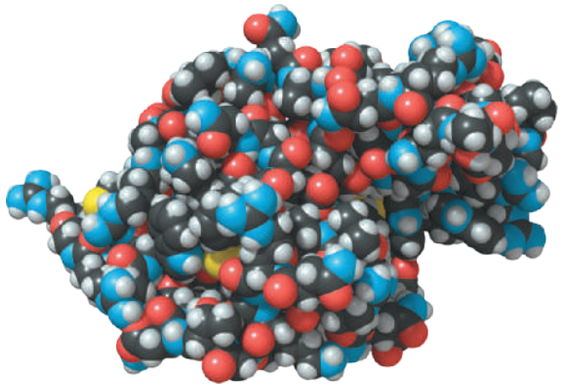
The amino acid sequence of lysozyme



Here you can see how the polypeptide folds into a compact shape.



▲ **Figure 3.17 Joining amino acids.** A dehydration reaction links adjacent amino acids by a peptide bond.



This model allows you to see the details of the protein's structure.

Changing a single letter can drastically affect the meaning of a word—"tasty" versus "nasty," for instance. Similarly, even a slight change in the amino acid sequence can affect a protein's ability to function. For example, the substitution of one amino acid for another at a particular position in hemoglobin, the blood protein that carries oxygen, causes sickle-cell disease, an inherited blood disorder **(Figure 3.19)**. Even though 145 out of the 146 amino acids in hemoglobin are correct, that one change is enough to cause the protein to fold into a different shape, which alters its function, which in turn causes disease. Misfolded proteins are associated with several severe brain disorders. For example, the diseases shown in **Figure 3.20** are all caused by prions, misfolded versions of normal brain proteins. Prions can infiltrate the brain, converting normally folded proteins into the abnormal shape. Clustering of the misfolded proteins eventually disrupts brain function.

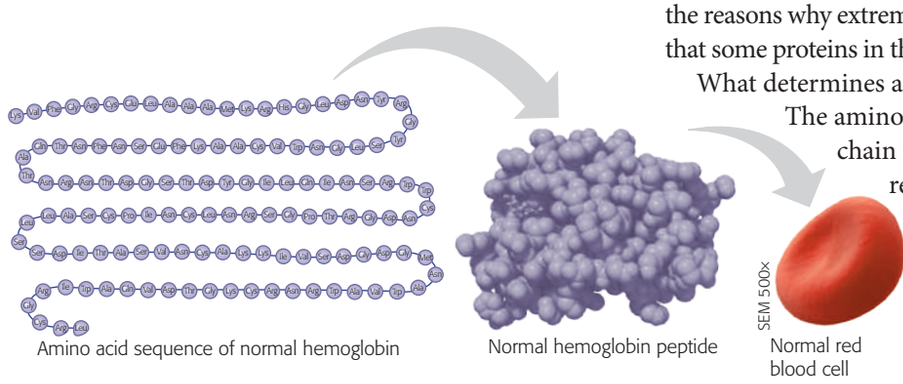
In addition to dependence on the amino acid sequence, a protein's shape is sensitive to the environment. An unfavorable change in temperature, pH, or some other factor can cause a protein to unravel. If you cook an egg, the transformation of the egg white from clear to opaque is caused by proteins in the egg white coming apart. One of the reasons why extremely high fevers are so dangerous is that some proteins in the body lose their shape above 104°F.

What determines a protein's amino acid sequence? The amino acid sequence of each polypeptide chain is specified by a gene. And this relationship between genes and proteins brings us to this chapter's last category of large biological molecules: nucleic acids. ☑

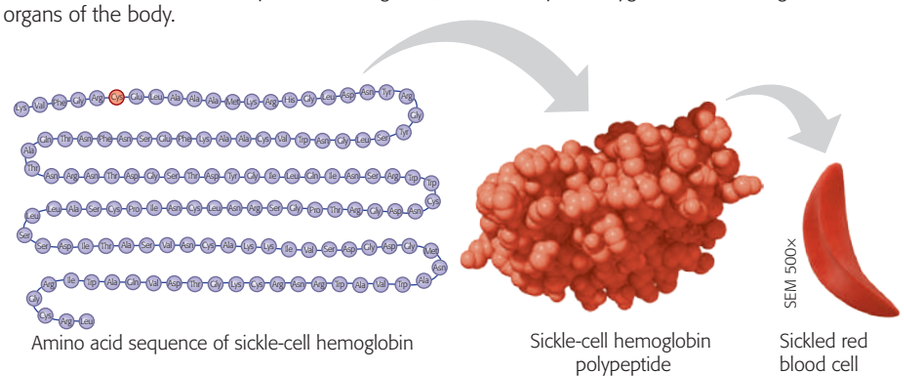
☑ CHECKPOINT

How can changing an amino acid alter the function of a protein?

Answer: Changing an amino acid may alter the shape of the protein, which changes its function.



(a) Normal hemoglobin. Red blood cells of humans are normally disk-shaped. Each cell contains millions of molecules of the protein hemoglobin, which transports oxygen from the lungs to other organs of the body.



(b) Sickle-cell hemoglobin. A slight change in the amino acid sequence of hemoglobin causes sickle-cell disease. The inherited substitution of one amino acid—valine in place of the amino acid glutamic acid—occurs at the 6th position of the polymer. Such abnormal hemoglobin molecules tend to crystallize, deforming some of the cells into a sickle shape. Someone with the disease suffers from dangerous episodes when the angular cells clog tiny blood vessels, impeding blood flow.

► **Figure 3.19 A single amino acid substitution in a protein causes sickle-cell disease.**



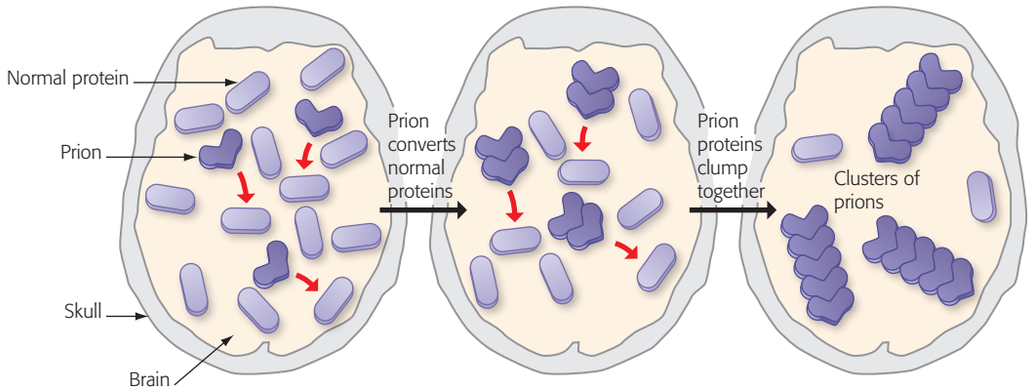
Prions in cattle cause mad cow disease, formally called bovine spongiform encephalopathy (BSE).



First described among one tribe in Papua New Guinea in the 1950s, kuru is a human brain disease transmitted by cannibalistic ingestion of an infected person's brain, which transfers the prion to a new host.



Prions cause fatal weight loss in several animals, including deer, elk, and moose.



► **Figure 3.20 How an improperly folded protein can lead to brain disease.** Here, you can see how prion proteins bring about the destruction of brain tissue, as well as several examples (right) of the resulting disease.

Nucleic Acids

Nucleic acids are macromolecules that store information and provide the instructions for building proteins. The name *nucleic* comes from the fact that DNA is found in the nuclei of eukaryotic cells. There are actually two types of nucleic acids: **DNA** (which stands for deoxyribonucleic acid) and **RNA** (for ribonucleic acid). The genetic material that humans and all other organisms inherit from their parents consists of giant molecules of DNA. The DNA resides in the cell as one or more very long fibers called chromosomes. A **gene** is a unit of inheritance encoded in a specific stretch of DNA that programs the amino acid sequence of a polypeptide. Those programmed instructions, however, are written in a chemical code that must be translated from "nucleic acid language" to "protein language" **(Figure 3.21)**. A cell's RNA molecules help make this translation (see Chapter 10).

Nucleic acids are polymers made from monomers called **nucleotides** **(Figure 3.22)**. Each nucleotide contains three parts. At the center of each nucleotide is a five-carbon sugar (blue in the figure), deoxyribose in DNA and ribose in RNA. Attached to the sugar is a negatively charged phosphate group (yellow) containing a phosphorus atom bonded to oxygen atoms (PO_4^-). Also attached to the sugar is a nitrogen-containing base (green) made of one or two rings. The sugar and phosphate are the same in all nucleotides; only the base varies. Each DNA nucleotide has one of four possible nitrogenous bases: adenine (abbreviated A), guanine (G), cytosine (C), or thymine (T) **(Figure 3.23)**. Thus, all genetic information is written in a four-letter alphabet.

Figure 3.22 A DNA nucleotide. A DNA nucleotide monomer consists of three parts: a sugar (deoxyribose), a phosphate, and a nitrogenous (nitrogen-containing) base.

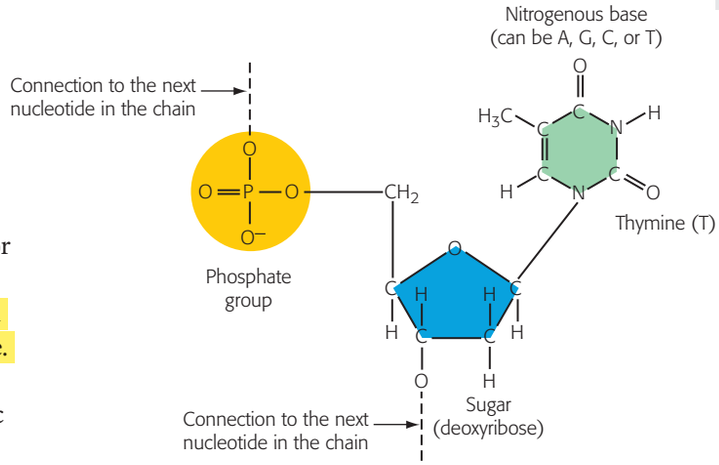


Figure 3.21 Building a protein. Within the cell, a gene (a segment of DNA) provides the directions to build a molecule of RNA, which can then be translated into a protein.

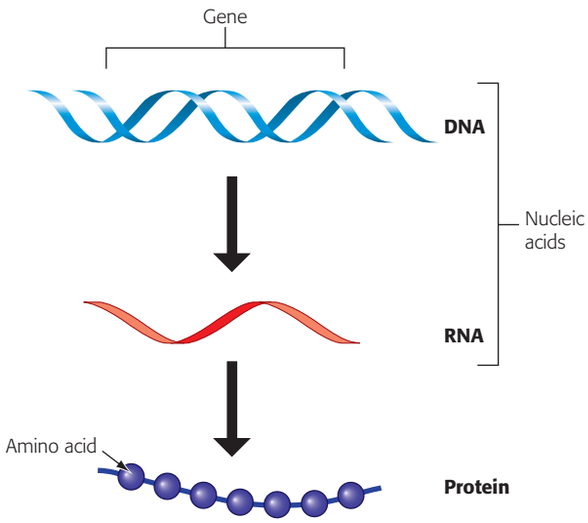
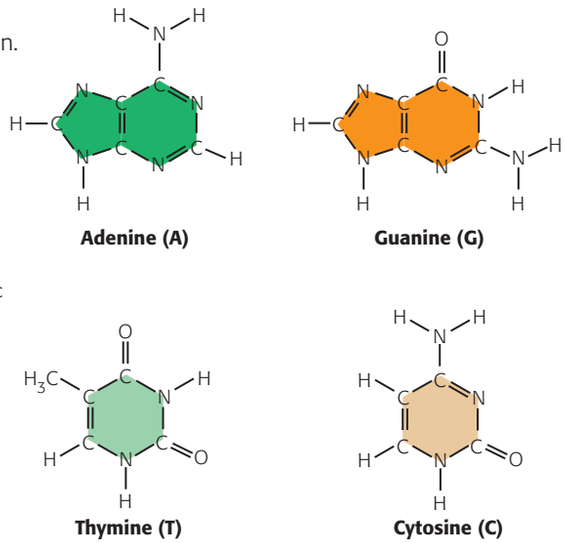
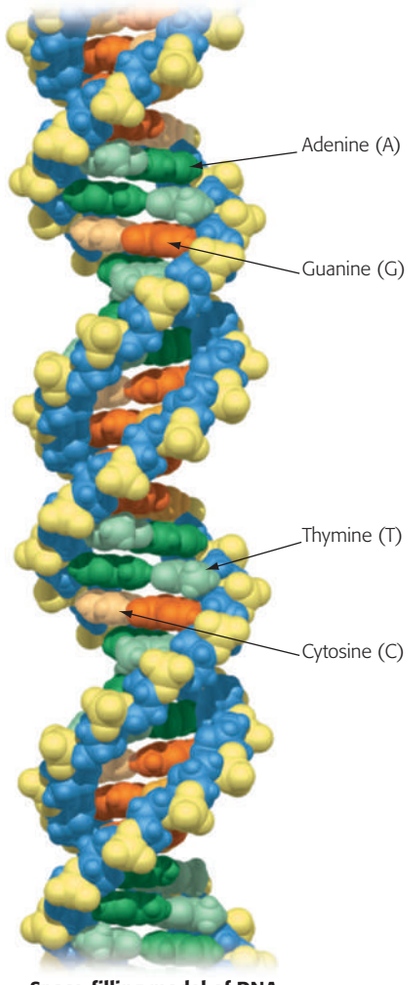


Figure 3.23 The nitrogenous bases of DNA. Notice that adenine and guanine have double-ring structures. Thymine and cytosine have single-ring structures.





Space-filling model of DNA
(showing the four bases in four different colors)

The structure of your DNA is nearly indistinguishable from the DNA of a mosquito or an elephant: The differences in animal species result from the way nucleotides are arranged.

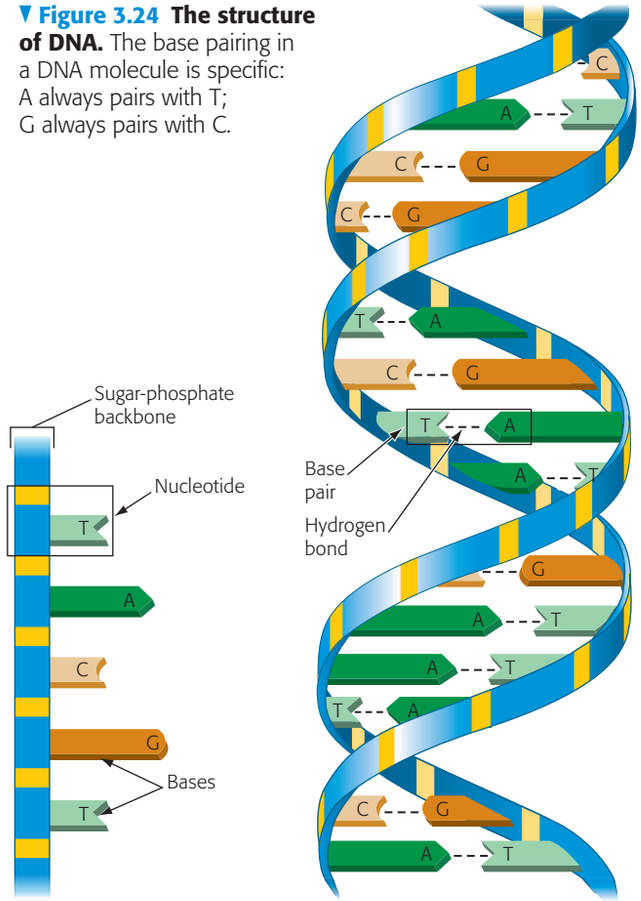
Dehydration reactions link nucleotide monomers into long chains called polynucleotides **(Figure 3.24a)**. In a polynucleotide, nucleotides are joined by covalent bonds between the sugar of one nucleotide and the phosphate of the next. This bonding results in a **sugar-phosphate backbone**, a repeating pattern of sugar-phosphate-sugar-phosphate, with the bases (A, T, C, or G) hanging off the backbone like appendages. With different combinations of the four bases, the number of possible polynucleotide sequences is vast. One long polynucleotide may contain many genes, each a specific series of hundreds or thousands of nucleotides. This sequence is a code that provides instructions for building a specific polypeptide from amino acids.

A molecule of cellular DNA is double-stranded, with two polynucleotide strands coiled around each other to form a **double helix** **(Figure 3.24b)**. Think of a candy cane or a barber pole that has two intertwined spirals, one red and one white. In the central core of the helix (corresponding to the interior of the candy cane), the bases along one DNA strand hydrogen-bond to bases along the other strand. The bonds are individually weak, but collectively they zip the two strands together into a very stable double helix formation. To understand how DNA strands are bonded, think of Velcro, in which two strips are held together by hook-and-loop bonds, each of which is weak but which collectively form a tight grip. Because of the way the functional groups hang off the bases, the base pairing in a DNA double helix is specific: The base A can pair only with T, and G can pair only with C. Thus, if you know the sequence of bases along one DNA strand, you also know the sequence along the complementary strand in the double helix. This unique base pairing is the basis of DNA's ability to act as the molecule of inheritance (as discussed in Chapter 10).

There are many similarities between DNA and RNA. Both are polymers of nucleotides, for example, and both are made of nucleotides consisting of a sugar, a phosphate, and a base. But there are three important differences. (1) As its name *ribonucleic acid* denotes, its sugar is ribose rather than deoxyribose. (2) Instead of the base thymine, RNA has a similar but distinct base called uracil (U) **(Figure 3.25)**. Except for the presence of ribose and uracil, an RNA polynucleotide chain is identical to a DNA polynucleotide chain. (3) RNA is usually found in living cells in single-stranded form, whereas DNA usually exists as a double helix.

Now that we've examined the structure of nucleic acids, we'll look at how a change in nucleotide sequence can affect protein production. To illustrate this point, we'll return to a familiar condition. ☑

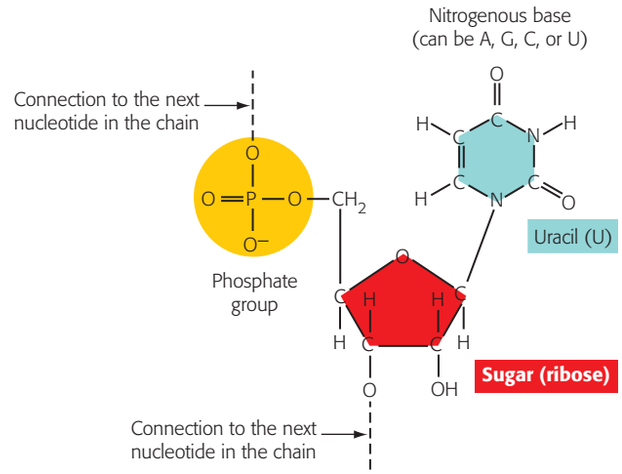
Figure 3.24 The structure of DNA. The base pairing in a DNA molecule is specific: A always pairs with T; G always pairs with C.



(a) DNA strand (polynucleotide)

(b) Double helix (two polynucleotide strands)

Figure 3.25 An RNA nucleotide. Notice that this RNA nucleotide differs from the DNA nucleotide in Figure 3.22 in two ways: The RNA sugar is ribose rather than deoxyribose, and the base is uracil (U) instead of thymine (T). The other three kinds of RNA nucleotides have the bases A, C, and G, as in DNA.



☑ CHECKPOINT

1. DNA contains _____ polynucleotide strands, each composed of _____ kinds of nucleotides. (Provide two numbers.)
2. If one DNA strand has the sequence GAATGC, what is the sequence of the other strand?

Answers: **1.** *two; four* **2.** *CTTACG*

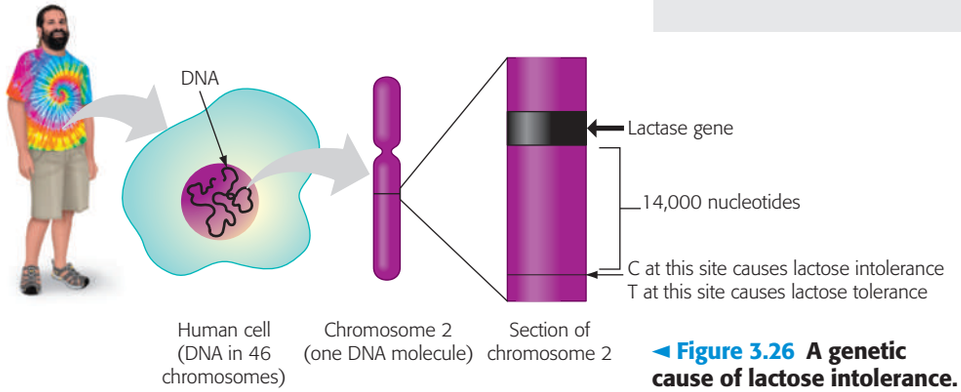


Lactose Intolerance THE PROCESS OF SCIENCE

Does Lactose Intolerance Have a Genetic Basis?

The enzyme lactase, like all proteins, is encoded by a DNA gene. A reasonable hypothesis is that lactose-intolerant people have a defect in their lactase gene. However, this hypothesis is not supported by **observation**. Even though lactose intolerance runs in families, most lactose-intolerant people have a normal version of the lactase gene. This raises the following **question**: What is the genetic basis for lactose intolerance?

A group of Finnish and American scientists proposed the **hypothesis** that lactose intolerance can be correlated with a single nucleotide at a particular site within one chromosome. They made the **prediction** that this site would be near, though not within, the lactase gene. In their **experiment**, they examined the genes of 196 lactose-intolerant people from nine Finnish families. Their **results** showed a 100% correlation between lactose intolerance and a nucleotide at a site approximately 14,000 nucleotides away from the lactase gene—a relatively short distance in terms of the whole chromosome **(Figure 3.26)**. Other experiments showed that depending on the nucleotide sequence within this region of the DNA molecule, the action of the lactase gene is ramped up or down (in a way that likely involves producing a regulatory protein that interacts with the nucleotides near the lactase gene). This study shows how a small change in a DNA nucleotide sequence can have a major effect on the production of a protein and the well-being of an organism.



◄ Figure 3.26 A genetic cause of lactose intolerance. A research study showed a correlation between lactose intolerance and a nucleotide at a specific location on one chromosome.



Lactose Intolerance EVOLUTION CONNECTION

The Evolution of Lactose Intolerance in Humans

As you'll recall from the Biology and Society section, most people in the world are lactose intolerant as adults and thus do not easily digest the milk sugar lactose. In fact, lactose intolerance is found in 80% of African Americans and Native Americans and 90% of Asian Americans but only in about 10% of Americans of northern European descent. And as discussed in the Process of Science section, lactose intolerance appears to have a genetic basis.

From an evolutionary perspective, it is reasonable to infer that lactose intolerance is rare among northern Europeans because the ability to tolerate lactose offered a survival advantage to their ancestors. In northern Europe's relatively cold climate, only one crop harvest a year is possible. Therefore, herd animals were a main source of food for early humans in that region. Cattle were first domesticated in northern Europe about 9,000 years ago **(Figure 3.27)**. With milk and other dairy products at hand year-round, natural selection would have favored anyone with a mutation that kept the lactase gene switched on beyond infanthood. In cultures where dairy products were not a staple in the diet, natural selection would not favor such a mutation.

Researchers wondered whether the genetic basis for lactose tolerance in northern Europeans might be present in other cultures that kept dairy herds. To find out, they compared the genetic makeup and lactose tolerance of 43 ethnic groups in East Africa. The researchers identified three other genetic changes that keep the lactase gene permanently active. These mutations appear to have occurred beginning around 7,000 years ago, about the time that archaeological evidence shows domestication of cattle in these African regions.

Genetic changes that confer a selective advantage, such as surviving cold winters or withstanding drought by drinking milk, spread rapidly in these early peoples. Whether or not you can digest milk is therefore an evolutionary record of the cultural history of your ancestors.



▲ Figure 3.27 A prehistoric cave painting in Lascaux, France, of wild cattle. The cow-like animal on the right is an auroch, the first species of domesticated cattle in Europe. Aurochs migrated from Asia about 250,000 years ago but became extinct in 1627.

Chapter Review

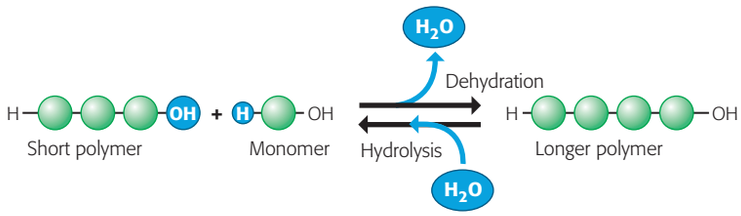
SUMMARY OF KEY CONCEPTS

Organic Compounds

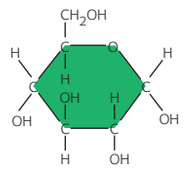
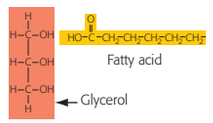
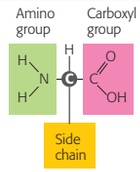
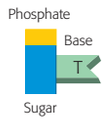
Carbon Chemistry

Carbon atoms can form large, complex, diverse molecules by bonding to four potential partners, including other carbon atoms. In addition to variations in the size and shape of carbon skeletons, organic compounds vary in the presence and locations of different functional groups.

Giant Molecules from Smaller Building Blocks



Large Biological Molecules

Large Biological Molecules	Functions	Components	Examples
Carbohydrates	Dietary energy; storage; plant structure	 Monosaccharide	Monosaccharides: glucose, fructose; disaccharides: lactose, sucrose; polysaccharides: starch, cellulose
Lipids	Long-term energy storage (fats); hormones (steroids)	 Fatty acid Glycerol Components of a triglyceride	Fats (triglycerides); steroids (testosterone, estrogen)
Proteins	Enzymes, structure, storage, contraction, transport, etc.	 Amino group Carboxyl group Side chain Amino acid	Lactase (an enzyme); hemoglobin (a transport protein)
Nucleic acids	Information storage	 Phosphate Base T Sugar Nucleotide	DNA, RNA

Carbohydrates

Simple sugars (monosaccharides) provide cells with energy and building materials. Double sugars (disaccharides), such as sucrose, consist of two monosaccharides joined by a dehydration reaction. Polysaccharides are long polymers of sugar monomers. Starch in plants and glycogen in animals are storage polysaccharides. The cellulose of plant cell walls, which is indigestible by animals, is an example of a structural polysaccharide.

Lipids

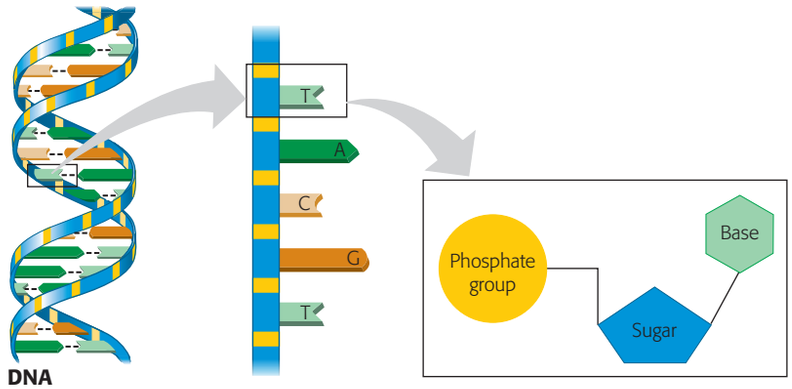
Lipids are hydrophobic. Fats, a type of lipid, are the major form of long-term energy storage in animals. A molecule of fat, or triglyceride, consists of three fatty acids joined by dehydration reactions to a molecule of glycerol. Most animal fats are saturated, meaning that their fatty acids have the maximum number of hydrogens. Plant oils contain mostly unsaturated fats, having fewer hydrogens in the fatty acids because of double bonding in the carbon skeletons. Steroids, including cholesterol and the sex hormones, are also lipids.

Proteins

There are 20 types of amino acids, the monomers of proteins. They are linked by dehydration reactions to form polymers called polypeptides. A protein consists of one or more polypeptides folded into a specific three-dimensional shape. The shape of a protein determines its function. Changing the amino acid sequence of a polypeptide may alter the shape and therefore the function of the protein. Shape is sensitive to environment, and if a protein loses its shape because of an unfavorable environment, its function may also be lost.

Nucleic Acids

Nucleic acids include RNA and DNA. DNA takes the form of a double helix, two DNA strands (polymers of nucleotides) held together by hydrogen bonds between nucleotide components called bases. There are four kinds of DNA bases: adenine (A), guanine (G), thymine (T), and cytosine (C). A always pairs with T, and G always pairs with C. These base-pairing rules enable DNA to act as the molecule of inheritance. RNA has U (uracil) instead of T.



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SELF-QUIZ

1. One isomer of methamphetamine is the addictive illegal drug known as "crank." Another isomer is a medicine for sinus congestion. How can you explain the differing effects of the two isomers?

Reason for Different Effects
The divergent effects occur because:

Biological molecules like enzymes and receptors recognize only one isomer more effectively, just like a lock fits only a specific key shape.
The d-isomer fits and activates brain receptors involved in mood and addiction very strongly, while the l-isomer mainly affects other pathways (like those for nasal congestion) with weak central effects.

2. Monomers are joined together to form larger polymers through _____. Such a reaction releases a molecule of _____.

3. Polymers are broken down into the monomers that make them up through the chemical reaction called _____.

4. Table sugar is _____.
- **a.** glucose, a monosaccharide
- **b.** glucose, a disaccharide
- **c.** sucrose, a monosaccharide
- **d.** sucrose, a disaccharide

5. When two molecules of glucose ($C_6H_{12}O_6$) are joined together by a dehydration reaction, what are the formulas of the two products? (*Hint*: No atoms are gained or lost.)

6. One molecule of dietary fat is made by joining three molecules of _____ to one molecule of _____. What is the formal name of the resulting molecule?

7. By definition, what type of fatty acid has double bonds?
- **a.** steroid
- **b.** triglyceride
- **c.** unsaturated
- **d.** saturated

8. Humans and other animals cannot digest wood because they
- **a.** cannot digest any carbohydrates.
- **b.** cannot chew it fine enough.
- **c.** lack the enzyme needed to break down cellulose.
- **d.** get no nutrients from it.

9. Explain how it could be possible to change an amino acid within a protein but not affect that protein's function.

10. Most proteins can easily dissolve in water. Knowing that, where within the overall three-dimensional shape of a protein would you most likely find hydrophobic amino acids?

11. A shortage of phosphorus in the soil would make it especially difficult for a plant to manufacture
- **a.** DNA.
- **b.** proteins.
- **c.** cellulose.
- **d.** fatty acids.

12. Nucleic acids are polymers of _____ monomers.

13. Name three similarities between DNA and RNA. Name three differences.

14. What is the structure of a gene? What is the function of a gene?

Answers to these questions can be found in Appendix: Self-Quiz Answers.

THE PROCESS OF SCIENCE

15. A food manufacturer is advertising a new cake mix as fat-free. Scientists at the Food and Drug Administration (FDA) are testing the product to see if it truly lacks fat. Hydrolysis of the cake mix yields glucose, fructose, glycerol, a number of amino acids, and several kinds of molecules with long chains. Further analysis shows that most of the chains have a carboxyl group at one end. What would you tell the food manufacturer if you were a spokesperson for the FDA?

16. Based on your knowledge of the different types of chemical bonds (see Chapter 2) and of the molecular structures involved, predict which of the families of macromolecules discussed in this chapter are soluble in water and which are not, and explain why.

17. **Interpreting Data** Below is a typical food label for one cookie. One gram of fat packs 9 Calories, and 1 gram of carbohydrates or protein packs 4 Calories. The top of the label shows that each cookie contains 140 Calories in total. What percentage of the Calories in this cookie are from fat, carbohydrates, and protein?

Nutrition Facts
Serving Size 1 Cookie (28 g /1 oz)
Servings Per Container 8

Amount Per Serving	
Calories 140	Calories from Fat 60
	% Daily Value*
Total Fat 7g	**11%**
Saturated Fat 3g	**15%**
Trans Fat 0g	
Cholesterol 10mg	**3%**
Sodium 80mg	**3%**
Total Carbohydrate 18g	**6%**
Dietary Fiber 1g	**4%**
Sugars 10g	
Protein 2g	

BIOLOGY AND SOCIETY

18. Some amateur and professional athletes take anabolic steroids to help them build strength ("bulk up"). The health risks of this practice are extensively documented. Apart from these health issues, what is your opinion about the ethics of athletes using chemicals to enhance performance? Is this a form of cheating, or is it just part of the preparation required to stay competitive in a sport where anabolic steroids are commonly used? Defend your opinion.

19. Heart disease is the leading cause of death among people in the United States and other industrialized nations. Fast food is a major source of unhealthy fats that contribute significantly to heart disease. Imagine you're a juror sitting on a trial where a fast-food manufacturer is being sued for producing a harmful product. To what extent do you think manufacturers of unhealthy foods should be held responsible for the health consequences of their products? As a jury member, how would you vote?

20. The production of genetically modified (GM) organisms is a controversial topic, and many consumers avoid products that contain GM food. Based on what you have learnt about the chemical nature of DNA and proteins, do you think there is a health risk associated with the consumption of GM food? What may be the wider implications of the production and consumption of such food?

4 A Tour of the Cell

Why Cells Matter



Mushrooms, amoebas, and you are all made up of the same kind of cells.

The caffeine that gives your cup of tea a kick also protects tea plants from herbivores.





Without the cytoskeleton, your cells would collapse in on themselves, much like a house collapses when the infrastructure fails.

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CHAPTER THREAD

Humans Versus Bacteria

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THE PROCESS OF SCIENCE **What Makes a Superbug?** 95
EVOLUTION CONNECTION **The Evolution of Bacterial Resistance in Humans** 105

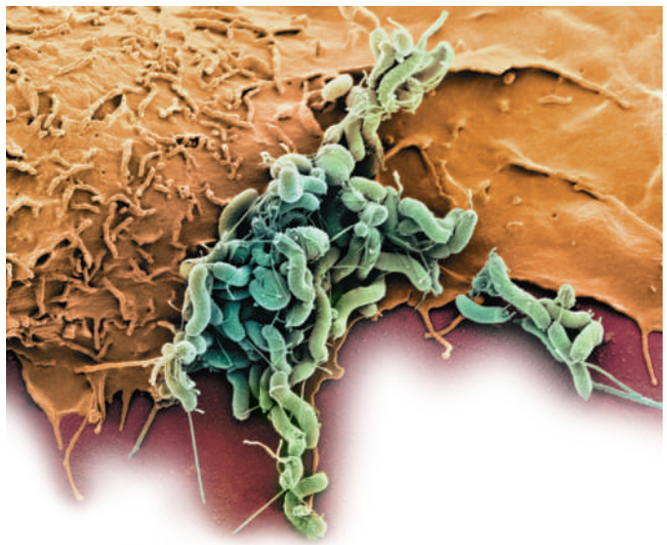


Humans Versus Bacteria BIOLOGY AND SOCIETY

Antibiotics: Drugs That Target Bacterial Cells

Antibiotics—drugs that disable or kill infectious bacteria—are a marvel of modern medicine. The first antibiotic to be discovered was penicillin in 1920. A revolution in human health rapidly followed. Fatality rates of many diseases (such as bacterial pneumonia and surgical infections) plummeted, saving millions of lives. In fact, human health care improved so quickly and so profoundly that some doctors in the early 1900s predicted the end of infectious diseases altogether. Alas, this did not come to pass (see the Evolution Connection section in Chapter 13 for a discussion of why infectious diseases were not so easily defeated).

The goal of antibiotic treatment is to knock out invading bacteria while doing no damage to the human host. So how does an antibiotic zero in on its target among trillions of human cells? Most antibiotics are so precise because they bind to structures found only in bacterial cells. For example, the common antibiotics erythromycin and streptomycin bind to the bacterial ribosome, a vital cellular structure responsible for the production of proteins. The ribosomes of humans are different enough from those of bacteria that the antibiotics bind only to bacterial ribosomes, leaving human ribosomes unaffected. Ciprofloxacin (commonly referred to as Cipro) is the antibiotic of choice to combat anthrax-causing bacteria. This drug targets an enzyme that bacteria need to maintain their chromosome structure. Your cells can survive just fine in the presence of Cipro because human chromosomes have a sufficiently different makeup than bacterial chromosomes. Other drugs, such as penicillin, ampicillin, and bacitracin, disrupt the synthesis of cell walls, a structure found in most bacteria that is absent from the cells of humans and other animals.



Colorized TEM 3,200×

Two kinds of cells. In this micrograph, *Helicobacter pylori* bacteria (green) can be seen mingling with cells within the human stomach. This species of bacteria causes stomach ulcers.

This discussion of how various antibiotics target bacteria underscores the main point of this chapter: To understand how life works—whether in bacteria or in your own body—you first need to learn about cells. On the scale of biological organization, cells occupy a special place: They are the simplest objects that can be alive. Nothing smaller than a cell is capable of displaying all of life's properties. In this chapter, we'll explore the microscopic structure and function of cells. Along the way, we'll further consider how the ongoing battle between humans and infectious bacteria is affected by the cellular structures present on both sides.

The Microscopic World of Cells



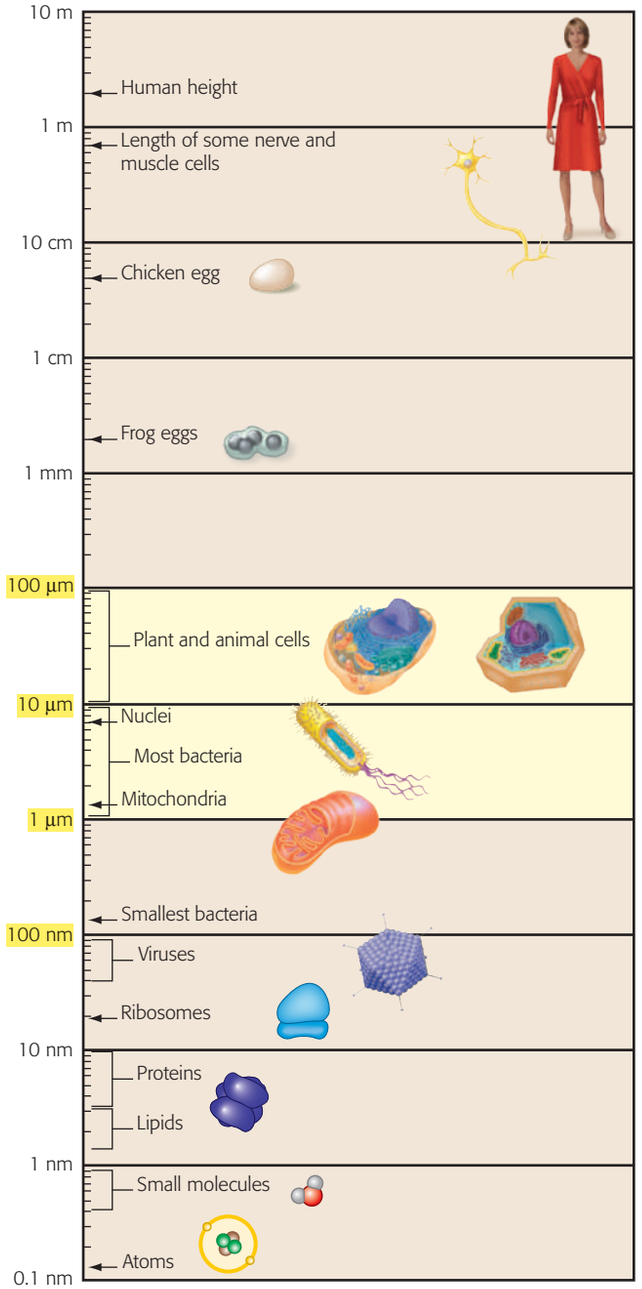
100 µm - 10 µm

Each cell in your body is a miniature marvel. If the world's most sophisticated jumbo jet was reduced to microscopic size, its intricacy would pale next to a living cell.

Organisms are either single-celled, such as most prokaryotes and protists, or multicelled, such as plants, animals, and most fungi. Your own body is a cooperative society of trillions of cells of many specialized types. As you read this page, muscle cells allow you to scan your eye across the words, while sensory cells in your eye gather information and send it to brain cells, which interpret the words. Everything you do—every action and every thought—is possible because of processes that occur at the cellular level.

Figure 4.1 shows the size range of cells compared with objects both larger and smaller. Notice that the scale along the left side of the figure increases by powers of 10 to accommodate the range of sizes shown. Starting at the top with 10 meters (m), each subsequent mark represents a tenfold decrease in length. Most cells are between 1 and 100 µm in diameter (yellow region of the figure) and are therefore visible only with a microscope. There are some interesting exceptions: an ostrich egg is a single cell about 6 inches across and weighing about 3 pounds; nerve cells in your body can stretch several feet long; and nerve cells in giant squid can be more than 30 feet long!

How do new living cells arise? First formulated in the 1800s, the **cell theory** states that all living things are composed of cells and that all cells come from earlier cells. So every cell in your body (and in every other living organism on Earth) was formed by division of a previously living cell. (That raises an obvious question: how did the first cell evolve? This fascinating topic will be addressed in Chapter 15.) With that introduction, let's begin to explore the variety of cells found among life on Earth.



Measurement Equivalents

1 meter (m) = 100 cm = 1,000 mm = about 39.4 inches

1 centimeter (cm) = 10^{-2} $\left(\frac{1}{100}\right)$ m = about 0.4 inch

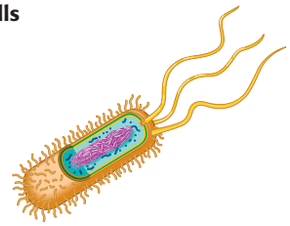
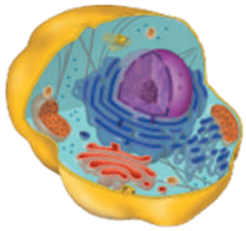
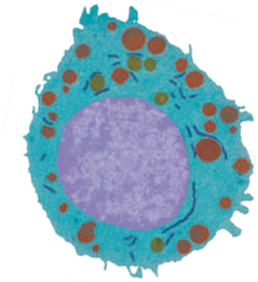
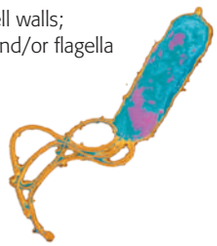
1 millimeter (mm) = 10^{-3} $\left(\frac{1}{1{,}000}\right)$ m = $\frac{1}{10}$ cm

1 micrometer (µm) = 10^{-6} m = 10^{-3} mm

1 nanometer (nm) = 10^{-9} m = 10^{-3} µm

▲ **Figure 4.1 The size range of cells.** Starting at the top of this scale with 10 m (10 meters) and going down, each measurement along the left side marks a tenfold decrease in size.

Table 4.1 Comparing Prokaryotic and Eukaryotic Cells

Prokaryotic cells	Eukaryotic cells
	
Plasma membrane of identical structure	
Cytoplasm occupies entire interior of cell	Cytoplasm occupies the region between the nucleus and the plasma membrane
Single circular chromosome in nucleoid region	One or more linear chromosomes in nucleus 
Both have ribosomes, but structure differs slightly	
First evolved approximately 3.5 billion years ago	First evolved approximately 2.1 billion years ago
Smaller, simpler	Larger, more complex
No membrane-bound organelles	Membrane-bound organelles (for example, nucleus, ER)
Most are surrounded by cell walls; some have capsules, pili, and/or flagella 	Plant cells surrounded by cell walls; animal cells surrounded by extracellular matrix

The Two Major Categories of Cells

The countless cells that exist on Earth can be placed into two basic categories: prokaryotic cells and eukaryotic cells (Table 4.1). **Prokaryotic cells** are found in organisms of the domains Bacteria and Archaea, known as prokaryotes (see Figure 1.7). Organisms of the domain Eukarya—including protists, plants, fungi, and animals—are composed of **eukaryotic cells** and are called eukaryotes.

All cells, whether prokaryotic or eukaryotic, have several features in common. They are all bounded by a barrier called a **plasma membrane**, which regulates the traffic of molecules between the cell and its surroundings. Inside all cells is a thick, jellylike fluid called the **cytosol**, in which cellular components are suspended. All cells have one or more **chromosomes** carrying genes made of DNA. And all cells have **ribosomes** that build proteins according to instructions from the genes. Because of structural differences between bacteria and eukaryotes, as mentioned in the Biology and Society section at the beginning of the chapter, some antibiotics—such as streptomycin—target prokaryotic ribosomes, crippling protein synthesis in the bacterial invaders but not in the eukaryotic host (you).

Mushrooms, amoebas, and you are all made up of the same kind of cells.

Although they have many similarities, prokaryotic and eukaryotic cells differ in several important ways. Fossil evidence shows that prokaryotes were the first life on Earth, appearing more than

☑ CHECKPOINT

1. Name four structures found in both prokaryotic and eukaryotic cells.
2. How is the nucleoid region of a prokaryotic cell different from the nucleus of a eukaryotic cell?

Answers: **1.** *plasma membrane, chromosomes, cytosol, and ribosomes* **2.** *There is no membrane enclosing the prokaryotic nucleoid region.*

3.5 billion years ago. In contrast, the first eukaryotes did not appear until around 2.1 billion years ago. Prokaryotic cells are usually much smaller—about one-tenth the length of a typical eukaryotic cell—and are simpler in structure. Think of a prokaryotic cell as being like a bicycle, whereas a eukaryotic cell is like a sports utility vehicle. Both a bike and an SUV get you from place to place, but one is much smaller and contains many fewer parts than the other. Similarly, prokaryotic cells and eukaryotic cells perform similar functions, but prokaryotes cells are much smaller and less complex. The most significant structural difference between the two types of cells is that eukaryotic cells have **organelles** ("little organs"), membrane-enclosed structures that perform specific functions, and prokaryotic cells do not. The most important organelle is the **nucleus**, which houses most of a eukaryotic cell's DNA. The nucleus is surrounded by a double membrane. A prokaryotic cell lacks a nucleus; its DNA is coiled into a "nucleus-like" region called the **nucleoid**, which is not partitioned from the rest of the cell by membranes.

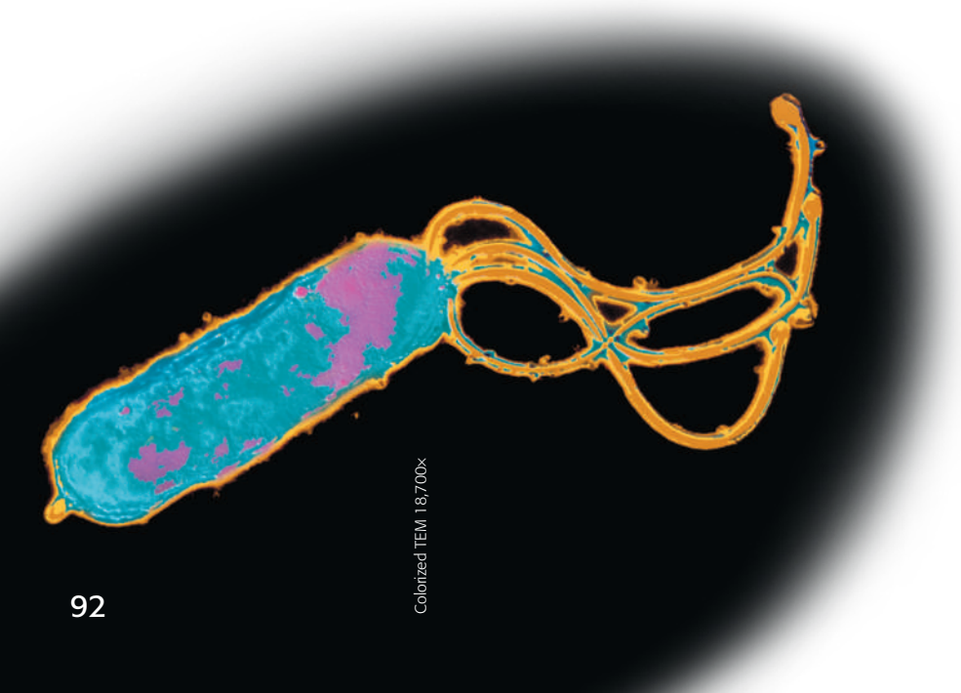
Consider this analogy: A eukaryotic cell is like an office building that is separated into cubicles. Within each cubicle, a specific function is performed, thus dividing the labor among many internal compartments. One cubicle may hold the accounting department, for example, while another is home to the sales force. The "cubicle walls" within eukaryotic cells are made from membranes that help maintain a unique chemical environment inside each cubicle. In contrast, the interior of a prokaryotic cell is like an open warehouse. The spaces for specific tasks within a "prokaryotic warehouse" are distinct but they are not separated by physical barriers.

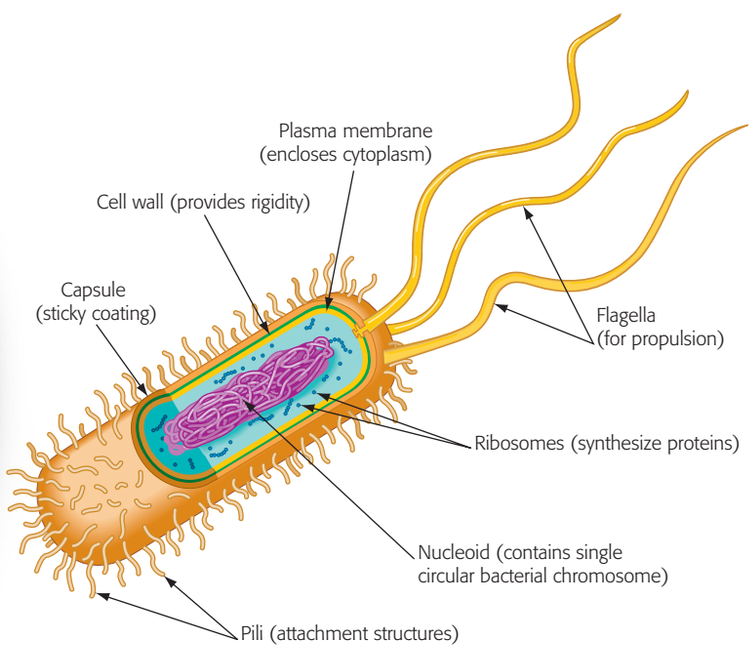
Figure 4.2 depicts an idealized prokaryotic cell and a micrograph of an actual bacterium. Surrounding the plasma membrane of most prokaryotic cells is a rigid cell wall, which protects the cell and helps maintain its shape. Recall from the Biology and Society section at the beginning of the chapter that bacterial cell walls are the targets of some antibiotics. In some prokaryotes, a sticky outer coat called a capsule surrounds the cell wall. Capsules provide protection and help prokaryotes stick to surfaces and to other cells in a colony. For example, capsules help bacteria in your mouth stick together to form harmful dental plaque. Prokaryotes can have short projections called pili, which can also attach to surfaces. Many prokaryotic cells have flagella, long projections that propel them through their liquid environment. ☑

An Overview of Eukaryotic Cells

All eukaryotic cells—whether from animals, plants, protists, or fungi—are fundamentally similar to one another and quite different from prokaryotic cells. **Figure 4.3** provides overviews of an idealized animal cell and plant cell. No real cell looks quite like these because living cells have many more copies of most of the structures shown; each of your cells have hundreds of mitochondria and millions of ribosomes, for example. To keep from getting lost on our tour of the cell, throughout this chapter we'll use miniature versions of the diagrams in Figure 4.3 as road maps, highlighting the structure we're

▼ **Figure 4.2 A prokaryotic cell.** A drawing of an idealized prokaryotic cell (right) is shown alongside a micrograph of *Helicobacter pylori* (left), a bacterium that causes stomach ulcers.

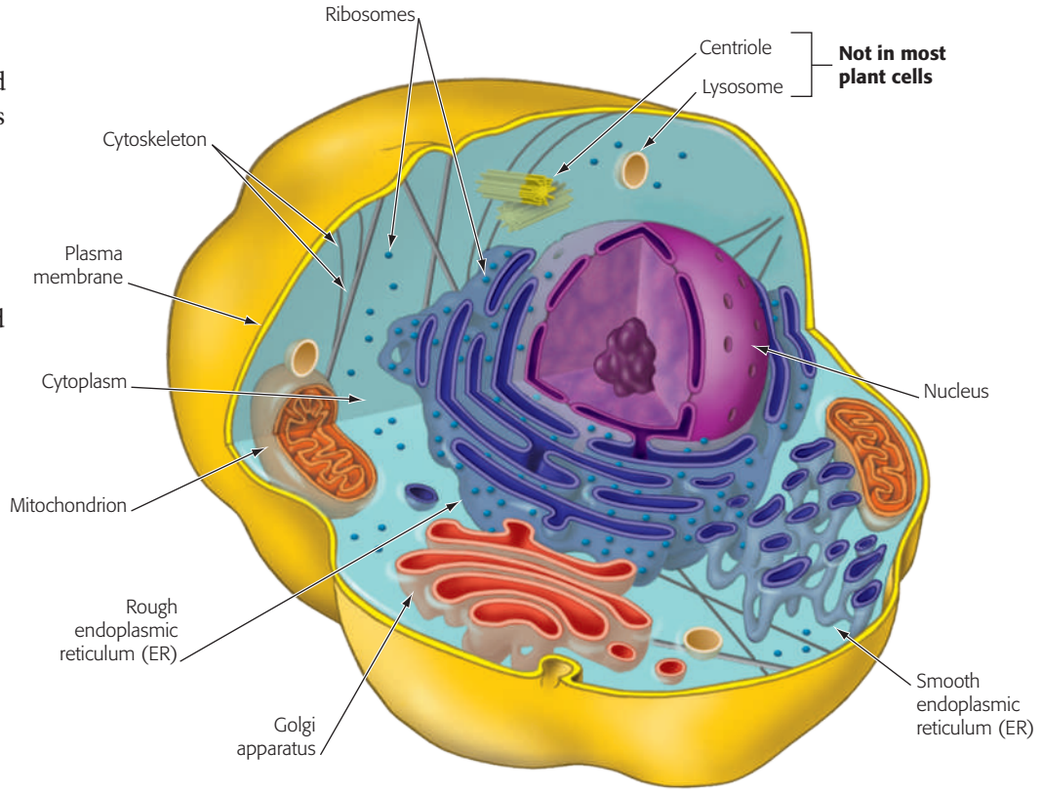




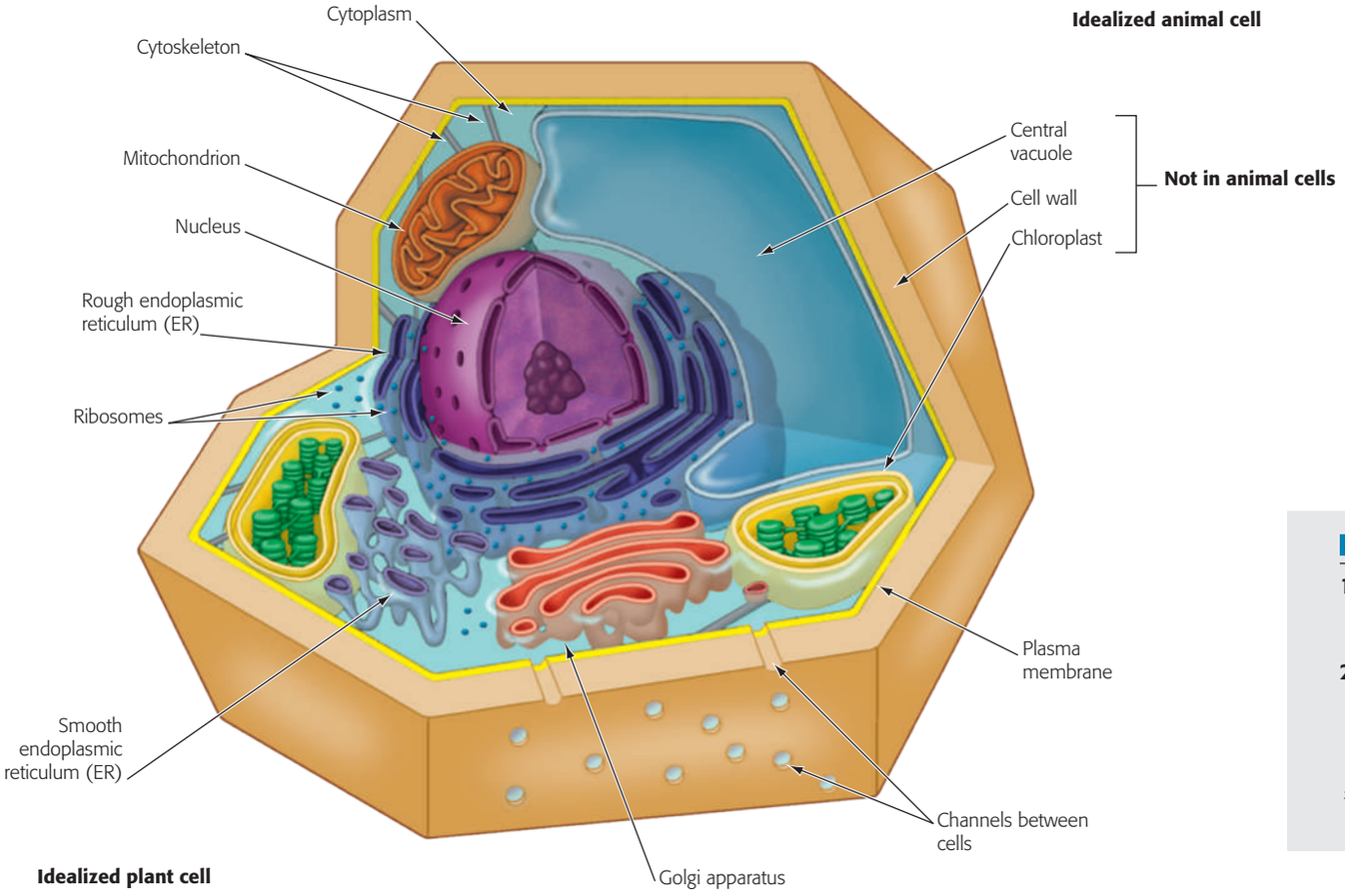
discussing. Notice that the structures are color-coded; we'll use this color scheme throughout this book.

The region of the cell outside the nucleus and within the plasma membrane is called the **cytoplasm**. (This term is also used to refer to the interior of a prokaryotic cell.) The cytoplasm of a eukaryotic cell consists of various organelles suspended in the liquid cytosol. As you can see in Figure 4.3, most organelles are found in both animal and plant cells. But you'll notice some important differences—for example, only plant cells have chloroplasts (where photosynthesis occurs) and a cell wall (which provides stiffness to plant structures), and only animal cells have lysosomes (bubbles of digestive enzymes surrounded by membranes). In the rest of this section, we'll take a closer look at the architecture of eukaryotic cells, beginning with the plasma membrane. ✓

▼ Figure 4.3 An idealized animal cell and plant cell. For now, the labels on the drawings are just words, but these organelles will come to life as we take a closer look at how each part of the cell functions.



Idealized animal cell



Idealized plant cell

✓ CHECKPOINT

1. Name three structures in plant cells that animal cells lack.
2. Name two structures that may be found in animal cells but not in plant cells.

Answers: **1.** *chloroplasts, a central vacuole, and a cell wall* **2.** *centrioles and lysosomes*

Membrane Structure

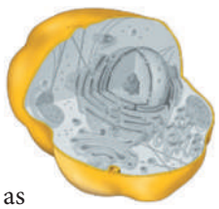
Before we enter the cell to explore the organelles, let's make a quick stop at the surface of this microscopic world: the plasma membrane. To help you understand the structure and function of the cell membrane, imagine you want to create a new homestead in the wilderness. You will probably want to start by fencing your property to protect it from the outside world. Similarly, the plasma membrane is the boundary that separates the living cell from its nonliving surroundings. The plasma membrane is a remarkable film, so thin that you would have to stack 8,000 of these membranes to equal the thickness of one piece of paper. Yet the plasma membrane can regulate the traffic of chemicals into and out of the cell. As with all things biological, the structure of the plasma membrane correlates with its function.

CHECKPOINT

What function does the organization of phospholipids into a bilayer in water serve?

Answer: The bilayer structure shields the hydrophobic tails of the phospholipids from water while exposing the hydrophilic heads to water.

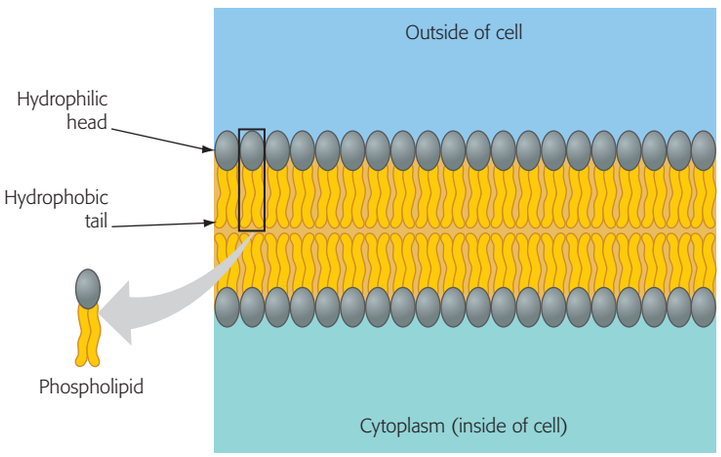
Structure/Function The Plasma Membrane



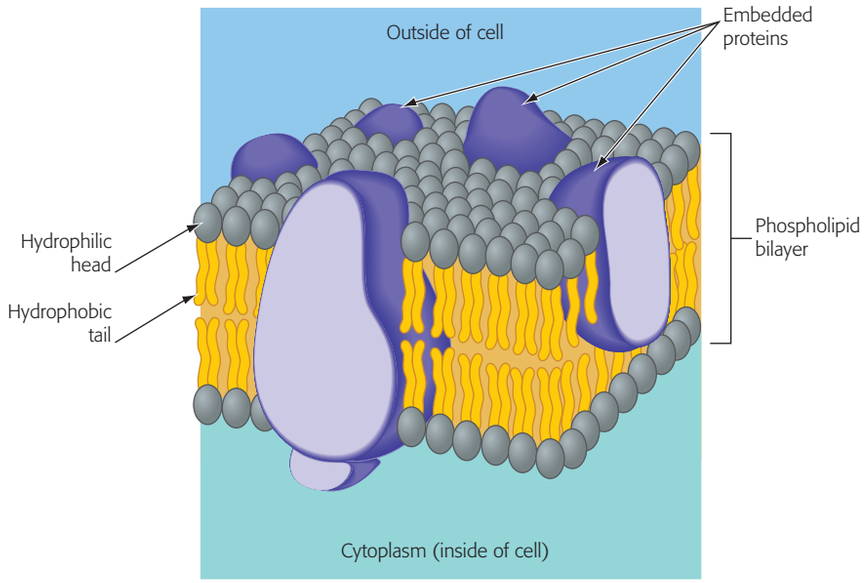
The plasma membrane and other membranes of the cell are composed mostly of **phospholipids**. The structure of phospholipid molecules is well suited to their function as a major constituent of biological membranes. Each phospholipid is composed of two distinct regions—a "head" with a negatively charged phosphate group and two nonpolar fatty acid "tails." Phospholipids group together to form a two-layer sheet called a **phospholipid bilayer**. As you can see in **Figure 4.4a**, the phospholipids' hydrophilic ("water-loving") heads are arranged to face outward, exposed to the aqueous solutions on both sides of a membrane. Their hydrophobic ("water-fearing") tails are arranged inward, mingling with each other and shielded from water. Suspended in the phospholipid bilayer of most membranes are proteins that help regulate traffic across the membrane; these proteins also perform other functions **(Figure 4.4b)**. (You'll learn more about membrane proteins in Chapter 5.)

Membranes are not static sheets of molecules locked rigidly in place, however. In fact, the texture of a cellular membrane is similar to salad oil. The phospholipids and most of the proteins can therefore drift about within the membrane. Thus, a membrane is a **fluid mosaic**—fluid because the molecules can move freely past one another and mosaic because of the diversity of proteins that float like icebergs in the phospholipid sea. Next, we'll see how some bacteria can cause illness by piercing the plasma membrane. ✓

▼ Figure 4.4 The plasma membrane structure.



(a) Phospholipid bilayer of membrane. In water, phospholipids arrange themselves into a bilayer. The symbol for a phospholipid that we'll use in this book looks like a lollipop with two wavy sticks. The "head" is the end with the phosphate group, and the two "tails" are chains of carbon and hydrogen. The bilayer arrangement keeps the heads exposed to water while keeping the tails in the oily interior of the membrane.



(b) Fluid mosaic model of membrane. Membrane proteins, like the phospholipids, have both hydrophilic and hydrophobic regions.



Humans Versus Bacteria THE PROCESS OF SCIENCE

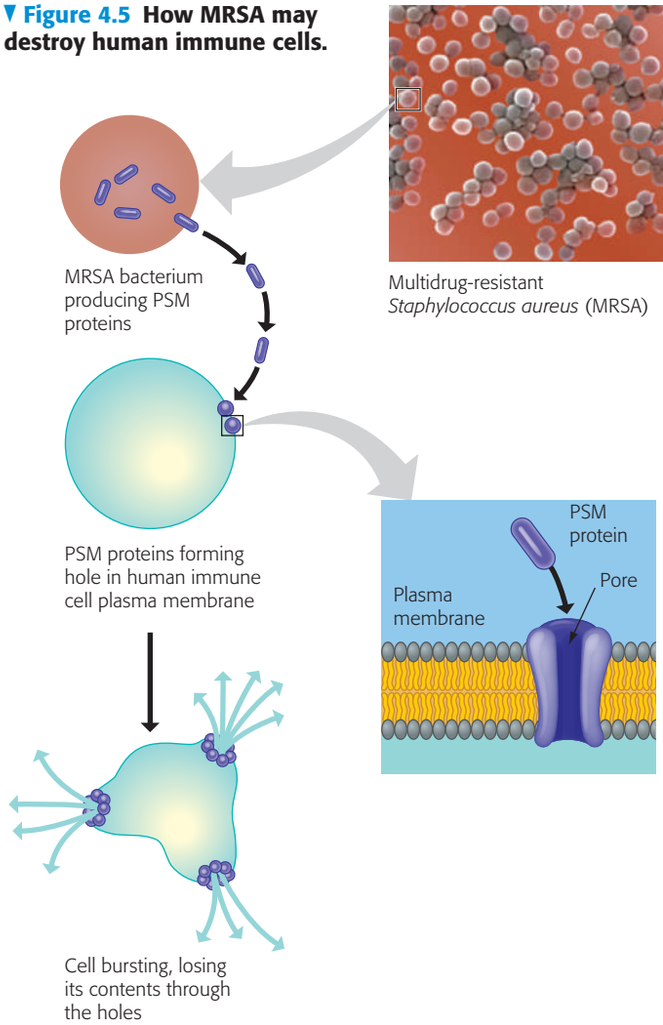
What Makes a Superbug?

Some bacteria cause disease by rupturing the plasma membrane of human immune cells. One example is a common bacteria called *Staphylococcus aureus* (commonly referred to as "staph" or SA). These bacteria, found on your skin, are usually harmless but may multiply and spread, causing a "staph infection." Staph infections typically occur in hospitals and can cause serious, even life-threatening conditions, such as pneumonia or necrotizing fasciitis ("flesh-eating disease").

Most staph infections can be treated with antibiotics. But particularly dangerous strains of *S. aureus*—known as MRSA (for multidrug-resistant SA)—are unaffected by all of the commonly used antibiotics. In recent years, MRSA infections have become more common in hospitals, gyms, and schools. In one study, scientists from the National Institutes of Health (NIH) studied a particular deadly MRSA strain. They began with the **observation** that other bacteria use a protein called PSM to disable human immune cells by forming holes that rip apart the plasma membrane. This observation led them to **question** whether PSM plays a role in MRSA infection **(Figure 4.5)**. Their **hypothesis** was that MRSA bacteria lacking the ability to produce PSM would be less deadly than normal MRSA strains that produced PSM.

In their **experiment**, scientists infected seven mice with a normal MRSA strain and eight mice with a MRSA strain genetically engineered to not produce PSM. The **results** were striking: All seven mice infected with the normal MRSA strain died, while five of the eight mice infected with the strain that did not produce PSM survived. Immune cells from all of the dead mice had holes within the plasma membrane. The researchers concluded that normal MRSA strains use the membrane-destroying PSM protein, but other factors must come into play because three mice died even in the absence of PSM. The deadly effects of MRSA are therefore a reminder of the critical role played by the plasma membrane and another example of the ongoing battle between humans and disease-causing bacteria.

▼ Figure 4.5 How MRSA may destroy human immune cells.



Cell Surfaces

Surrounding their plasma membranes, plant cells have a cell wall made from cellulose fibers, which are long chains of polysaccharide (see Figure 3.9c). The walls protect the cells, maintain cell shape, and keep cells from absorbing so much water that they burst. Plant cells are connected to each other via channels that pass through the cell walls, joining the cytoplasm of each cell to that of its neighbors. These channels allow water and other small molecules to move between cells, integrating the activities of a tissue.

Animal cells lack a cell wall, but most animal cells secrete a sticky coat called the **extracellular matrix**. Fibers made of the protein collagen (also found in your skin, cartilage, bones, and tendons) hold cells together in tissues and can also have protective and supportive functions. In addition, the surfaces of most animal cells contain cell junctions, structures that connect cells together into tissues, allowing the cells to function in a coordinated way. ✓

✓ CHECKPOINT

What polysaccharide is the primary component of plant cell walls?

Answer: cellulose

The Nucleus and Ribosomes: Genetic Control of the Cell

If you think of the cell as a factory, then the nucleus is its control center. Here, the master plans are stored, orders are given, changes are made in response to external factors, and the process of making new factories is initiated. The factory supervisors are the genes, the inherited DNA molecules that direct almost all the business of the cell. Each gene is a stretch of DNA that stores the information necessary to produce a particular protein. Proteins can be likened to workers on the factory floor because they do most of the actual work of the cell.

CHECKPOINT

What is the relationship between chromosomes, chromatin, and DNA?

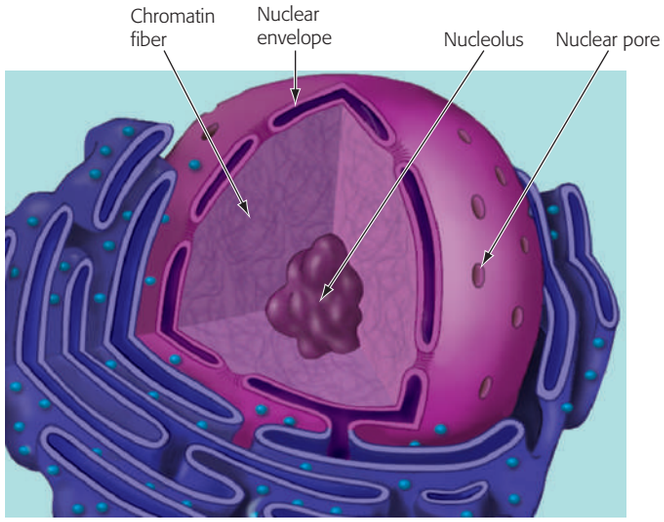
Answer: Chromosomes are made of chromatin, which is a combination of DNA and proteins.

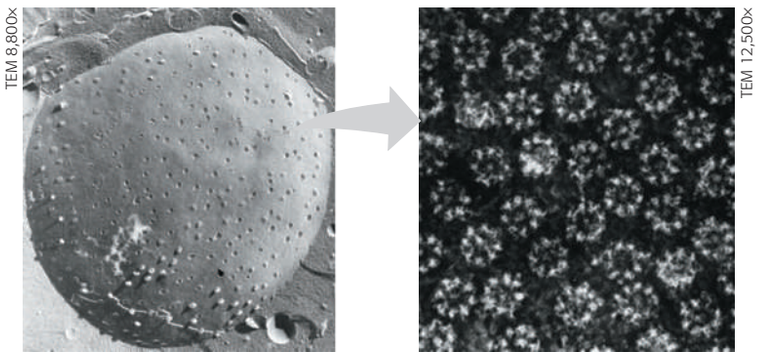
The Nucleus



The nucleus is separated from the cytoplasm by a double membrane called the **nuclear envelope** (Figure 4.6). Each membrane of the nuclear envelope is similar in structure to the plasma membrane: a phospholipid bilayer with associated proteins. Pores in the envelope allow certain materials to pass between the nucleus and the surrounding cytoplasm. (As you'll soon see, among the most important materials that passes between the nucleus and cytoplasm through the nuclear pores are molecules of RNA that carry instructions for building proteins.) Within the nucleus, long DNA molecules and associated proteins form fibers called **chromatin**. Each long chromatin fiber constitutes one chromosome (Figure 4.7). The number of chromosomes in a cell depends on the species; for example, each human body cell has 46 chromosomes, whereas rice cells have 24 and dog cells have 78 (see Figure 8.2 for more examples). The **nucleolus** (shown in Figure 4.6), a prominent structure within the nucleus, is the site where the components of ribosomes are made. We'll examine ribosomes next. ☑

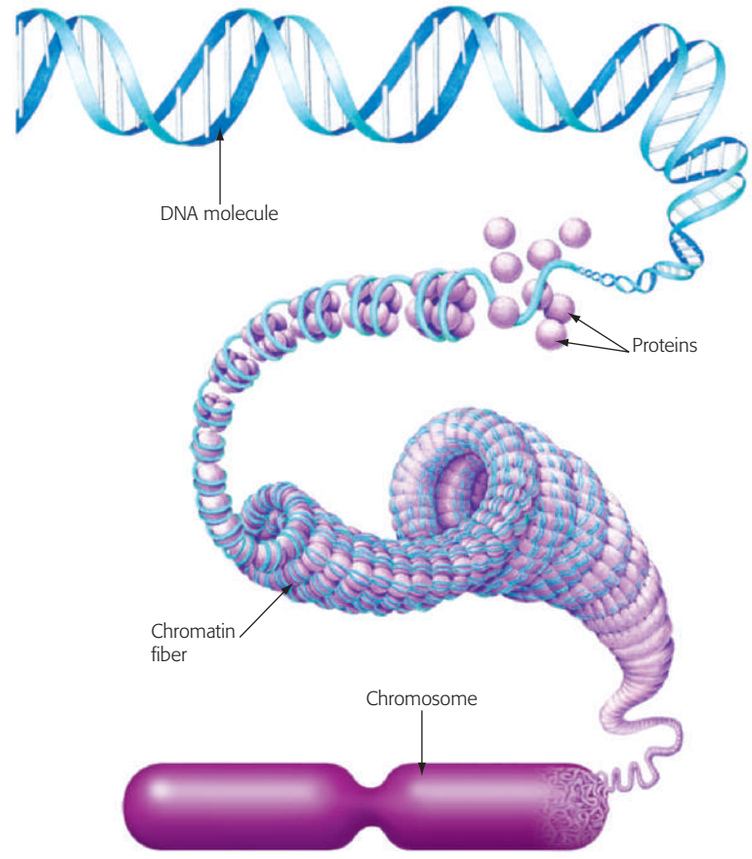
▼ Figure 4.6 The nucleus.





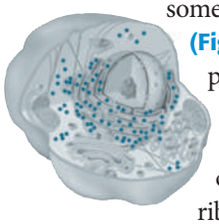
Surface of nuclear envelope **Nuclear pores**

▼ Figure 4.7 The relationship between DNA, chromatin, and a chromosome.



Ribosomes

The small blue dots in the cells in Figure 4.3 and outside the nucleus in Figure 4.6 represent the ribosomes. Ribosomes are responsible for protein synthesis **(Figure 4.8)**. In eukaryotic cells, the components of ribosomes are made in the nucleus and then transported through the pores of the nuclear envelope into the cytoplasm. It is in the cytoplasm that the ribosomes begin their work. Some ribosomes are suspended in the cytosol, making proteins that remain within the fluid of the cell. Other ribosomes are attached to the outside of the nucleus or an organelle called the endoplasmic reticulum **(Figure 4.9)**, making proteins that are incorporated into membranes or secreted by the cell. Free and bound ribosomes are structurally identical, and ribosomes can switch locations, moving between the endoplasmic reticulum and the cytosol. Cells that make a lot of proteins have a large number of ribosomes. For example, each cell in your pancreas that produces digestive enzymes may contain a few million ribosomes.



How DNA Directs Protein Production

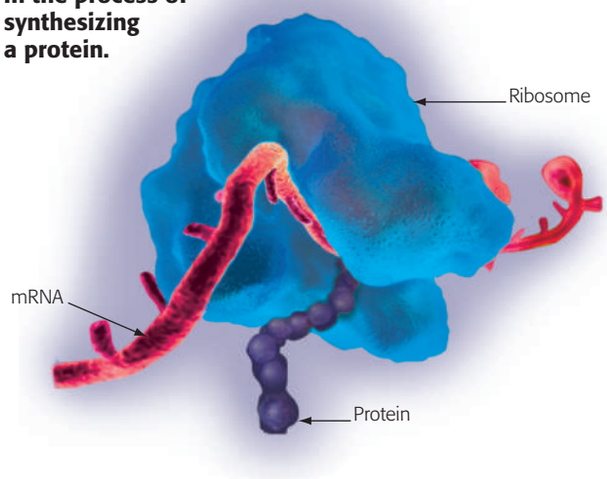
Like a company executive, the DNA doesn't actually do any of the work of the cell. Instead, the DNA "executive" issues orders that result in work being done by the protein "workers." **Figure 4.10** shows the sequence of events during protein production in a eukaryotic cell (with the DNA and other structures being shown disproportionately large in relation to the nucleus). 1 DNA transfers its coded information to a molecule called messenger RNA (mRNA). Like a middle manager, the mRNA molecule carries the order to "build this type of protein." 2 The mRNA exits the nucleus through pores in the nuclear envelope and travels to the cytoplasm, where it binds to a ribosome. 3 The ribosome moves along the mRNA, translating the genetic message into a protein with a specific amino acid sequence. (You'll learn how the message is translated in Chapter 10.) In this way, information carried by the DNA can direct the work of the entire cell without the DNA ever leaving the protective confines of the nucleus. ✓

✓ CHECKPOINT

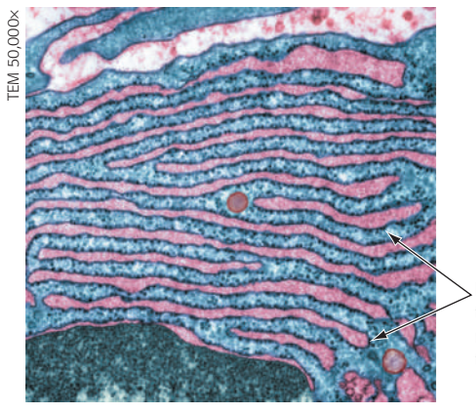
1. What is the function of ribosomes?
2. What is the role of mRNA in making a protein?

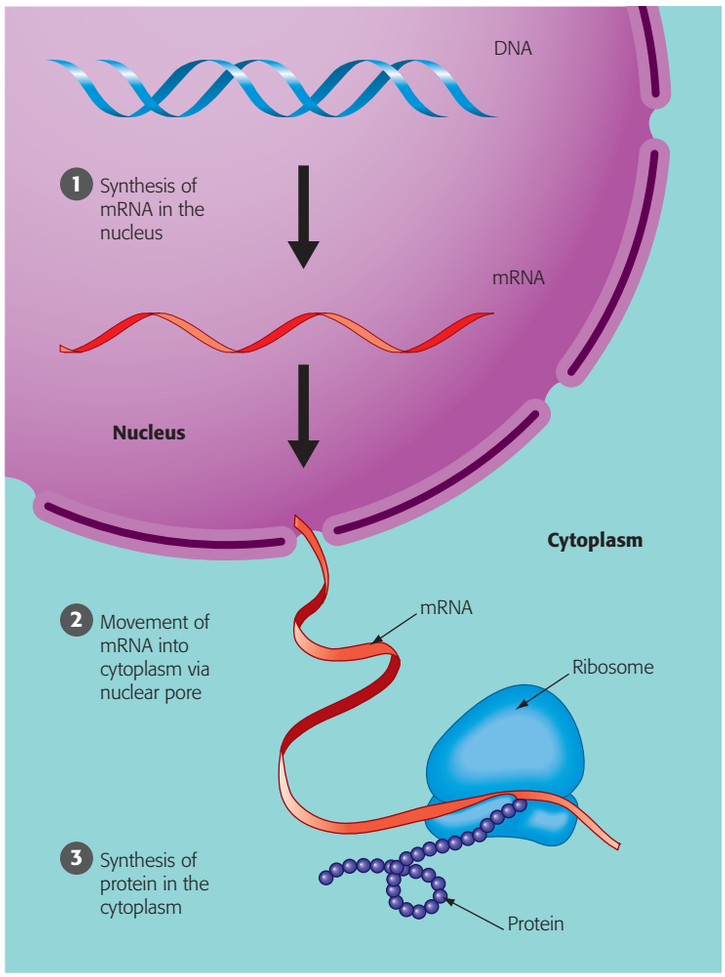
Answers: **1.** *protein synthesis* **2.** *A molecule of mRNA carries the genetic message from a gene (DNA) to ribosomes that translate it into protein.*

► **Figure 4.8 A computer model of a ribosome in the process of synthesizing a protein.**



▼ **Figure 4.9 ER-bound ribosomes.**





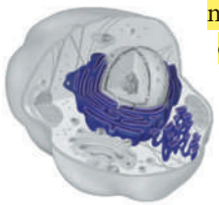
◄ **Figure 4.10 DNA → RNA → protein.** Inherited genes in the nucleus control protein production and hence the activities of the cell.

The Endomembrane System: Manufacturing and Distributing Cellular Products

Like an office partitioned into cubicles, the cytoplasm of a eukaryotic cell is partitioned by organelle membranes (see Figure 4.3). Some of the membranes are physically connected, and others are linked by **vesicles** (sacs made of membrane) that transfer membrane segments between organelles. Together, these organelles form the **endomembrane system**. This system includes the nuclear envelope, the endoplasmic reticulum, the Golgi apparatus, lysosomes, and vacuoles.

The Endoplasmic Reticulum

The **endoplasmic reticulum (ER)** is one of the main manufacturing facilities within a cell. It produces an enormous variety of molecules.

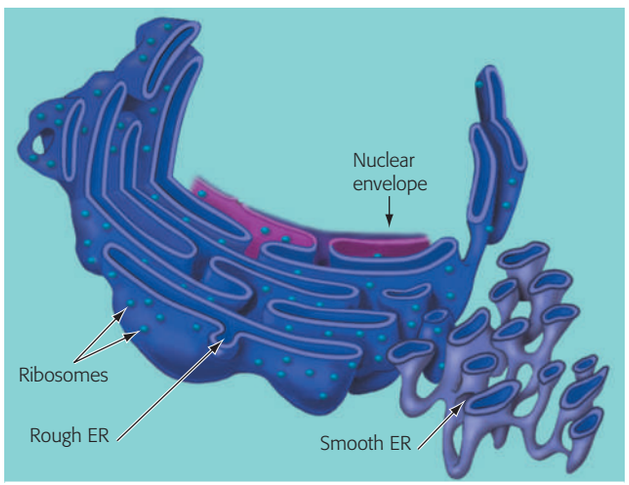


Connected to the nuclear envelope, the ER forms an extensive labyrinth of tubes and sacs running throughout the cytoplasm (Figure 4.11). A membrane separates the internal ER compartment from the cytosol. There are two components that make up the ER: rough ER and smooth ER. These two types of ER are physically connected but differ in structure and function.

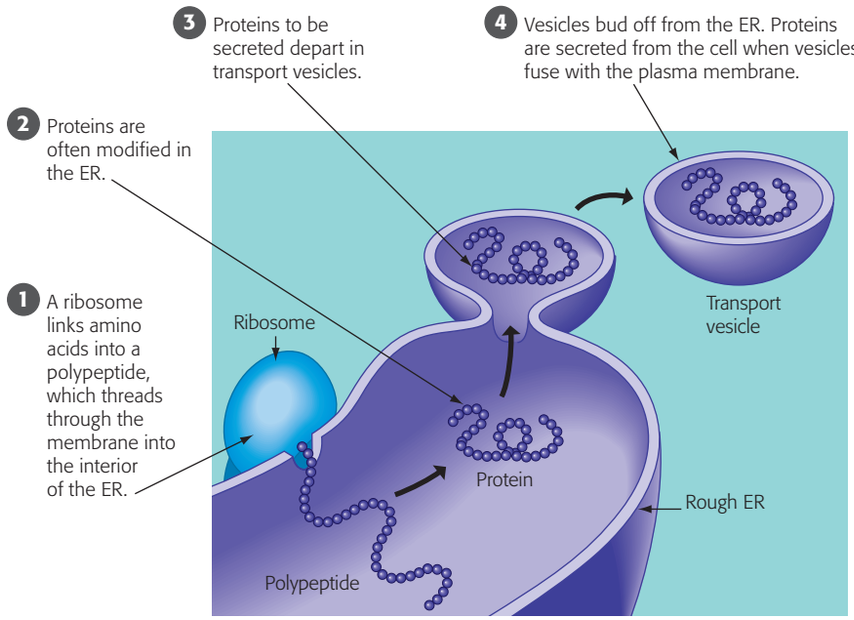
Rough ER

The "rough" in **rough ER** refers to ribosomes that stud the outside of its membrane. One of the functions of rough ER is to make more membrane. Phospholipids made by enzymes of the rough ER are inserted into the ER membrane. In this way, the ER membrane grows, and portions of it can bubble off and be transferred to other parts of the cell. The ribosomes attached to the rough ER produce proteins that will be inserted into the growing ER membrane, transported to other organelles, and eventually exported. Cells that secrete a lot of protein—such as the cells of your salivary glands, which secrete enzymes into your mouth—are especially rich in rough ER. As shown in Figure 4.12, 1 some products manufactured by rough ER are 2 chemically modified and then 3 packaged into **transport vesicles**, sacs made of membrane that bud off from the rough ER. The transport vesicles may then be 4 dispatched to other locations in the cell.

▼ Figure 4.11 The endoplasmic reticulum (ER). In this drawing, the flattened sacs of rough ER and the tubes of smooth ER are connected. Notice that the ER is also connected to the nuclear envelope (the nucleus has been omitted from the illustration for clarity).



▼ Figure 4.12 How rough ER manufactures and packages secretory proteins.



Smooth ER

The "smooth" in **smooth ER** refers to the fact that this organelle lacks the ribosomes that populate the surface of rough ER (see Figure 4.11). A diversity of enzymes built into the smooth ER membrane enables this organelle to perform many functions. One is the synthesis of lipids, including steroids (see Figure 3.13). For example, the cells in ovaries or testes that produce the steroid sex hormones are enriched with smooth ER. In liver cells, enzymes of the smooth ER detoxify circulating drugs such as barbiturates, amphetamines, and some antibiotics (which is why antibiotics don't persist in the bloodstream after combating an infection). As liver cells are exposed to a drug, the amounts of smooth ER and its detoxifying enzymes increase. This can strengthen the body's tolerance of the drug, meaning that higher doses will be required in the future to achieve the desired effect. The growth of smooth ER in response to one drug can also increase tolerance of other drugs. For example, barbiturate use, such as frequently taking sleeping pills, may make certain antibiotics less effective by accelerating their breakdown in the liver.

The Golgi Apparatus



Working in close partnership with the ER, the **Golgi apparatus**, an organelle named for its discoverer (Italian scientist Camillo Golgi), who first described this structure in 1898, receives, refines, stores, and distributes chemical products of the cell (**Figure 4.13**). You can think of the Golgi apparatus as a detailing facility that receives shipments of newly manufactured cars (proteins), puts on the finishing touches, stores the completed cars, and then ships them out when needed.

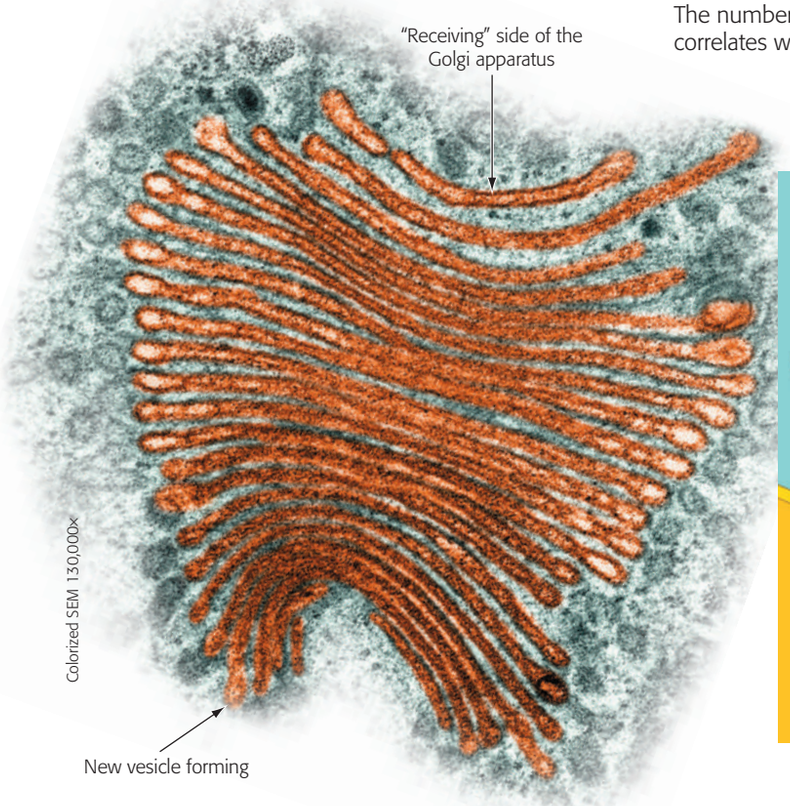
Products made in the ER reach the Golgi apparatus in transport vesicles. The Golgi apparatus consists of a stack of membrane plates, looking much like a pile of pita bread. (1) One side of a Golgi stack serves as a receiving dock for vesicles from the ER. (2) Proteins within a vesicle are usually modified by enzymes during their transit from the receiving to the shipping side of the Golgi apparatus. For example, molecular identification tags may be added that serve to mark and sort protein molecules into different batches for different destinations. (3) The shipping side of a Golgi stack is a depot from which finished products can be carried in transport vesicles to other organelles or to the plasma membrane. Vesicles that bind with the plasma membrane transfer proteins to it or secrete finished products to the outside of the cell. ☑

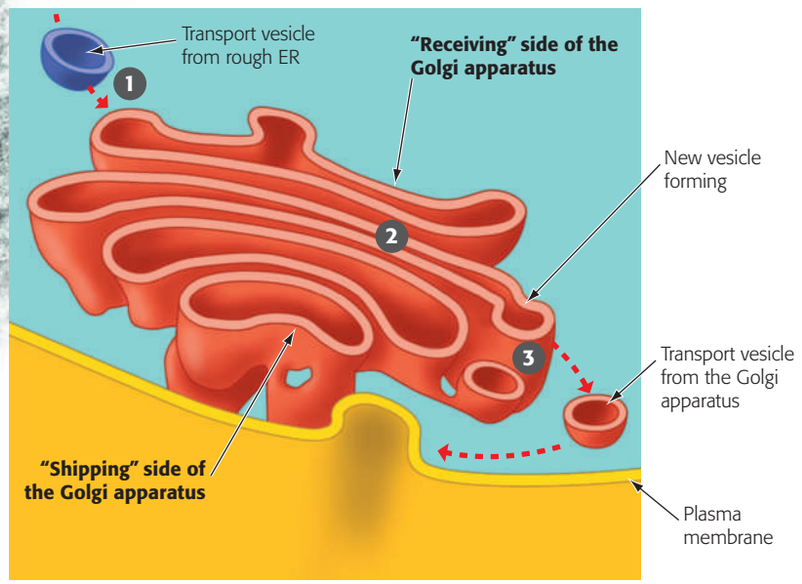
☑ CHECKPOINT

1. What makes rough ER rough?
2. What is the relationship between the Golgi apparatus and the ER in a protein-secreting cell?

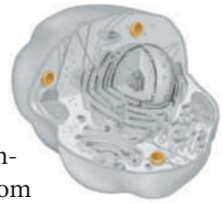
Answers: 1. ribosomes attached to the membrane 2. The Golgi apparatus receives proteins from the ER via vesicles, processes the proteins, and then dispatches them in vesicles.

▼ **Figure 4.13 The Golgi apparatus.** The Golgi consists of flattened sacs arranged something like a stack of pita bread. The number of stacks in a cell (from a few to hundreds) correlates with how active the cell is in secreting proteins.





Lysosomes



A **lysosome** is a membrane-enclosed sac of digestive enzymes found in animal cells. Most plant cells don't contain lysosomes. Lysosomes develop from vesicles that bud off from the Golgi apparatus. Enzymes within a lysosome can break down large molecules such as proteins, polysaccharides, fats, and nucleic acids. The lysosome provides a compartment where the cell can digest these molecules safely, without unleashing the digestive enzymes on the cell itself.

Lysosomes have several types of digestive functions. Many single-celled protists engulf nutrients into tiny cytoplasmic sacs called food vacuoles. Lysosomes fuse with the food vacuoles, exposing the food to digestive enzymes **(Figure 4.14a)**. Small molecules that result from this digestion, such as amino acids, leave the lysosome and nourish the cell. Lysosomes also help destroy harmful bacteria. For example, your white blood cells ingest bacteria into vacuoles, and lysosomal enzymes that are emptied into these vacuoles rupture the bacterial cell walls. In addition, without harming the cell, a lysosome can engulf and digest parts of another organelle, essentially recycling it by making its molecules available for the construction of new organelles **(Figure 4.14b)**. With the help of lysosomes, a cell can thereby continually renew itself. Lysosomes also have sculpting functions in embryonic development. In an early human embryo, lysosomes release enzymes that digest webbing between fingers of the developing hand.

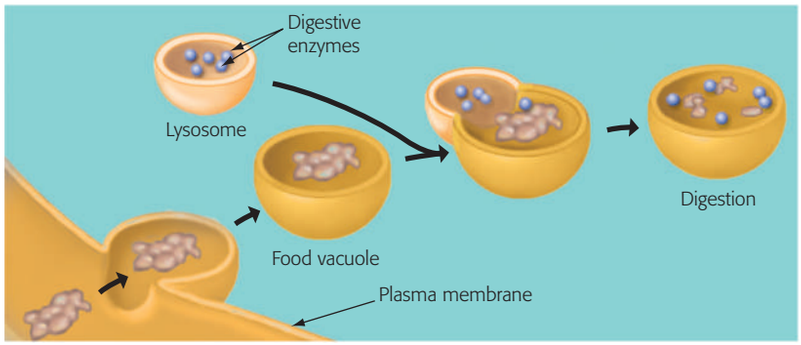
The importance of lysosomes to cell function and human health is made clear by hereditary disorders called lysosomal storage diseases. A person with such a disease is missing one or more of the digestive enzymes normally found within lysosomes. The abnormal lysosomes become engorged with indigestible substances, and this eventually interferes with other cellular functions. Most of these diseases are fatal in early childhood. In Tay-Sachs disease, for example, lysosomes lack a lipid-digesting enzyme. As a result, nerve cells die as they accumulate excess lipids, ravaging the nervous system. Fortunately, lysosomal storage diseases are rare. ☑

☑ CHECKPOINT

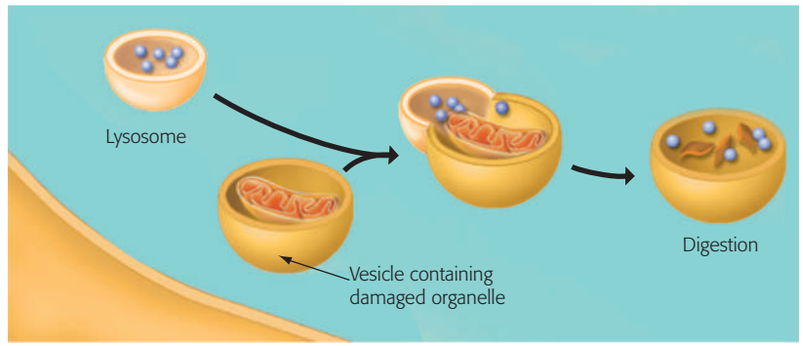
How can defective lysosomes result in excess accumulation of a particular chemical compound in a cell?

Answer: If the lysosomes lack an enzyme needed to break down the compound, the cell will accumulate an excess of that compound.

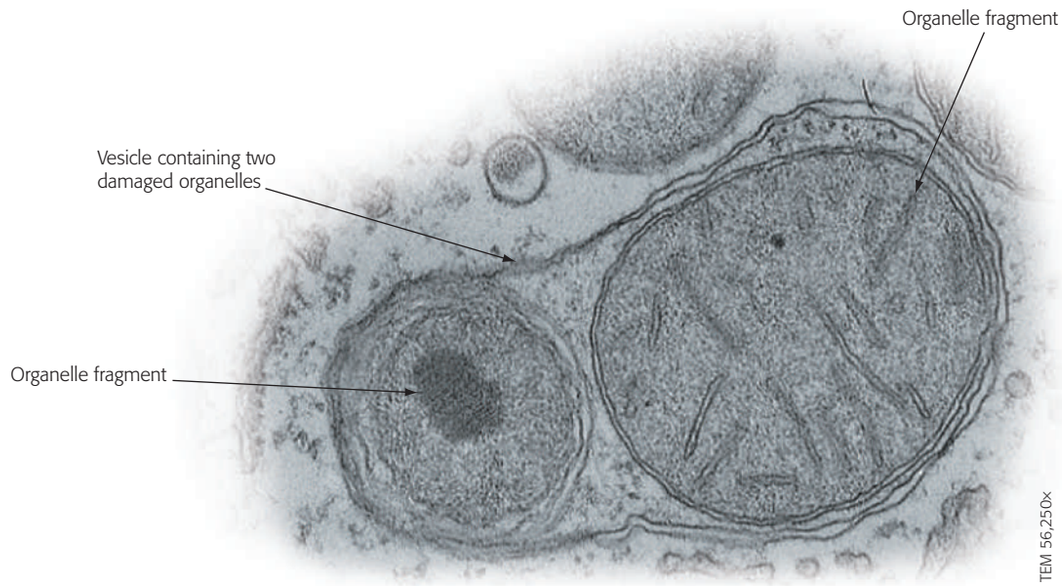
▼ Figure 4.14 Two functions of lysosomes.



(a) A lysosome digesting food



(b) A lysosome breaking down the molecules of damaged organelles



Vacuoles

Vacuoles are large sacs made of membrane that bud off from the ER or Golgi apparatus. Vacuoles have a variety of functions. For example, Figure 4.14a shows a food vacuole budding from the plasma membrane. Certain freshwater protists have contractile vacuoles that pump out excess water that flows into the cell **(Figure 4.15a)**.



Another type of vacuole is a **central vacuole**, a versatile compartment that can account for more than half the volume of a mature plant cell **(Figure 4.15b)**. A central vacuole stores organic nutrients, such as proteins stockpiled in the vacuoles of seed cells. It also contributes to plant growth by absorbing water and causing cells to expand. In the cells of flower petals, central vacuoles may contain pigments that attract pollinating insects. Central vacuoles may also contain poisons that protect against plant-eating animals. Some important crop plants produce and store large amounts of toxic chemicals—harmful to animals that might graze on the plant but useful to us—such as tobacco plants (which store nicotine) and coffee and tea plants (which store caffeine).

The caffeine that gives your cup of tea a kick also protects tea plants from herbivores.

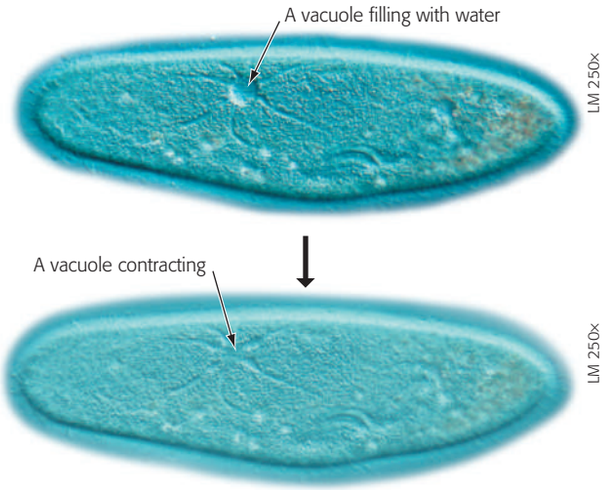
Figure 4.16 will help you review how organelles of the endomembrane system are related. Note that a product made in one part of the endomembrane system may exit the cell or become part of another organelle without crossing a membrane. Also note that membrane made by the ER can become part of the plasma membrane through the fusion of a transport vesicle. In this way, even the plasma membrane is related to the endomembrane system. ☑

☑ CHECKPOINT

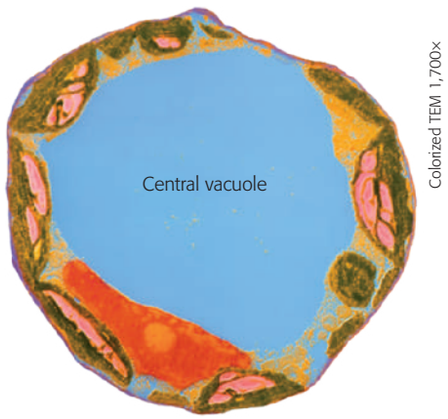
Place the following cellular structures in the order they would be used in the production and secretion of a protein: Golgi apparatus, nucleus, plasma membrane, ribosome, transport vesicle.

Answer: *nucleus, ribosome, transport vesicle, Golgi apparatus, plasma membrane*

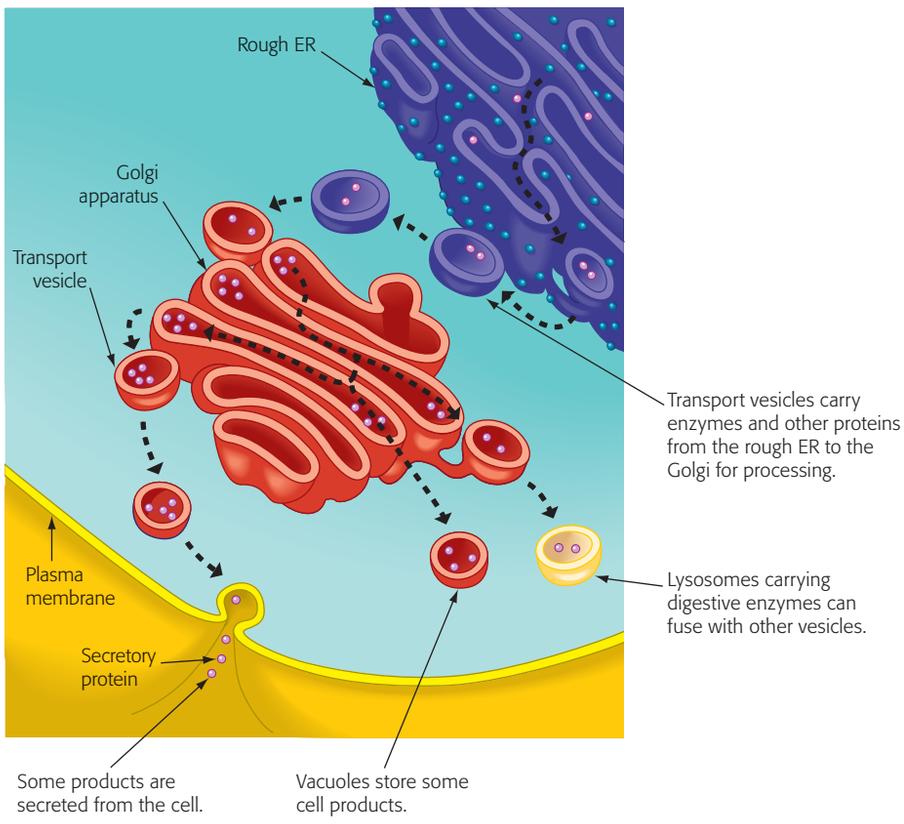
▼ Figure 4.15 Two types of vacuoles.



(a) Contractile vacuole in *Paramecium*. A contractile vacuole fills with water and then contracts to pump the water out of the cell.



(b) Central vacuole in a plant cell. The central vacuole (colorized blue in this micrograph) is often the largest organelle in a mature plant cell.



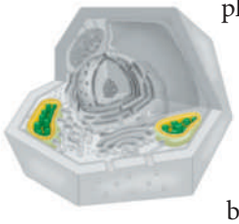
▲ Figure 4.16 Review of the endomembrane system. The dashed arrows show some of the pathways of cell product distribution and membrane migration via transport vesicles.

Energy Transformations

Chloroplasts and Mitochondria

One of the central themes of biology is the transformation of energy: how it enters living systems, is converted from one form to another, and is eventually given off as heat. To follow the energy through living systems, we must consider the two organelles that act as cellular power stations: chloroplasts and mitochondria.

Chloroplasts



Most of the living world runs on the energy provided by photosynthesis, the conversion of light energy from the sun to the chemical energy of sugar and other organic molecules. **Chloroplasts**, which are unique to the photosynthetic cells of plants and algae, are the organelles that perform photosynthesis.

A chloroplast is partitioned into compartments by two membranes, one inside the other **(Figure 4.17)**. The stroma is a thick fluid found inside the innermost membrane. Suspended in that fluid, the interior of a network of membrane-enclosed disks and tubes forms another compartment. Notice in Figure 4.17 that the disks occur in interconnected stacks called grana (singular, *granum*) that resemble stacks of poker chips. The grana are a chloroplast's solar power packs, the structures that trap light energy and convert it to chemical energy (as detailed in Chapter 7).

CHECKPOINT

1. What does photosynthesis accomplish?
2. What is cellular respiration?

Answers: **1.** the conversion of light energy to chemical energy stored in food molecules **2.** a process that converts the chemical energy of sugars and other food molecules to chemical energy in the form of ATP

Mitochondria



In contrast to chloroplasts (found only in plant cells), mitochondria are found in almost all eukaryotic cells, including those of plants and animals. **Mitochondria** (singular, *mitochondrion*) are the organelles in which cellular respiration takes place; during cellular respiration, energy is harvested from sugars and transformed into another form of chemical energy called ATP (adenosine triphosphate). Cells use molecules of ATP as a direct energy source.

An envelope of two membranes encloses the mitochondrion, and the inner membrane encloses a thick fluid called the mitochondrial matrix **(Figure 4.18)**. The inner membrane of the envelope has numerous infoldings called cristae. The cristae create a large surface area in which many of the enzymes and other molecules that function in cellular respiration are embedded, thereby maximizing ATP output. (You'll learn more about how mitochondria convert food energy to ATP energy in Chapter 6.)

Besides their ability to provide cellular energy, mitochondria and chloroplasts share another feature: They contain their own DNA that encodes some of their own proteins made by their own ribosomes. Each chloroplast and mitochondrion contains a single circular DNA chromosome that resembles a prokaryotic chromosome. In fact, mitochondria and chloroplasts can grow and pinch in two, reproducing themselves. This is evidence that mitochondria and chloroplasts evolved from ancient free-living prokaryotes that established residence within other, larger host prokaryotes. This phenomenon, where one species lives inside a host species, is a special type of symbiosis (see Chapter 16 for further information). Over time, mitochondria and chloroplasts likely became increasingly interdependent with the host prokaryote, eventually evolving into a single organism with inseparable parts. The DNA found within mitochondria and chloroplasts is therefore likely remnants of this ancient evolutionary event. ☑

Figure 4.17 The chloroplast: site of photosynthesis.

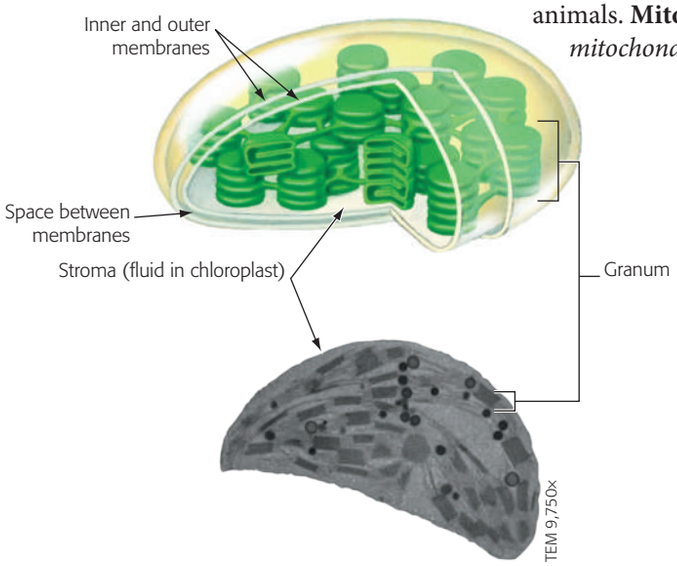
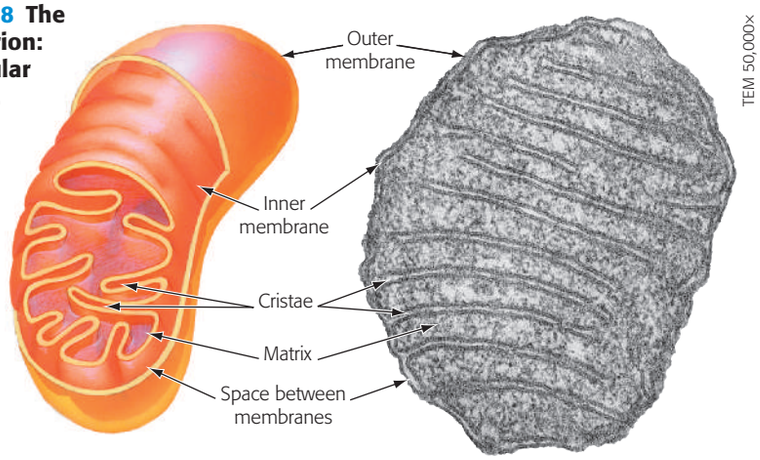


Figure 4.18 The mitochondrion: site of cellular respiration.

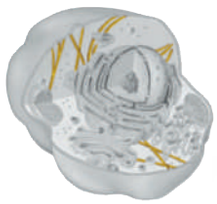


The Cytoskeleton: Cell Shape and Movement

If someone asked you to describe a house, you would most likely mention the various rooms and their locations. You probably would not think to mention the foundation and beams that support the house. Yet these structures perform an extremely important function. Similarly, cells have an infrastructure called the **cytoskeleton**, a network of protein fibers extending throughout the cytoplasm. The cytoskeleton serves as both skeleton and "muscles" for the cell, functioning in support and movement.

Without the cytoskeleton, your cells would collapse in on themselves, much like a house collapses when the infrastructure fails.

Maintaining Cell Shape



One function of the cytoskeleton is to give mechanical support to the cell and maintain its shape. This is especially important for animal cells, which lack rigid cell walls. The cytoskeleton contains several types of fibers made from different types of protein. One important type of fiber forms **microtubules**, hollow tubes of protein **(Figure 4.19a)**. The other kinds of cytoskeletal fibers, called intermediate filaments and microfilaments, are thinner and solid.

Just as the bony skeleton of your body helps fix the positions of your organs, the cytoskeleton provides anchorage and reinforcement for many organelles in a cell. For instance, the nucleus is held in place by a "cage" of cytoskeletal filaments.

Other organelles use the cytoskeleton for movement. For example, a lysosome might reach a food vacuole by gliding along a microtubule track. Microtubules also guide the movement of chromosomes when cells divide (via the mitotic spindle—see Chapter 8).

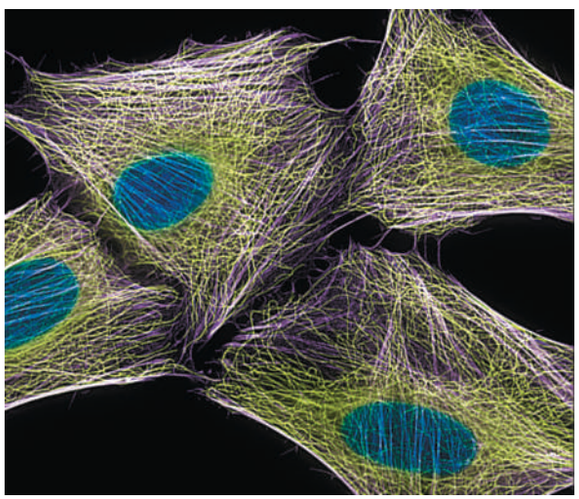
A cell's cytoskeleton is dynamic: It can be quickly dismantled in one part of the cell by removing protein subunits and re-formed in a new location by reattaching the subunits. Such rearrangement can provide rigidity in a new location, change the shape of the cell, or even cause the whole cell or some of its parts to move. This process contributes to the amoeboid (crawling) movements of the protist *Amoeba* **(Figure 4.19b)** and movement of some of our white blood cells. ☑

☑ CHECKPOINT

From which important class of biological molecules are the microtubules of the cytoskeleton made?

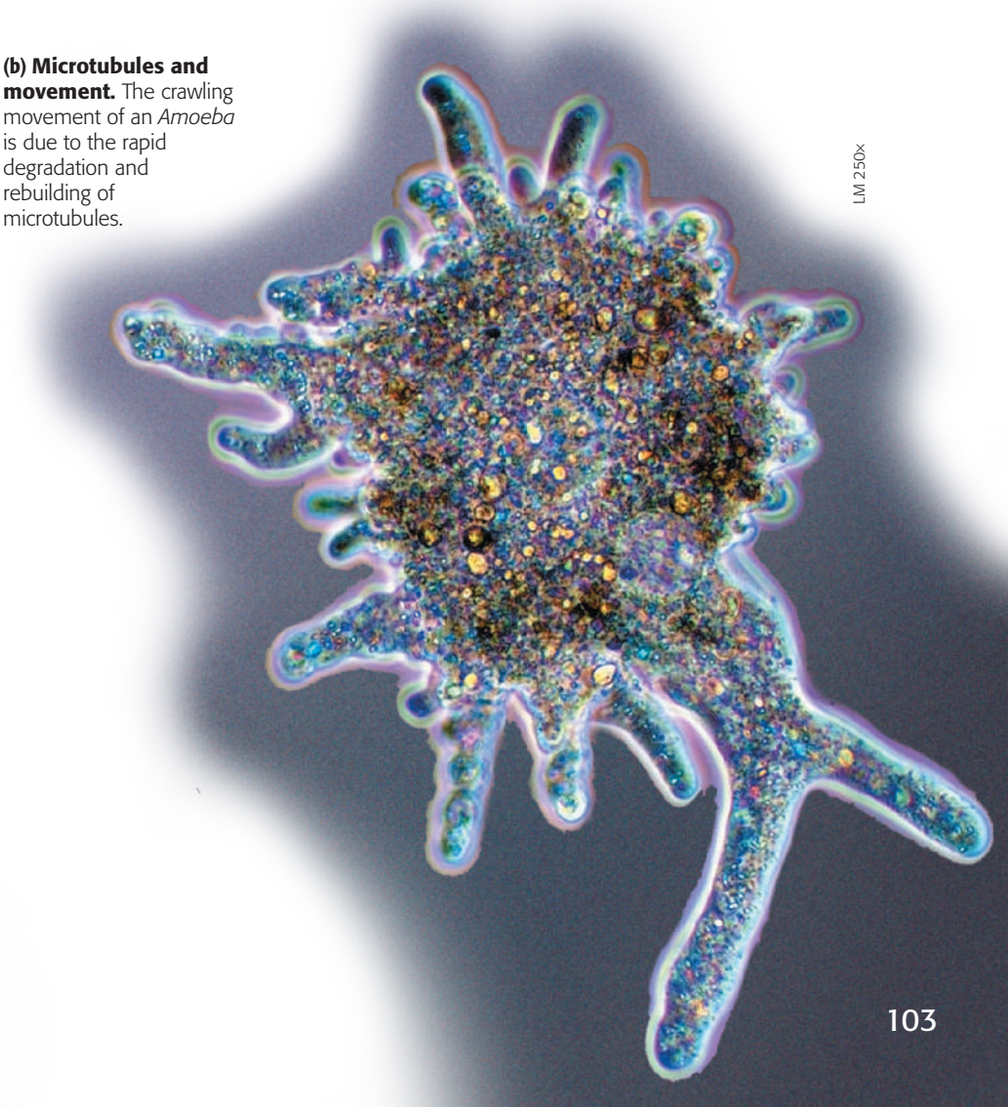
Answer: *protein*

▼ Figure 4.19 The cytoskeleton.



LM 2,500×

(a) Microtubules in the cytoskeleton. In this micrograph of animal cells, the cytoskeleton microtubules are labeled with a fluorescent yellow dye.



LM 250×

(b) Microtubules and movement. The crawling movement of an *Amoeba* is due to the rapid degradation and rebuilding of microtubules.

Cilia and Flagella

In some eukaryotic cells, microtubules are arranged into structures called flagella and cilia, extensions from a cell that aid in movement. Eukaryotic **flagella** (singular, *flagellum*) propel cells with an undulating, whiplike motion. They often occur singly, such as in human sperm cells **(Figure 4.20a)**, but may also appear in groups on the outer surface of protists. **Cilia** (singular, *cilium*) are generally shorter and more numerous than flagella and move in a coordinated back-and-forth motion, like the rhythmic oars of a crew team. Both cilia and flagella propel various protists through water **(Figure 4.20b)**. Though different in length, number per cell, and beating pattern, cilia and flagella have the same basic architecture. Not all animals have cilia or flagella—many do not—and they are almost never found on plant cells.

Some cilia extend from nonmoving cells that are part of a tissue layer. There, they move fluid over the tissue's surface. For example, cilia lining your windpipe clean your respiratory system by sweeping mucus with trapped debris out of your lungs **(Figure 4.20c)**. Tobacco smoke can inhibit or destroy these cilia, interfering with the normal cleansing mechanisms and allowing more toxin-laden smoke particles to reach the lungs. Frequent coughing—common in heavy smokers—then becomes the body's attempt to cleanse the respiratory system.

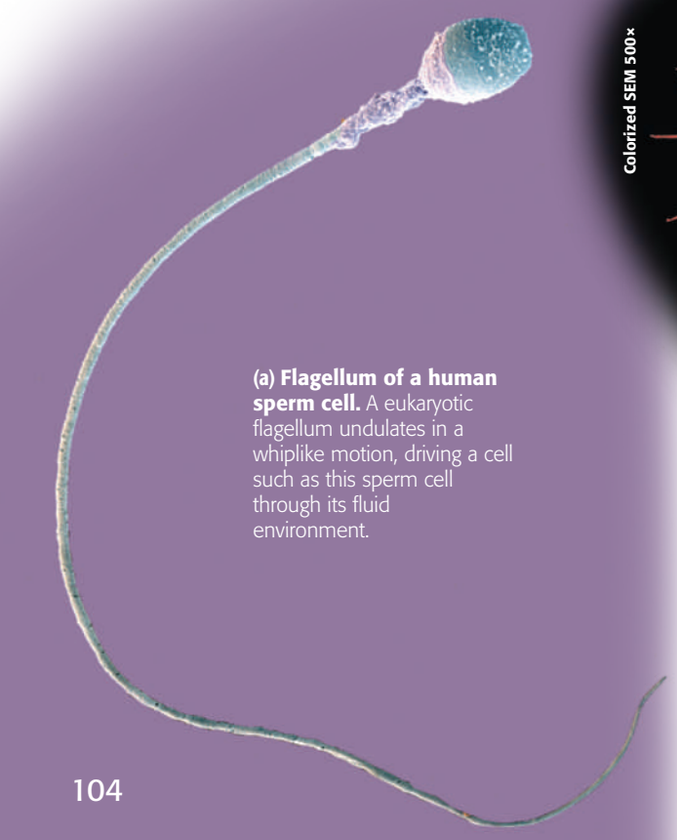
Because human sperm rely on flagella for movement, it's easy to understand why problems with flagella can lead to male infertility. Interestingly, some men with a type of hereditary sterility also suffer from respiratory problems. Because of a defect in the structure of their flagella and cilia, the sperm of men afflicted with this disorder cannot swim normally within the female reproductive tract to fertilize an egg (causing sterility), and their cilia do not sweep mucus out of their lungs (causing recurrent respiratory infections). ☑

☑ CHECKPOINT

Compare and contrast cilia and flagella.

Answer: Cilia and flagella have the same basic structure, are made from microtubules, and help move cells or move fluid over cells. Cilia are short and numerous and move back and forth. Flagella are longer, often occurring singly, and they undulate.

▼ Figure 4.20 Examples of flagella and cilia.



(a) Flagellum of a human sperm cell. A eukaryotic flagellum undulates in a whiplike motion, driving a cell such as this sperm cell through its fluid environment.



(b) Cilia on a protist. Cilia are shorter and more numerous than flagella and move with a back-and-forth motion. As shown here, a layer of beating cilia covers *Paramecium*, a freshwater protist that can dart rapidly through its watery environment.



(c) Cilia lining the respiratory tract. The cilia lining your respiratory tract sweep mucus with trapped debris out of your lungs. This helps keep your airway clear and prevents infections.



Humans Versus Bacteria EVOLUTION CONNECTION

The Evolution of Bacterial Resistance in Humans

Individuals with variations that make them better suited for the local environment will survive and reproduce more often (on average) than those who lack such variations. When the advantageous variations have a genetic basis, the offspring of individuals with the variations will more often also have the favorable adaptions, giving them a survival and reproductive advantage. In this way, repeated over many generations, natural selection promotes evolution of the population.

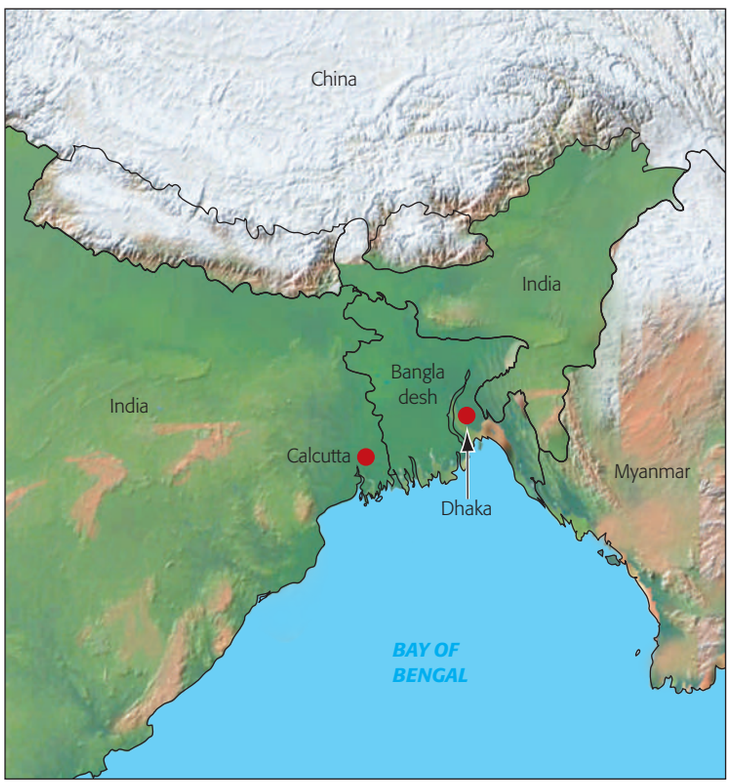
Within a human population, the persistent presence of a disease can provide a new basis for measuring those individuals who are best suited for survival in the local environment. For example, a recent evolutionary study examined people living in Bangladesh. This population has been exposed to the disease cholera—caused by an infectious bacterium—for millennia **(Figure 4.21)**. After cholera bacteria enter a victim's digestive tract (usually through contaminated drinking water), the bacteria produce a toxin that binds to intestinal cells. There, the toxin alters proteins in the plasma membrane, causing the cells to excrete fluid. The resulting diarrhea, which spreads the bacteria by shedding it back into the environment, can cause severe dehydration and death if untreated.

Because Bangladeshis have lived for so long in an environment that teems with cholera bacteria, one might expect that natural selection would favor those individuals who have some resistance to the bacteria. Indeed, recent studies of people from Bangladesh revealed mutations in several genes that appear to confer an increased resistance to cholera. Researchers discovered one mutation in a gene that encodes for the plasma membrane proteins that are the targets of the cholera bacteria. Although the mechanism is not yet understood, these genes appear to offer a survival advantage by making the proteins somewhat more resistant to attack by the cholera toxin. Because such genes offer a survival advantage within this population, they have slowly spread through the Bangladeshi population over the past 30,000 years. In other words, the Bangladeshi population is evolving increased resistance to cholera.

In addition to providing insight into the recent evolutionary past, data from this study reveal potential ways that humans might thwart the cholera bacterium. Perhaps pharmaceutical companies can exploit the proteins produced by the identified mutations to create a new generation of antibiotics. If so, this will represent another way that biologists have applied lessons learned from our understanding of evolution to improve human health. It also reminds us that we humans, like all life on Earth, are shaped by evolution due to changes in our environment, including the presence of infectious microorganisms that live around us.

▼ Figure 4.21 People living in Bangladesh appear to have evolved resistance to *Vibrio cholerae*, the bacterium that causes the deadly disease cholera.





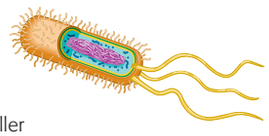
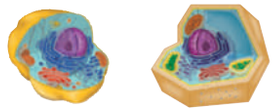
Chapter Review

SUMMARY OF KEY CONCEPTS

The Microscopic World of Cells

The Two Major Categories of Cells

CATEGORIES OF CELLS

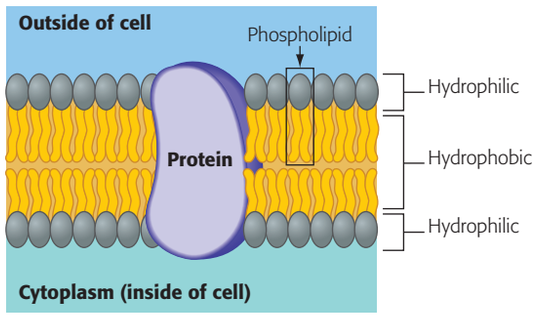
Prokaryotic Cells	Eukaryotic Cells
	
▪ Smaller ▪ Simpler ▪ Lack membrane-bound organelles ▪ Found in bacteria and archaea	▪ Larger ▪ More complex ▪ Have membrane-bound organelles ▪ Found in protists, plants, fungi, animals

An Overview of Eukaryotic Cells

Membranes partition eukaryotic cells into a number of functional compartments. The largest organelle is usually the nucleus. Other organelles are located in the cytoplasm, the region outside the nucleus and within the plasma membrane.

Membrane Structure

The Plasma Membrane



Cell Surfaces

The walls that encase plant cells support plants against the pull of gravity and also prevent cells from absorbing too much water. Animal cells are coated by a sticky extracellular matrix.

The Nucleus and Ribosomes: Genetic Control of the Cell

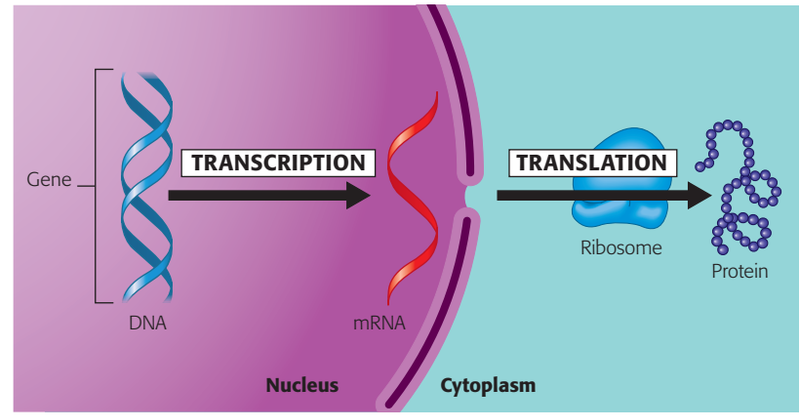
The Nucleus

An envelope consisting of two membranes encloses the nucleus. Within the nucleus, DNA and proteins make up chromatin fibers; each very long fiber is a single chromosome. The nucleus also contains the nucleolus, which produces components of ribosomes.

Ribosomes

Ribosomes produce proteins in the cytoplasm using messages produced by the DNA.

How DNA Directs Protein Production



The Endomembrane System: Manufacturing and Distributing Cellular Products

The Endoplasmic Reticulum

The ER consists of membrane-enclosed tubes and sacs within the cytoplasm. Rough ER, named because of the ribosomes attached to its surface, makes membrane and secretory proteins. The functions of smooth ER include lipid synthesis and detoxification.

The Golgi Apparatus

The Golgi apparatus refines certain ER products and packages them in transport vesicles targeted for other organelles or export from the cell.

Lysosomes

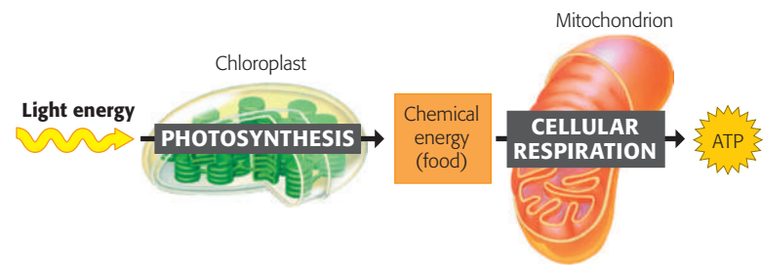
Lysosomes, sacs containing digestive enzymes, aid digestion and recycling within the cell.

Vacuoles

Vacuoles include the contractile vacuoles that expel water from certain freshwater protists and the large, multifunctional central vacuoles of plant cells.

Energy Transformations: Chloroplasts and Mitochondria

Chloroplasts and Mitochondria



The Cytoskeleton: Cell Shape and Movement

Maintaining Cell Shape

Microtubules are an important component of the cytoskeleton, an organelle that gives support to, and maintains the shape of, cells.

Cilia and Flagella

Cilia and eukaryotic flagella are appendages that aid in movement, and they are made primarily of microtubules. Cilia are short and numerous and move the cell via coordinated beating. Flagella are long, often occur singly, and propel a cell with whiplike movements.

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SELF-QUIZ

1. You look into a microscope and view an unknown cell. What might you see that would tell you that the cell is eukaryotic?
   a. DNA
   b. a nucleoid region
   c. a plasma membrane
   d. membrane-enclosed structures called organelles
2. Explain how each word in the term *fluid mosaic* describes the structure of a membrane.
3. Identify which of the following structures includes all the others in the list: rough ER, smooth ER, endomembrane system, the Golgi apparatus.
4. Based on its function in detoxifying drugs, a large amount of _____ ER can be found in liver cells.
5. A type of cell called a lymphocyte makes proteins that are exported from the cell. You can track the path of these proteins within the cell from production through export by labeling them with radioactive isotopes. Identify which of the following structures would be radioactively labeled in your experiment, listing them in the order in which they would be labeled: chloroplasts, Golgi apparatus, plasma membrane, smooth ER, rough ER, nucleus, mitochondria.
6. Name two similarities in the structure or function of chloroplasts and mitochondria. Name two differences.
7. Match the following organelles with their functions:

a. nucleus	1. locomotion
b. flagella	2. protein export
c. mitochondria	3. gene control
d. Golgi apparatus	4. digestion
e. lysosomes	5. cellular respiration

8. DNA controls the cell by transmitting genetic messages that result in protein production. Place the following organelles in the order that represents the flow of genetic information from the DNA through the cell: nuclear pores, ribosomes, nucleus, rough ER, Golgi apparatus.
9. Compare and contrast cilia and flagella.

Answers to these questions can be found in Appendix: Self-Quiz Answers.

THE PROCESS OF SCIENCE

10. The cells of plant seeds store oils in the form of droplets enclosed by membranes. Unlike the membranes you learned about in this chapter, the oil droplet membrane consists of a single layer of phospholipids rather than a bilayer. Draw a model for a membrane around an oil droplet. Explain why this arrangement is more stable than a bilayer.
11. Both chloroplasts and mitochondria contain rudimentary genomes, are able to divide independently from their host cells, and are bounded by double membranes. These similarities led to the idea that these organelles could have been derived from ancient prokaryotes that took up residence in eukaryotic ancestor cells. Which of the two organelles is more likely to have invaded the eukaryotic ancestor first and why?
12. **Interpreting Data** A population of bacteria may evolve resistance to a drug over time. Draw a graph that represents such a change. Label the ***x***-axis as "time" and the ***y***-axis as "population size." Draw a line on the graph that represents how the size of the population might change over time after the introduction of a new antibiotic. Label the point on your line where the new drug was introduced, and then indicate how the population size might change over time after that introduction.

BIOLOGY AND SOCIETY

13. In 2010, Craig Venter's group announced that they have assembled a completely synthetic genome based on the DNA sequence of the bacterium *Mycoplasma mycoides*. They inserted it into a living *Mycoplasma capricolum* cell, whose own genetic material was removed. This experiment resulted in a living and replicating cell that Venter referred to as "the first synthetic cell." In your opinion and based on what you have learnt in the first four chapters, is this description correct? Do you think it is feasible to create an entirely synthetic cell? What are the potential benefits of this type of synthetic biology? What are the dangers associated?
14. Scientists are learning to manipulate living cells in various ways, changing their genetic composition and the way they function. Some biotechnology companies have sought to patent their unique engineered cell lines. Do you think it is in society's best interest to allow cells to be patented? Why or why not? Do you think the same patent rules should apply whether the cells originate from a human or from a bacterium?

5 The Working Cell

Why Cellular Functions Matter



Both nerve gas and insecticides work by crippling a vital enzyme.

For thousands of years, people have used osmosis to preserve food through salt and sugar curing.







You'd have to walk more than 2 hours to burn the calories in half a pepperoni pizza.

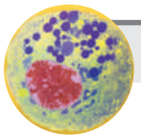
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CHAPTER THREAD

Nanotechnology

BIOLOGY AND SOCIETY **Harnessing Cellular Structures** 109
THE PROCESS OF SCIENCE **Can Enzymes Be Engineered?** 115
EVOLUTION CONNECTION **The Origin of Membranes** 121

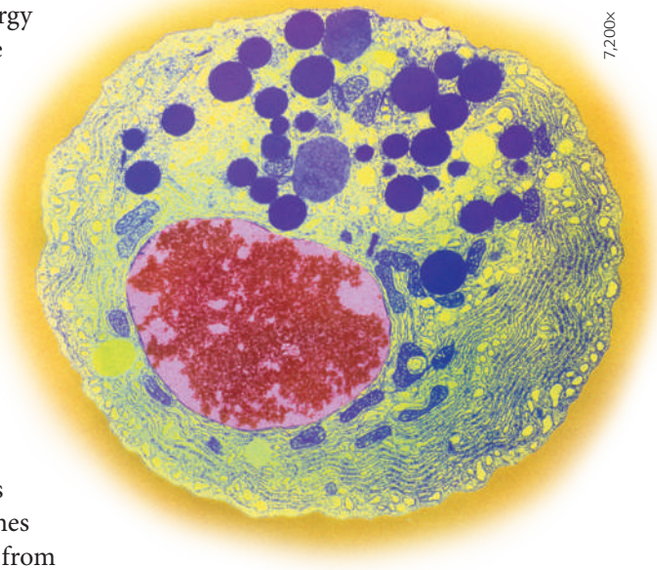


Nanotechnology BIOLOGY AND SOCIETY

Harnessing Cellular Structures

Imagine a tiny movable "car" with balls of carbon atoms for wheels, or a three-dimensional relief map of the world carved onto an object 1,000 times smaller than a grain of sand. These are real-world examples of nanotechnology, the manipulation of materials at the molecular scale. When designing devices of such small size, researchers often turn to living cells for inspiration. After all, you can think of a cell as a machine that continuously and efficiently performs a variety of functions, such as movement, energy processing, and production of various products. Let's consider one example of cell-based nanotechnology and see how it relates to working cells.

Researchers at Cornell University are attempting to harvest the energy-producing capability of human sperm cells. Like other cells, a sperm cell generates energy by breaking down sugars and other molecules that pass through its plasma membrane. Enzymes within the cell carry out a process called glycolysis. During glycolysis, the energy released from the breakdown of glucose is used to produce molecules of ATP. Within a living sperm, the ATP produced during glycolysis and other processes provides the energy that propels the sperm through the female reproductive tract. In an attempt to harness this energy-producing system, the Cornell researchers attached three glycolysis enzymes to a computer chip. The enzymes continued to function in this artificial system, producing energy from sugar. The hope is that a larger set of enzymes can eventually be used to power microscopic robots. Such nanorobots could use glucose from the bloodstream to power the delivery of drugs to body tissues, among many other possible tasks. This example is only a glimpse into the incredible potential of new technologies inspired by working cells.



Cellular structures. Even the smallest cell, such as this one from a human pancreas, is a miniature machine of startling complexity.

In this chapter, we'll explore three processes common to all living cells: energy metabolism, the use of enzymes to speed chemical reactions, and transport regulation by the plasma membrane. Along the way, we'll further consider nanotechnologies that mimic the natural activities of living cells.

Some Basic Energy Concepts

Energy makes the world go round—both on a planetary scale and on a cellular scale. But what exactly is energy? Our first step in understanding the working cell is to learn a few basic concepts about energy.

Conservation of Energy

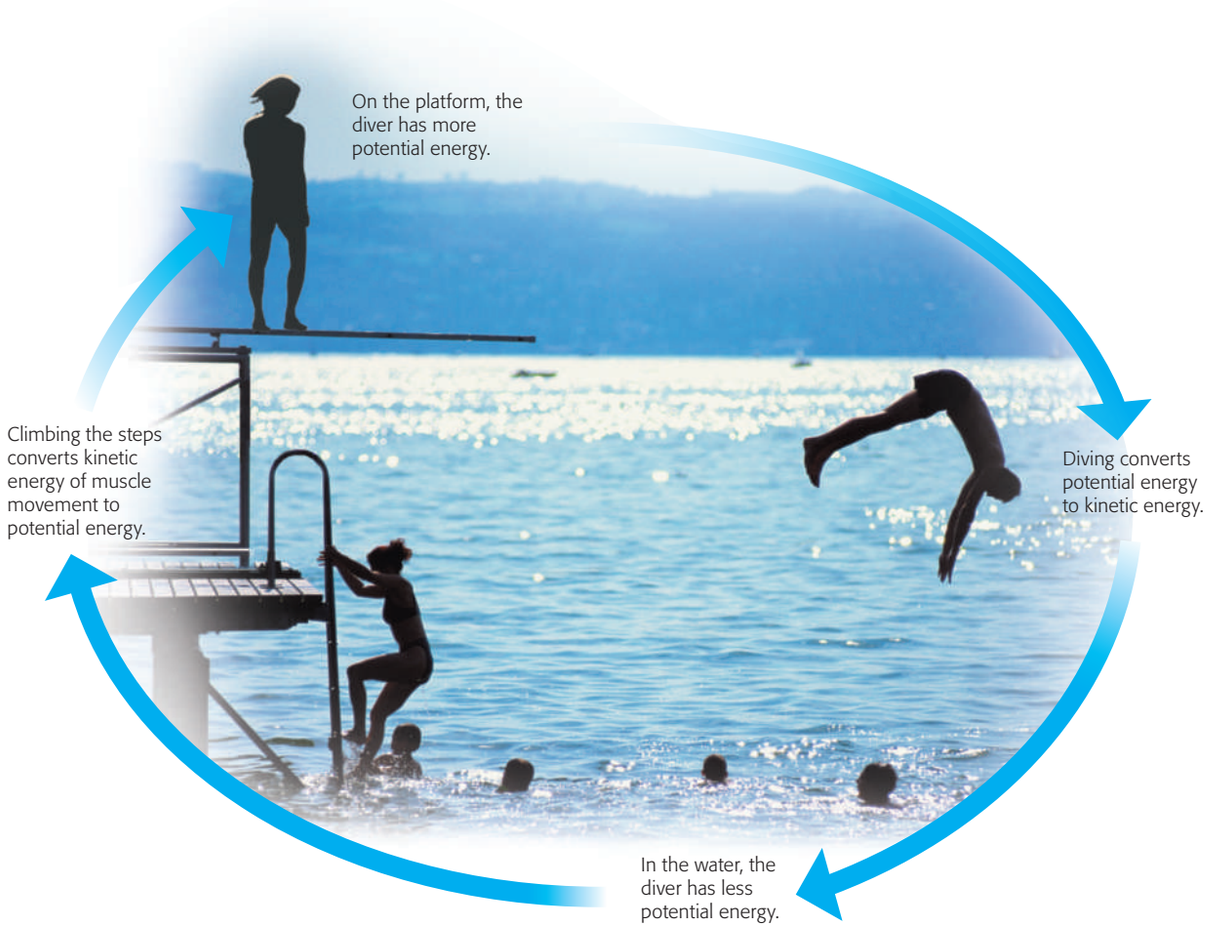
Energy is defined as the capacity to cause change. Some forms of energy are used to perform work, such as moving an object against an opposing force—for example, lifting a barbell against the force of gravity. Imagine a diver climbing to the top of a platform and then diving off **(Figure 5.1)**. To get to the top of the platform, the diver must perform work to overcome the opposing force of gravity. Specifically, chemical energy from food is converted to **kinetic energy**, the energy of motion. In this case, the kinetic energy takes the form of muscle movement propelling the diver to the top of the platform.

What happens to the kinetic energy when the diver reaches the top of the platform? Does it disappear at that point? In fact, it does not. A physical principle known as **conservation of energy** explains that it is not possible to destroy or create energy. Energy can only be converted from one form to another. A power plant, for example, does not make energy; it merely converts it from one form (such as energy stored in coal) to a more convenient form (such as electricity). That's what happens in the diver's climb up the steps. The kinetic energy of muscle movement is stored as **potential energy**, the energy an object has because of its location or structure. The energy contained by water behind a dam or by a compressed spring are examples of potential energy. In our example, the diver at the top of the platform has potential energy because of his elevated location. Then the act of diving off the platform into the water converts the potential energy back to kinetic energy. Life depends on countless similar conversions of energy from one form to another. ☑

☑ CHECKPOINT

Can an object at rest have energy?

Answer: Yes; it can have potential energy because of its location or structure.



► **Figure 5.1 Energy conversions during a dive.**

Heat

If energy cannot be destroyed, where has the energy gone in our example when the diver hits the water? The energy has been converted to **heat**, a type of kinetic energy contained in the random motion of atoms and molecules. The friction between the body and its surroundings generated heat in the air and then in the water.

All energy conversions generate some heat. Although releasing heat does not destroy energy, it does make it more difficult to harness for useful work. Heat is energy in its most disordered, chaotic form, the energy of aimless molecular movement.

Entropy is a measure of the amount of disorder, or randomness, in a system. Consider an analogy from your own room. It's easy to increase the chaos—in fact, it seems to happen spontaneously! But it requires the expenditure of significant energy to restore order once again.

Every time energy is converted from one form to another, entropy increases. The energy conversions during the climb up the ladder and the dive from the platform increased entropy as the diver emitted heat to the surroundings. To climb up the steps again for another dive, the diver must use additional stored food energy. This conversion will also create heat and therefore increase entropy. ☑

Chemical Energy

How can molecules derived from the food we eat provide energy for our working cells? The molecules of food, gasoline, and other fuels have a form of potential energy called **chemical energy**, which arises from the arrangement of atoms and can be released by a chemical reaction. Carbohydrates, fats, and gasoline have structures that make them especially rich in chemical energy.

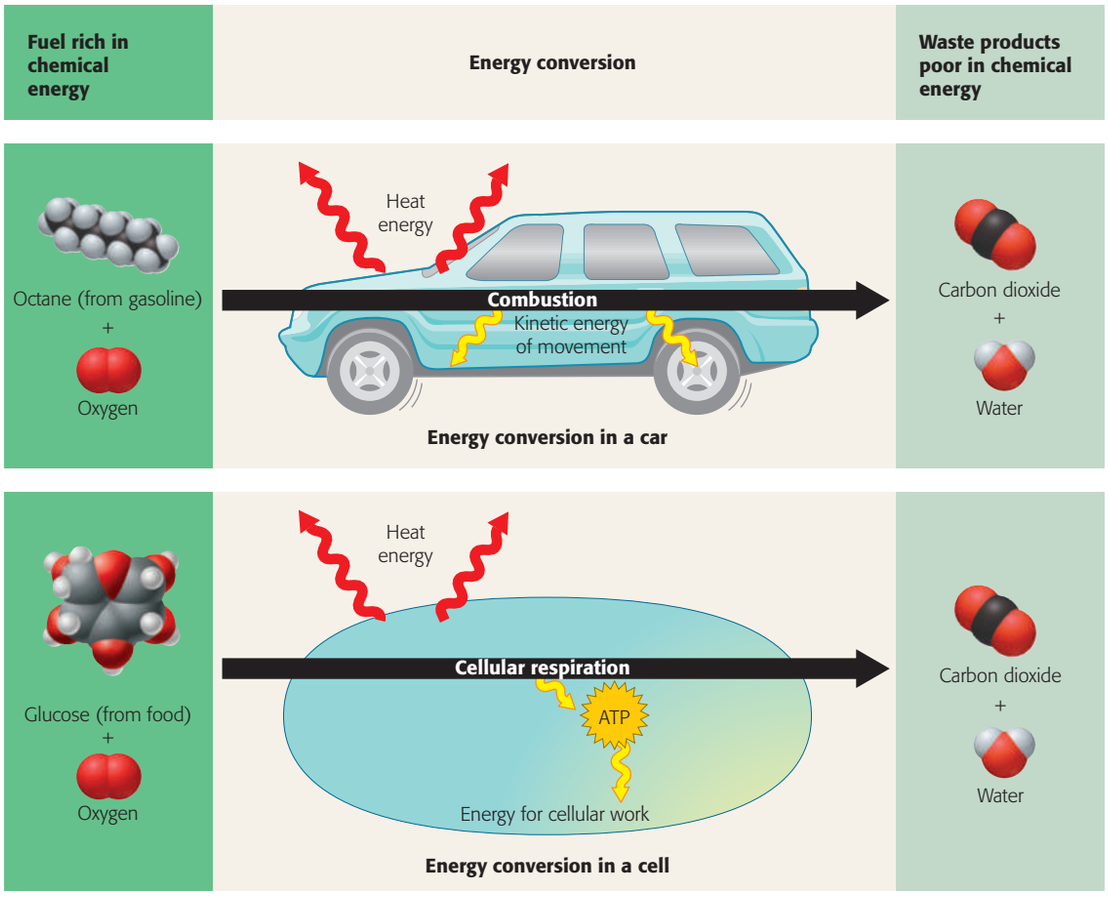
Living cells and automobile engines use the same basic process to make the chemical energy stored in their fuels available for work **(Figure 5.2)**. In both cases, this process breaks organic fuel into smaller waste molecules that have much less chemical energy than the fuel molecules did, thereby releasing energy that can be used to perform work.

For example, the engine of an automobile mixes oxygen with gasoline (which is why all cars require an air intake system) in an explosive chemical reaction that breaks down the fuel molecules and pushes the pistons that eventually move the wheels. The waste products emitted from the car's exhaust pipe are mostly carbon dioxide and water. Only about 25% of the energy that an automobile engine extracts from its fuel is converted to the kinetic energy of the car's movement. Most of

☑ CHECKPOINT

Which form of energy is most randomized and difficult to put to work?

Answer: *heat*



◄ Figure 5.2 Energy conversions in a car and a cell. In both a car and a cell, the chemical energy of organic fuel molecules is harvested using oxygen. This chemical breakdown releases energy stored in the fuel molecules and produces carbon dioxide and water. The released energy can be used to perform work.

the rest is converted to heat—so much that the engine would melt if the car's radiator did not disperse heat into the atmosphere. That is why high-end performance cars need sophisticated air flow systems to avoid overheating.

Cells also use oxygen in reactions that release energy from fuel molecules. As in a car engine, the "exhaust" from such reactions in cells is mostly carbon dioxide and water. The combustion of fuel in cells is called cellular respiration, which is a more gradual and efficient "burning" of fuel compared with the explosive combustion in an automobile engine. Cellular respiration is the energy-releasing chemical breakdown of fuel molecules and the storage of that energy in a form the cell can use to perform work. (We will discuss the details of cellular respiration in Chapter 6.) You convert about 34% of your food energy to useful work, such as movement of your muscles. The rest of the energy released by the breakdown of fuel molecules generates body heat. Humans and many other animals can use this heat to keep the body at an almost constant temperature (37°C, or 98.6°F, in the case of humans), even when the surrounding air is much colder. You've probably noticed how quickly a crowded room warms up—it's all that released metabolic heat energy! The liberation of heat energy also explains why you feel hot after exercise. Sweating and other cooling mechanisms enable your body to lose the excess heat, much as a car's radiator keeps the engine from overheating.

✓ CHECKPOINT

According to Figure 5.3, how long would you have to ride your bicycle to burn off the energy in a cheeseburger?

Answer: a little more than a half hour (1 cheeseburger = 295 Calories; 1 hour of cycling consumes 490 Calories, so 295/490 = 0.6 hours, or about 36 minutes)

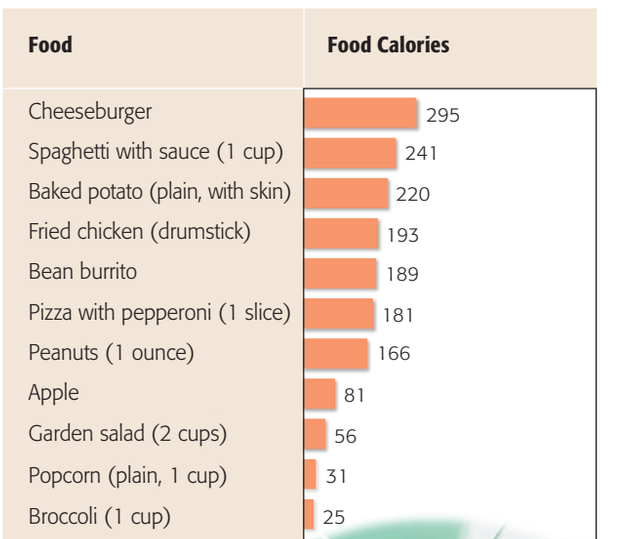
You'd have to walk more than 2 hours to burn the calories in half a pepperoni pizza.

Food Calories

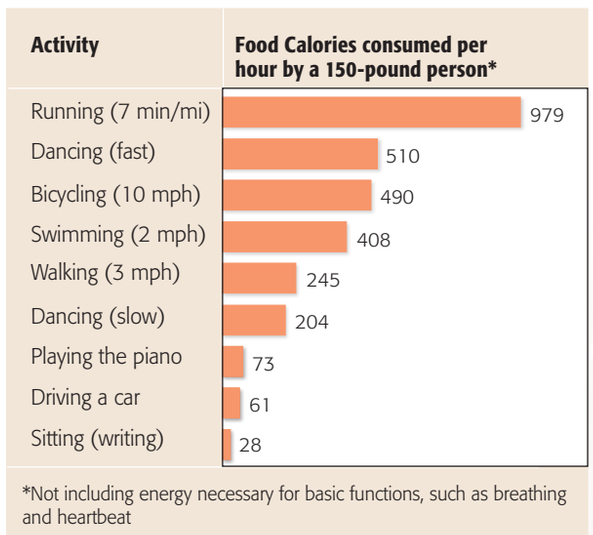
Read any packaged food label and you'll find the number of calories in each serving of that food. Calories are units of energy. A **calorie** (cal) is the amount of energy that can raise the temperature of 1 gram (g) of water by 1°C. You could actually measure the caloric content of a peanut by burning it under a container of water to convert all of the stored chemical energy to heat and then measuring the temperature increase of the water.

Calories are tiny units of energy, so using them to describe the fuel content of foods is not practical. Instead, it's conventional to use kilocalories (kcal), units of 1,000 calories. In fact, the Calories (capital C) on a food package are actually kilocalories. For example, one peanut has about 5 Calories. That's a lot of energy, enough to increase the temperature of 1 kg (a little more than a quart) of water by 5°C. And just a handful of peanuts contains enough Calories, if converted to heat, to boil 1 kg of water. In living organisms, of course, food isn't used to boil water but instead used to fuel the activities of life. **Figure 5.3** shows the number of Calories in several foods and how many Calories are burned by some typical activities. ✓

▼ Figure 5.3 Some caloric accounting.



(a) Food Calories (kilocalories) in various foods



(b) Food Calories (kilocalories) we burn in various activities





Energy Transformations

ATP and Cellular Work

The carbohydrates, fats, and other fuel molecules we obtain from food cannot be used directly as fuel for our cells. Instead, the chemical energy released by the breakdown of organic molecules during cellular respiration is used to generate molecules of ATP. These molecules of ATP then power cellular work. ATP acts like an energy shuttle, storing energy obtained from food and then releasing it as needed at a later time. Such energy transformations are essential for all life on Earth.

The Structure of ATP

The abbreviation ATP stands for adenosine triphosphate. **ATP** consists of an organic molecule called adenosine plus a tail of three phosphate groups **(Figure 5.4)**. The triphosphate tail is the "business" end of ATP, the part that provides energy for cellular work. Each phosphate group is negatively charged. Negative charges repel each other. The crowding of negative charges in the triphosphate tail contributes to the potential energy of ATP. It's analogous to storing energy by compressing a spring; if you release the spring, it will relax, and you can use that springiness to do some useful work. For ATP power, it is release of the phosphate at the tip of the triphosphate tail that makes energy available to working cells. What remains is **ADP**, adenosine diphosphate (two phosphate groups instead of three, shown on the right side of Figure 5.4).

Phosphate Transfer

When ATP drives work in cells by being converted to ADP, the released phosphate groups don't just fly off into space. ATP energizes other molecules in cells by transferring phosphate groups to those molecules. When a target molecule accepts the third phosphate group, it becomes energized and can then perform work in the cell. Imagine a bicyclist pedaling up a hill. In the muscle cells of the rider's legs, ATP transfers phosphate groups to motor proteins. The proteins then change shape, causing the muscle cells to contract **(Figure 5.5a)**. This contraction provides the mechanical energy needed to propel the rider. ATP also enables the transport of ions and other dissolved substances across the membranes of the rider's nerve cells **(Figure 5.5b)**, helping them send signals to her legs. And ATP drives the production of a cell's large molecules from smaller molecular building blocks **(Figure 5.5c)**.

Figure 5.4 ATP power. Each P in the triphosphate tail of ATP represents a phosphate group, a phosphorus atom bonded to oxygen atoms. The transfer of a phosphate from the triphosphate tail to other molecules provides energy for cellular work.

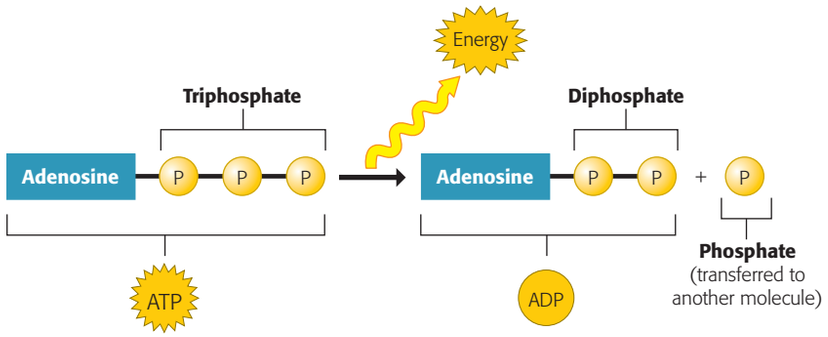
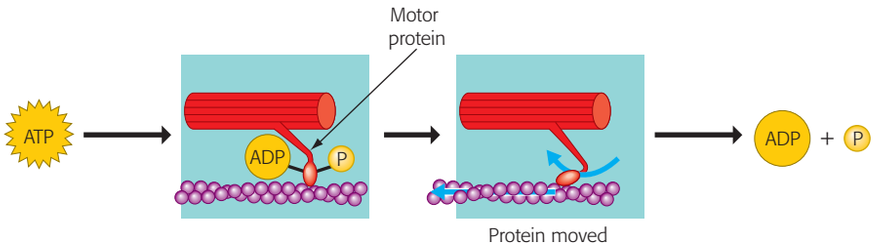
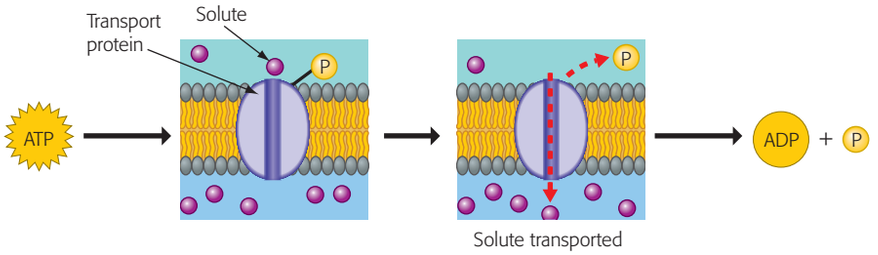


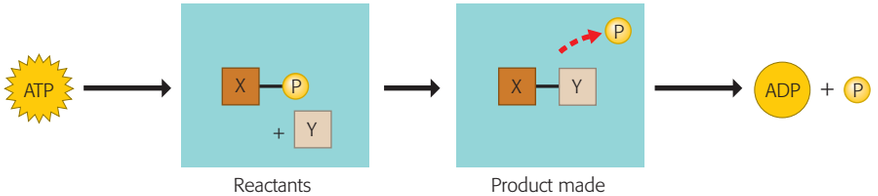
Figure 5.5 How ATP drives cellular work. Each type of work shown here is powered when an enzyme transfers phosphate from ATP to a recipient molecule.



(a) Motor protein performing mechanical work (moving a muscle fiber)



(b) Transport protein performing transport work (importing a solute)

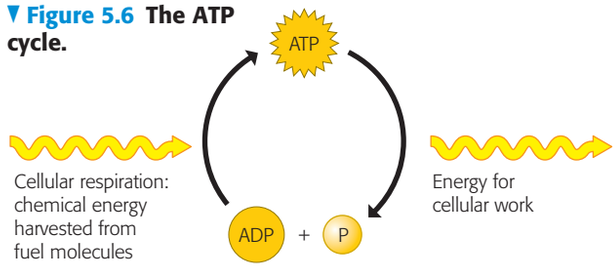


(c) Chemical reactants performing chemical work (promoting a chemical reaction)

The ATP Cycle

Your cells spend ATP continuously. Fortunately, it is a renewable resource. ATP can be restored by adding a phosphate group back to ADP. That takes energy, like recompressing a spring. And that's where food enters the picture. The chemical energy that cellular respiration harvests from sugars and other organic fuels is put to work regenerating a cell's supply of ATP. Cellular work spends ATP, which is recycled when ADP and phosphate are combined using energy released by cellular respiration **(Figure 5.6)**. Thus, energy from processes that yield energy, such as the breakdown of organic fuels, is transferred to processes that consume energy, such as muscle contraction and other cellular work. The ATP cycle can run at an astonishing pace: Up to 10 million ATPs are consumed and recycled each second in a working muscle cell. ☑

▼ Figure 5.6 The ATP cycle.



☑ CHECKPOINT

1. Explain how ATP powers cellular work.
2. What is the source of energy for regenerating ATP from ADP?

Answers: **1.** *ATP transfers a phosphate to another molecule, increasing that molecule's energy;* **2.** *chemical energy harvested from sugars and other organic fuels via cellular respiration*

Enzymes

A living organism contains a vast collection of chemicals, and countless chemical reactions constantly change the organism's molecular makeup. In a sense, a living organism is a complex "chemical square dance," with the molecular "dancers" continually changing partners via chemical reactions. The total of all the chemical reactions in an organism is called **metabolism**. But almost no metabolic reactions occur without help. Most require the assistance of **enzymes**, proteins that speed up chemical reactions without being consumed by those reactions. All living cells contain thousands of different enzymes, each promoting a different chemical reaction.

☑ CHECKPOINT

How does an enzyme affect the activation energy of a chemical reaction?

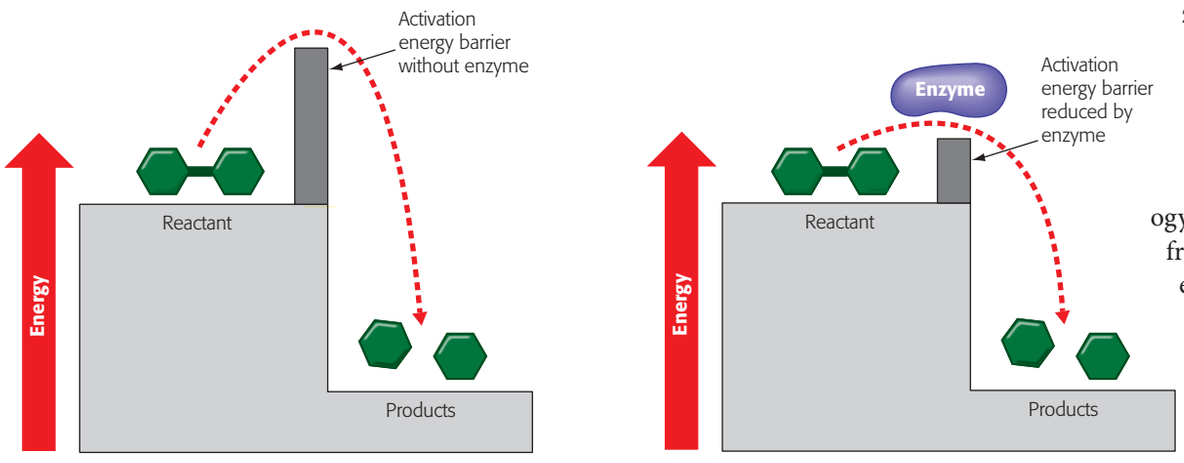
Answer: An enzyme lowers the activation energy.

Activation Energy

For a chemical reaction to begin, chemical bonds in the reactant molecules must be broken. (The first step in swapping partners during a square dance is to let go of your current partner's hand.) This process requires that the molecules absorb energy from their surroundings. In other words, for most chemical reactions, a cell has to spend a little energy to make more. You can easily relate this concept to your own life: it takes effort to clean your room, but this will save you more energy in the long run because you won't have to hunt for your belongings. The energy that must be invested to start a reaction is called **activation energy** because it activates the reactants and triggers the chemical reaction.

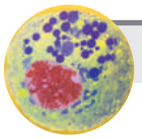
Enzymes enable metabolism to occur by reducing the amount of activation energy required to break the bonds of reactant molecules. If you think of the activation energy as a barrier to a chemical reaction, an enzyme's function is to lower that barrier **(Figure 5.7)**. It does so by binding to reactant molecules and putting them under physical or chemical stress, making it easier to break their bonds and start a reaction. In our analogy of cleaning your room, this is like a friend offering to help you. You start and end in the same place whether solo or assisted, but your friend's help lowers your activation energy, making it more likely that you'll proceed. Next, we'll return to our theme of nanotechnology to see how enzymes can be engineered to be even more efficient. ☑

▼ Figure 5.7 Enzymes and activation energy.



(a) Without enzyme. A reactant molecule must overcome the activation energy barrier before a chemical reaction can break the molecule into products.

(b) With enzyme. An enzyme speeds the chemical reaction by lowering the activation energy barrier.



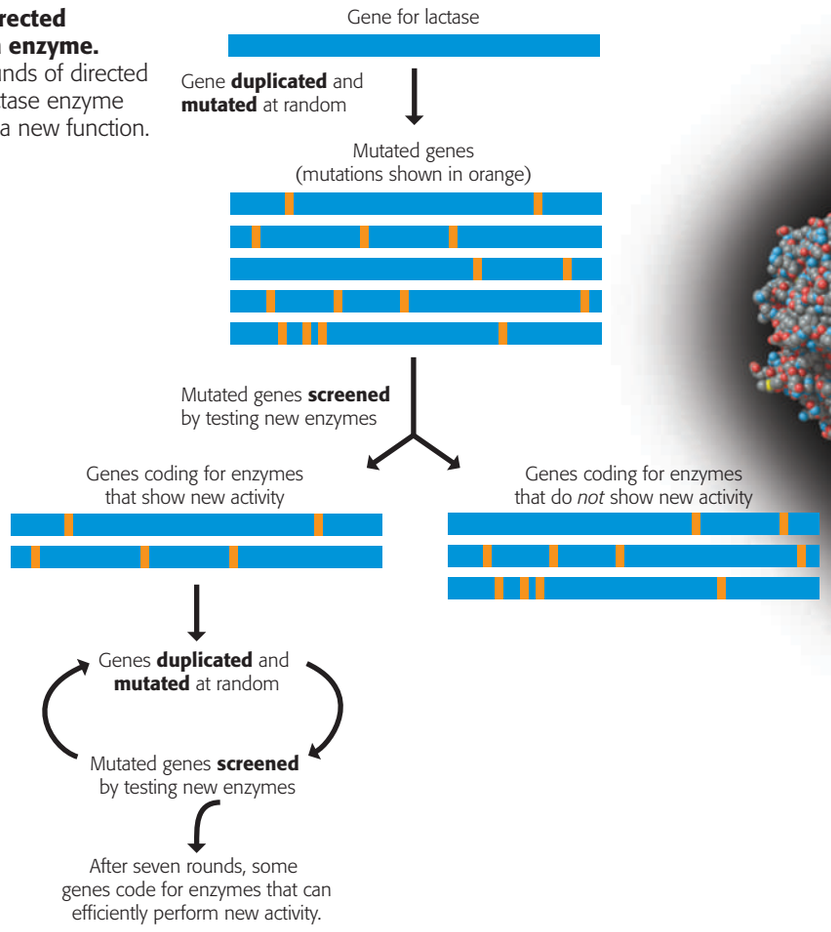
Nanotechnology THE PROCESS OF SCIENCE

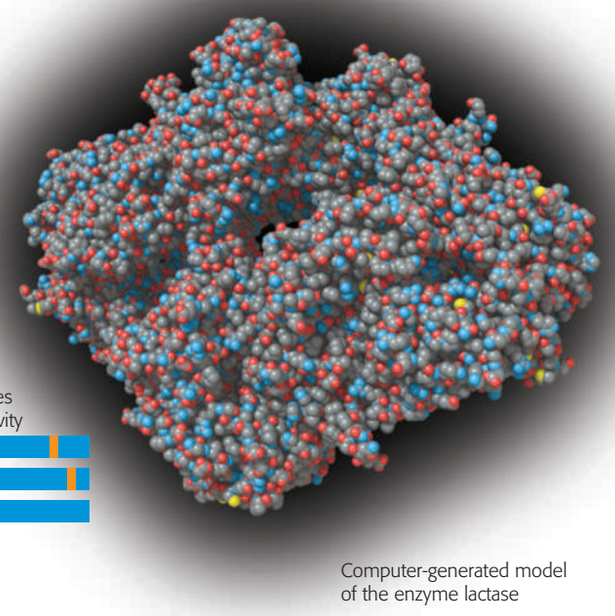
Can Enzymes Be Engineered?

Like all other proteins, enzymes are encoded by genes. **Observations** of genetic sequences suggest that many of our genes were formed through a type of molecular evolution: One ancestral gene duplicated, and the two copies diverged over time via random genetic changes, eventually becoming distinct genes for enzymes with different functions.

The natural evolution of enzymes raises a **question**: Can laboratory methods mimic this process through artificial selection? A group of researchers at two California biotechnology companies formed the **hypothesis** that an artificial process could be used to modify the gene that codes for the enzyme lactase (which breaks down the sugar lactose) into a new gene coding for a new enzyme with a new function. Their **experiment** used a procedure called directed evolution. In this process, many copies of the gene for the starting lactase enzyme were mutated at random **(Figure 5.8)**. The researchers tested the enzymes resulting from these mutated genes to determine which enzymes best displayed a new activity (in this case, breaking down a different sugar). The genes for the enzymes that did show the new activity were then subjected to several more rounds of duplication, mutation, and screening.

After seven rounds, the **results** indicated that directed evolution had produced a new enzyme with a novel function. Researchers have used similar methods to produce many artificial enzymes with desired properties, such as one that produces an antibiotic with tenfold greater efficiency, ones that remain stable and productive under high-heat industrial conditions, and one that greatly improves the production of cholesterol-lowering drugs. These results show that directed evolution is another example of how scientists can mimic the natural processes of cells for their own purposes.





Computer-generated model of the enzyme lactase

▼ Figure 5.8 Directed evolution of an enzyme. During seven rounds of directed evolution, the lactase enzyme gradually gained a new function.

Structure/ Function

Enzyme Activity

An enzyme is very selective in the reaction it catalyzes. This selectivity is based on the enzyme's ability to recognize a certain reactant molecule, which is called the enzyme's **substrate**. A region of the enzyme called the **active site** has a shape and chemistry that fits the substrate molecule. The active site is typically a pocket or groove on the surface of the enzyme. When a substrate slips into this docking station, the active site changes shape slightly to embrace the substrate and catalyze the reaction. This interaction is called **induced fit** because the entry of the substrate induces the enzyme to change shape slightly, making the fit between substrate and active site snugger. Think of a handshake: As your hand makes contact with another hand, it changes shape slightly to make a better fit.

After the products are released from the active site, the enzyme can accept another molecule of substrate. In fact, the ability to function repeatedly is a key characteristic of enzymes. **Figure 5.9** follows the action of the enzyme lactase, which breaks down the disaccharide lactose (the substrate). This enzyme is underproduced or defective in lactose-intolerant people. Like lactase, many enzymes are named for their substrates, with an *-ase* ending. ☑

☑ CHECKPOINT

How does an enzyme recognize its substrate?

Answer: The substrate and the enzyme's active site are complementary in shape and chemistry.

Enzyme Inhibitors

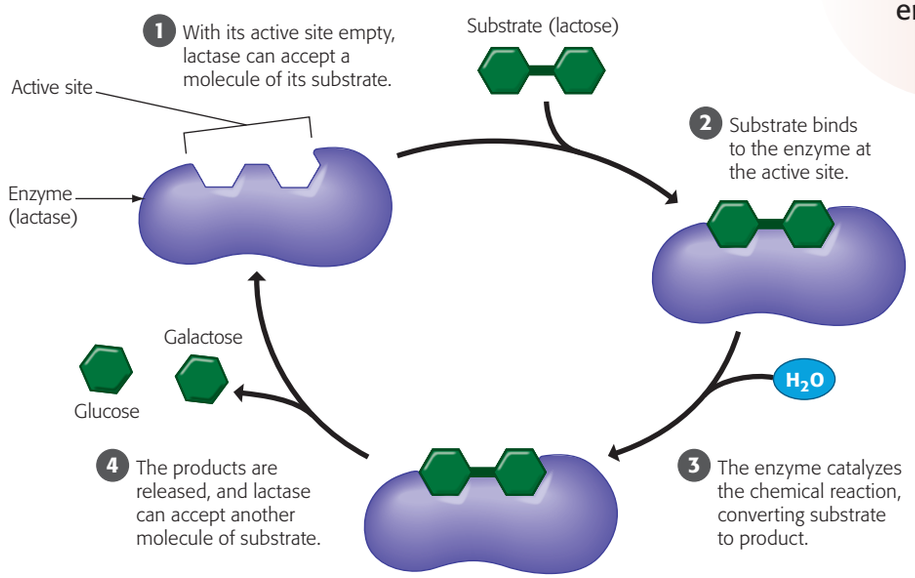
Certain molecules can inhibit a metabolic reaction by binding to an enzyme and disrupting its function **(Figure 5.10)**. Some of these **enzyme inhibitors** are substrate imposters that plug up the active site. (You can't shake a person's hand if someone else puts a banana in it first!) Other inhibitors bind to the enzyme at a site remote from the active site, but the binding changes the enzyme's shape. (Imagine trying to shake hands when someone is tickling your ribs, causing you to clench your hand.) In each case, an inhibitor disrupts the enzyme by altering its shape—a clear example of the link between structure and function.

Both nerve gas and insecticides work by crippling a vital enzyme.

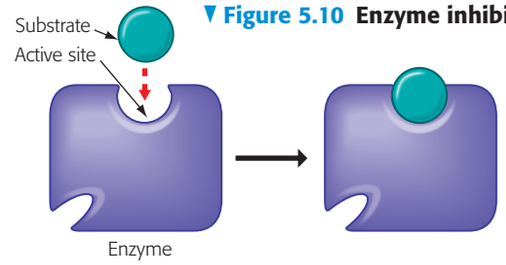
In some cases, the binding of an inhibitor is reversible. For example, if a cell is producing more of a certain product than it needs, that product may reversibly inhibit an enzyme required for its production. This feedback regulation keeps the cell from wasting resources that could be put to better use.

Many beneficial drugs work by inhibiting enzymes. Penicillin blocks the active site of an enzyme that bacteria use in making cell walls. Ibuprofen inhibits an enzyme involved in sending pain signals. Many cancer drugs inhibit enzymes that promote cell division. Many toxins and poisons also work as inhibitors. Nerve gases (a form of chemical warfare) irreversibly bind to the active site of an enzyme vital to transmitting nerve impulses, leading to rapid paralysis and death. Many pesticides are toxic to insects because they inhibit this same enzyme.

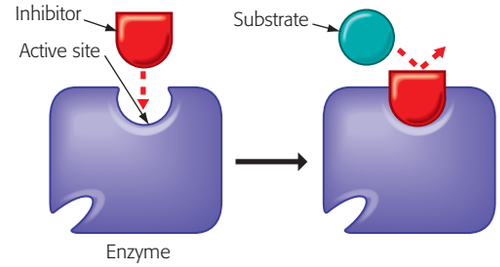
▼ Figure 5.9 How an enzyme works. Our example is the enzyme lactase, named for its substrate, lactose.



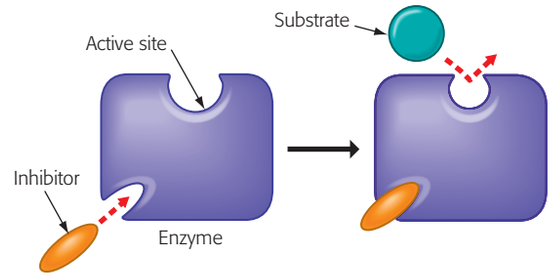
▼ Figure 5.10 Enzyme inhibitors.



(a) Enzyme and substrate binding normally



(b) Enzyme inhibition by a substrate imposter



(c) Inhibition of an enzyme by a molecule that causes the active site to change shape

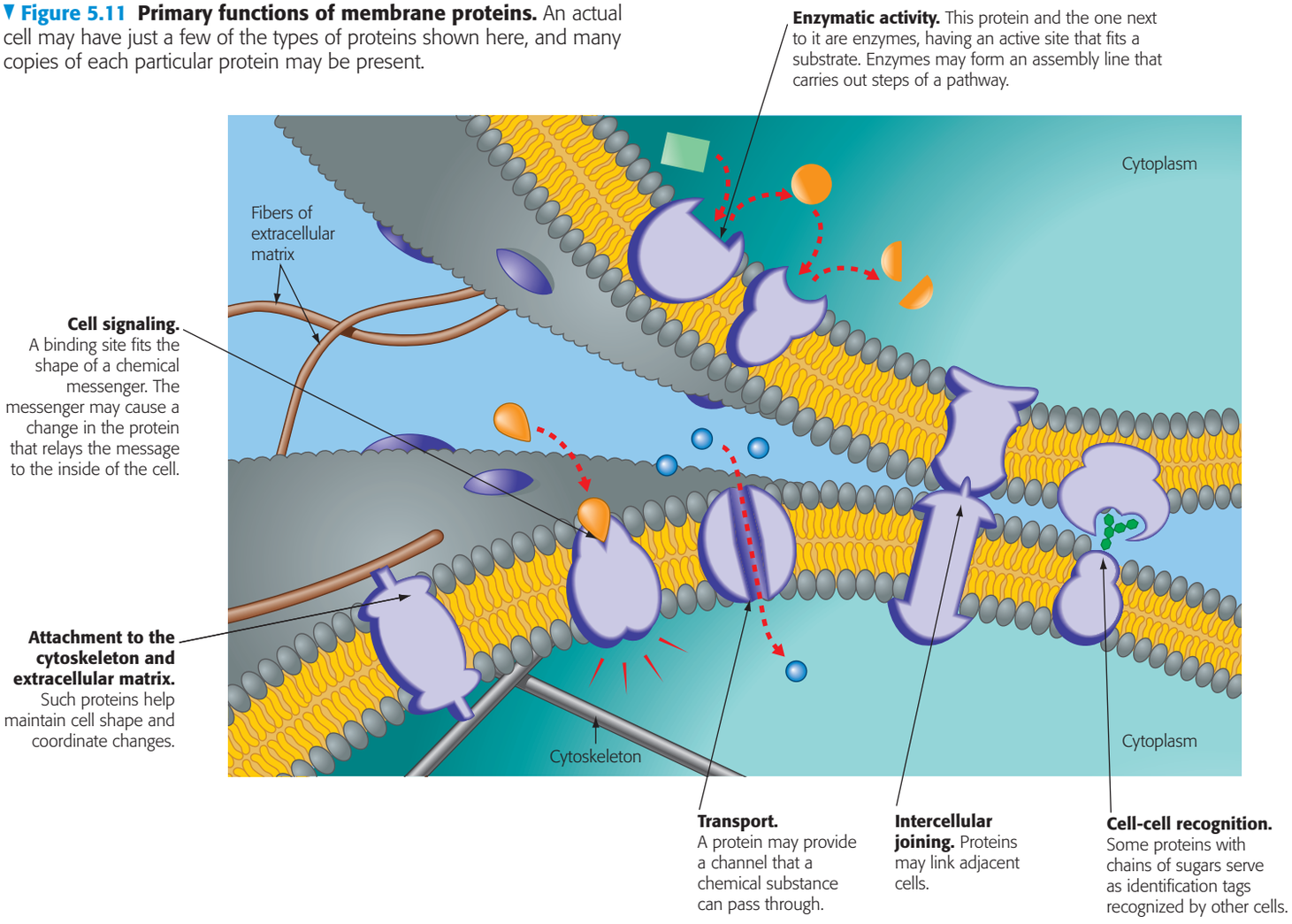
Membrane Function

So far, we have discussed how cells control the flow of energy and how enzymes affect the pace of chemical reactions. In addition to these vital processes, cells must also regulate the flow of materials to and from the environment. The plasma membrane consists of a double layer of fat (a phospholipid bilayer) with embedded proteins (see Figure 4.4). **Figure 5.11** describes the major functions of these membrane proteins. Of all the functions shown in the figure, one of the most important is the regulation of transport in and out of the cell. A steady traffic of small molecules moves across a cell's plasma membrane in both directions. But this traffic flow is never willy-nilly. Instead, all biological membranes are selectively permeable—that is, they only allow certain molecules to pass. Let's explore this in more detail.

Passive Transport: Diffusion across Membranes

Molecules are restless. They constantly vibrate and wander randomly. One result of this motion is **diffusion**, the movement of molecules spreading out evenly into the available space. Each molecule moves randomly, but the overall diffusion of a population of molecules is usually directional, from a region where the molecules are more concentrated to a region where they are less concentrated. For example, imagine many molecules of perfume inside a bottle. If you remove the bottle top, every molecule of perfume will move randomly about, but the overall movement will be out of the bottle, and the room will eventually smell of the perfume. You could, with great effort, return the perfume molecules to its bottle, but the molecules would never all return spontaneously.

Figure 5.11 **Primary functions of membrane proteins.** An actual cell may have just a few of the types of proteins shown here, and many copies of each particular protein may be present.



For an example closer to a living cell, imagine a membrane separating pure water from a mixture of dye dissolved in water **(Figure 5.12)**. Assume that this membrane has tiny holes that allow dye molecules to pass. Although each dye molecule moves randomly, there will be a net migration across the membrane to the side that began as pure water. Movement of the dye will continue until both solutions have equal concentrations. After that, there will be a dynamic equilibrium: Molecules will still be moving, but at that point as many dye molecules move in one direction as in the other.

Diffusion of dye across a membrane is an example of **passive transport**—*passive* because the cell does not expend any energy for the diffusion to happen. But remember that the cell membrane is selectively permeable. For example, small molecules such as oxygen (O_2) generally pass through more readily than larger molecules such as amino acids. But the membrane is relatively impermeable to even some very small substances, such as most ions, which are too hydrophilic to pass through the phospholipid bilayer. In passive transport, a substance diffuses down its **concentration gradient**, from where the substance is more concentrated to where it is less concentrated. In our lungs, for example, there is more oxygen gas (O_2) in the air than in the blood. Therefore, oxygen moves by passive transport from the air into the bloodstream.

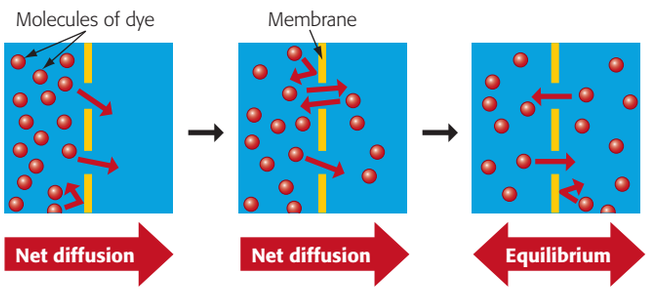
Substances that do not cross membranes spontaneously—or otherwise cross very slowly—can be transported via proteins that act as corridors for specific molecules (see Figure 5.11). This assisted transport is called **facilitated diffusion**. For example, water molecules can move through the plasma membrane of some cells via transport proteins—each of which can help 3 billion water molecules per second pass through! People with a rare mutation in the gene that encodes these water-transport proteins have defective kidneys that cannot reabsorb water; such people must drink 20 liters of water every day to prevent dehydration. On the flip side, a common complication of pregnancy is fluid retention, the culprit responsible for swollen ankles and feet, often caused by increased synthesis of water channel proteins. Other specific transport proteins move glucose across cell membranes 50,000 times faster than diffusion. Even at this rate, facilitated diffusion is a type of passive transport because it does not require the cell to expend energy. As in all passive transport, the driving force is the concentration gradient. ☑

☑ CHECKPOINT

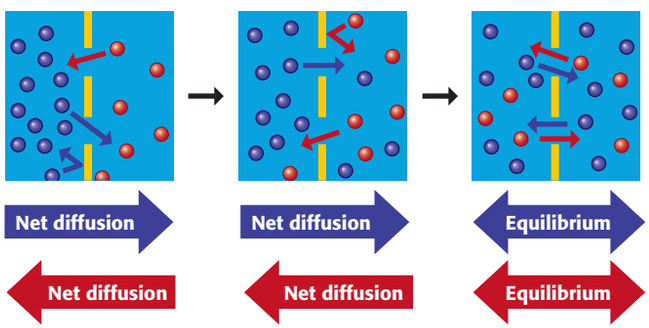
Why is facilitated diffusion a form of passive transport?

Answer: It uses proteins to transport materials down a concentration gradient without expending energy.

▼ Figure 5.12 Passive transport: diffusion across a membrane. A substance will diffuse from where it is more concentrated to where it is less concentrated. Put another way, a substance tends to diffuse down its concentration gradient.



(a) Passive transport of one type of molecule. The membrane is permeable to these dye molecules, which diffuse down the concentration gradient. At equilibrium, the molecules are still restless, but the rate of transport is equal in both directions.

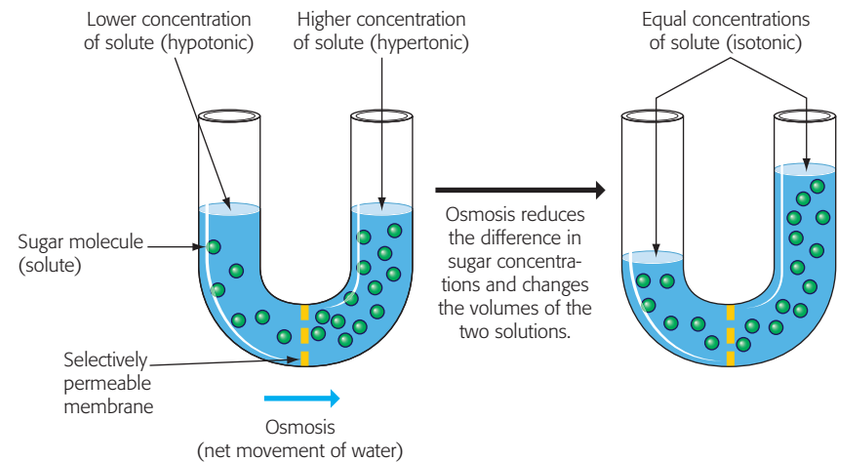


(b) Passive transport of two types of molecules. If solutions have two or more solutes, each will diffuse down its own concentration gradient.

Osmosis and Water Balance

The diffusion of water across a selectively permeable membrane is called **osmosis (Figure 5.13)**. A **solute** is a substance that is dissolved in a liquid solvent, and the resulting mixture is called a solution. For example, a solution of salt water contains salt (the solute) dissolved in water (the solvent). Imagine a membrane separating two solutions with different concentrations of a solute. The solution with a higher concentration of solute is

▼ Figure 5.13 Osmosis. A membrane separates two solutions with different sugar concentrations. Water molecules can pass through the membrane, but sugar molecules cannot.



said to be **hypertonic** to the other solution. The solution with the lower solute concentration is said to be **hypotonic** to the other. Note that the hypotonic solution, by having the lower solute concentration, has the higher water concentration (less solute = more water). Therefore, water will diffuse across the membrane along its concentration gradient from an area of higher water concentration (hypotonic solution) to one of lower water concentration (hypertonic solution). This reduces the difference in solute concentrations and changes the volumes of the two solutions.

People can take advantage of osmosis to preserve foods. Salt is often applied to meats—like pork and cod—to cure them; the salt causes water to move out of food-spoiling bacteria and fungi. Food can also be preserved in honey because a high sugar concentration draws water out of food.

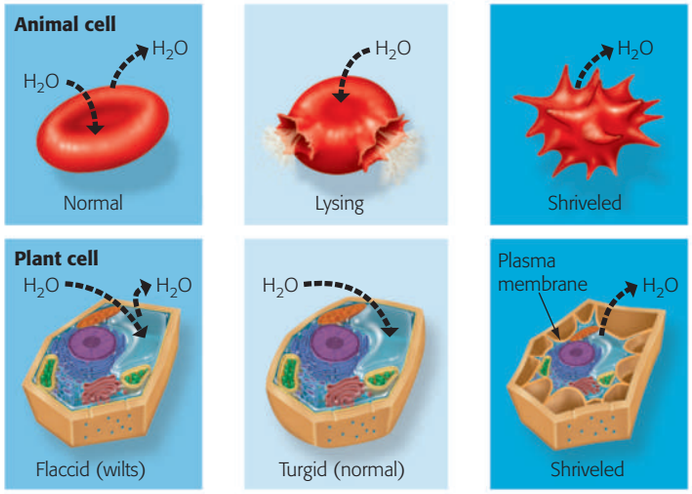
When the solute concentrations are the same on both sides of a membrane, water molecules will move at the same rate in both directions, so there will be no net change in solute concentration. Solutions of equal solute concentration are said to be **isotonic**. For example, many marine animals, such as sea stars and crabs, are isotonic to seawater, so that overall they neither gain nor lose water from the environment. In hospitals, intravenous (IV) fluids administered to patients must be isotonic to blood cells to avoid harm.

For thousands of years, people have used osmosis to preserve food through salt and sugar curing.

Water Balance in Animal Cells

The survival of a cell depends on its ability to balance water uptake and loss. When an animal cell is immersed in an isotonic solution, the cell's volume remains constant because the cell gains water at the same rate that it loses water (**Figure 5.14a**, top). But what happens if an animal cell is in contact with a hypotonic solution, which has a lower solute concentration than the cell? Due to osmosis, the cell would gain water, swell, and possibly burst (lyse) like an overfilled water balloon (**Figure 5.14b**, top). A hypertonic environment is also harsh on an animal cell; the cell shrivels from water loss (**Figure 5.14c**, top).

For an animal to survive a hypotonic or hypertonic environment, the animal must have a way to balance the uptake and loss of water. The control of water balance is called **osmoregulation**. For example, a freshwater fish has kidneys and gills that work constantly to prevent an excessive buildup of water in the body. Humans can suffer consequences of osmoregulation failure. Dehydration (consumption of too little water) can cause fatigue and even death. Drinking too much water—called hyponatremia or "water intoxication"—can also cause death by overdiluting necessary ions.



(a) Isotonic solution **(b) Hypotonic solution** **(c) Hypertonic solution**

◄ **Figure 5.14 Osmotic environments.** Animal cells (such as a red blood cell) and plant cells behave differently in different osmotic environments.

Water Balance in Plant Cells

Problems of water balance are somewhat different for cells that have rigid cell walls, such as those from plants, fungi, many prokaryotes, and some protists. A plant cell immersed in an isotonic solution is flaccid (floppy), and the plant wilts (Figure 5.14a, bottom). In contrast, a plant cell is turgid (very firm) and healthiest in a hypotonic environment, with a net inflow of water (Figure 5.14b, bottom). Although the elastic cell wall expands a bit, the back pressure it exerts prevents the cell from taking in too much water and bursting.

Turgor is necessary for plants to retain their upright posture and the extended state of their leaves (**Figure 5.15**). However, in a hypertonic environment, a plant cell is no better off than an animal cell. As a plant cell loses water, it shrivels, and its plasma membrane pulls away from the cell wall (Figure 5.14c, bottom). This usually kills the cell. Thus, plant cells thrive in a hypotonic environment, whereas animal cells thrive in an isotonic one. ☑



▼ **Figure 5.15 Plant turgor.** Watering a wilted plant will make it regain its turgor.

☑ CHECKPOINT

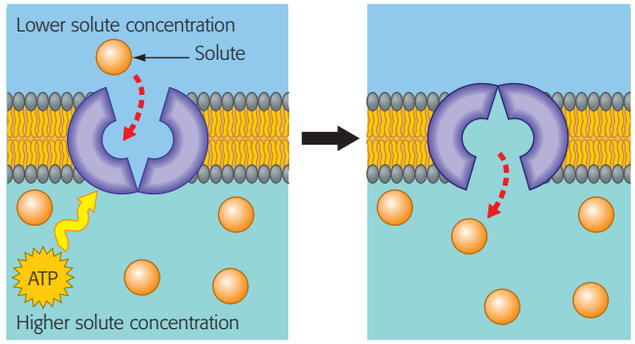
1. An animal cell shrivels when it is _____ compared with its environment.
2. The cells of a wilted plant are _____ compared with their environment.

Answers: 1. *hypotonic* 2. *isotonic*

Active Transport: The Pumping of Molecules across Membranes

In contrast to passive transport, **active transport** requires that a cell expend energy to move molecules across a membrane. Cellular energy (usually provided by ATP) is used to drive a transport protein that pumps a solute *against* the concentration gradient—that is, in the direction that is opposite the way it would naturally flow **(Figure 5.16)**. Movement against a force, like rolling a boulder uphill against gravity, requires a considerable expenditure of energy.

Active transport allows cells to maintain internal concentrations of small solutes that differ from environmental concentrations. For example, compared with its surroundings, an animal nerve cell has a much higher concentration of potassium ions and a much lower concentration of sodium ions. The plasma membrane helps maintain these differences by pumping sodium out of the cell and potassium into the cell. This particular case of active transport (called the sodium-potassium pump) is vital to the nervous system of most animals. ☑



▲ **Figure 5.16 Active transport.** Transport proteins are specific in their recognition of atoms or molecules. This transport protein (purple) has a binding site that accepts only a certain solute. Using energy from ATP, the protein pumps the solute against its concentration gradient.

☑ CHECKPOINT

What molecule is the usual energy source for active transport?

Answer: ATP

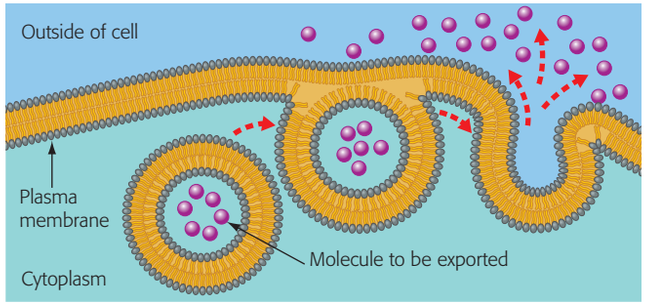
Exocytosis and Endocytosis: Traffic of Large Molecules

So far, we've focused on how water and small solutes enter and leave cells by moving through the plasma membrane. The story is different for large molecules such as proteins, which are much too big to fit through the membrane. Their traffic into and out of the cell depends on the ability of the cell to package large molecules inside sacs called vesicles. You have already seen an example of this: During protein production by the cell, secretory proteins exit the cell from transport vesicles that fuse with the plasma membrane, spilling the contents outside the cell (see Figures 4.12 and 4.16). That process is called **exocytosis** **(Figure 5.17)**. When you cry, for example, cells in your tear glands use exocytosis to export the salty tears. In your brain, the exocytosis of neurotransmitter chemicals such as dopamine helps neurons communicate.

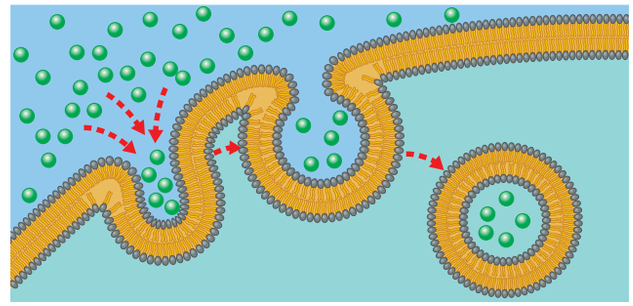
In **endocytosis**, a cell takes material in via vesicles that bud inward **(Figure 5.18)**. For example, in a process called **phagocytosis** ("cellular eating"), a cell engulfs a particle and packages it within a food vacuole. Other times, a cell "gulps" droplets of fluid into vesicles. Endocytosis can also be triggered by the binding of certain external molecules to specific receptor proteins built into the plasma membrane. This binding causes the local region of the membrane to form a vesicle that transports the specific substance into the cell. In human liver cells, this process is used to take up cholesterol from the blood. An inherited defect in the receptors on liver cells can lead to an inability to process cholesterol, which can lead to heart attacks at ages as young as 5. Cells of your immune system use endocytosis to engulf and destroy invading bacteria and viruses.

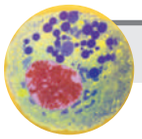
Because all cells have a plasma membrane, it is logical to infer that membranes first formed early in the evolution of life on Earth. In the final section of this chapter, we'll consider the evolution of membranes.

▼ **Figure 5.17 Exocytosis.**



▼ **Figure 5.18 Endocytosis.**





Nanotechnology EVOLUTION CONNECTION

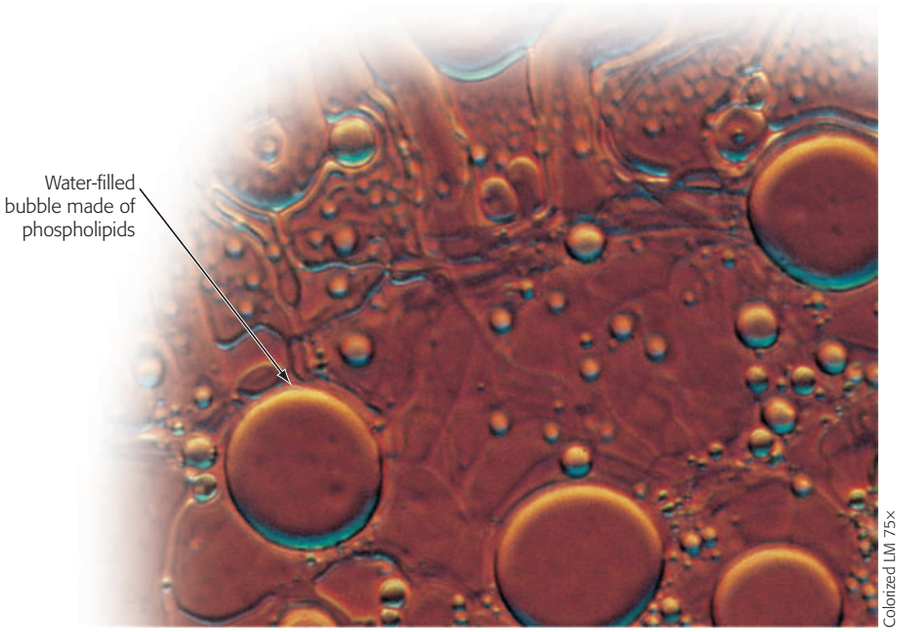
The Origin of Membranes

By simulating conditions found on the early Earth, scientists have been able to demonstrate that many of the molecules important to life can form spontaneously. (See Figure 15.3 and the accompanying text for a description of one such experiment.) Such results suggest that phospholipids, the key ingredients in all membranes, were probably among the first organic compounds that formed from chemical reactions on the early Earth. Once formed, they could self-assemble into simple membranes. When a mixture of phospholipids and water is shaken, for example, the phospholipids organize into bilayers, forming water-filled bubbles of membrane **(Figure 5.19)**. This assembly requires neither genes nor other information beyond the properties of the phospholipids themselves.

The tendency of lipids in water to spontaneously form membranes has led biomedical engineers to produce liposomes (a type of artificial vesicle) that can encase particular chemicals. In the future, these engineered liposomes may be used to deliver nutrients or medications to specific sites within the body. In fact, as of 2012, 12 drugs have been approved for delivery via liposomes, including ones that target fungal infections, influenza, and hepatitis. Thus, membranes—like the other cellular components discussed in the Biology and Society and the Process of Science sections—have inspired novel nanotechnologies.

The formation of membrane-enclosed collections of molecules would have been a critical step in the evolution of the first cells. A membrane can enclose a solution that is different in composition from its surroundings. A plasma membrane that allows cells to regulate their chemical exchanges with the environment is a basic requirement for life. Indeed, all cells are enclosed by a plasma membrane that is similar in structure and function—illustrating the evolutionary unity of life.

▼ Figure 5.19 The spontaneous formation of membranes: a key step in the origin of life.



Chapter Review

SUMMARY OF KEY CONCEPTS

Some Basic Energy Concepts

Conservation of Energy

Machines and organisms can transform kinetic energy (energy of motion) to potential energy (stored energy) and vice versa. In all such energy transformations, total energy is conserved. Energy cannot be created or destroyed.

Heat

Every energy conversion releases some randomized energy in the form of heat. Entropy is a measure of disorder, or randomness.

Chemical Energy

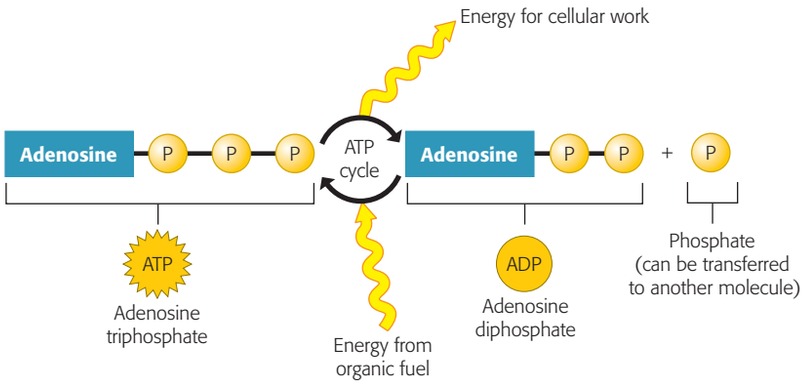
Molecules store varying amounts of potential energy in the arrangement of their atoms. Organic compounds are relatively rich in such chemical energy. The combustion of gasoline within a car's engine and the breakdown of glucose via cellular respiration within living cells are both examples of how the chemical energy stored in molecules can be converted to useful work.

Food Calories

Food Calories, actually kilocalories, are units used to measure the amount of energy in our foods and the amount of energy we expend in various activities.

ATP and Cellular Work

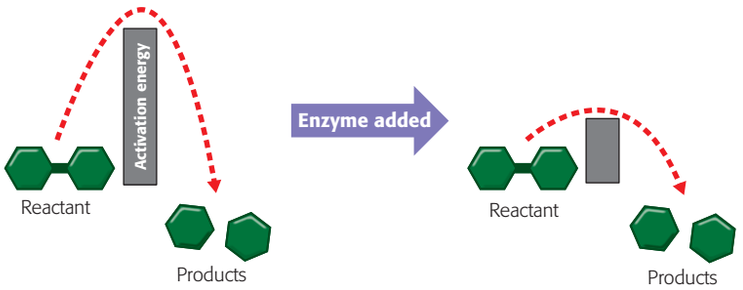
Your cells recycle ATP: as ATP is broken down to ADP to drive cellular work, new molecules of ATP are built from ADP using energy obtained from food.



Enzymes

Activation Energy

Enzymes are biological catalysts that speed up metabolic reactions by lowering the activation energy required to break the bonds of reactant molecules.



Enzyme Activity

The entry of a substrate into the active site of an enzyme causes the enzyme to change shape slightly, allowing for a better fit and thereby promoting the interaction of enzyme with substrate.

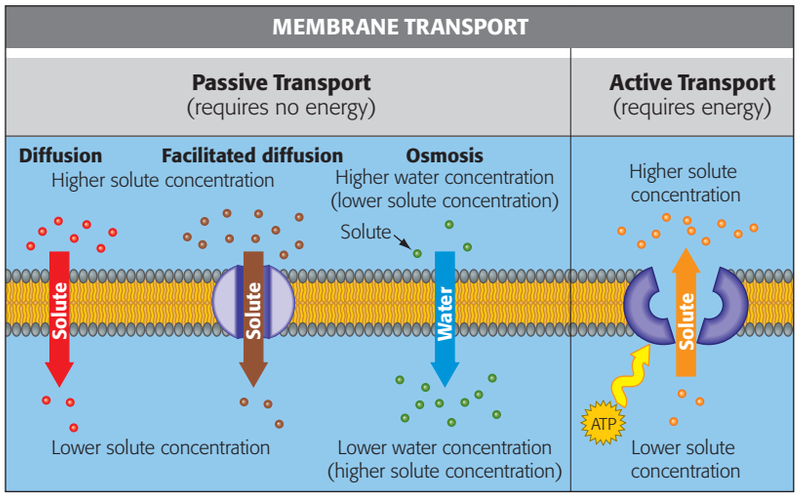
Enzyme Inhibitors

Enzyme inhibitors are molecules that can disrupt metabolic reactions by binding to enzymes, either at the active site or elsewhere.

Membrane Function

Proteins embedded in the plasma membrane perform a wide variety of functions, including regulating transport, anchoring to other cells or substances, promoting enzymatic reactions, and recognizing other cells.

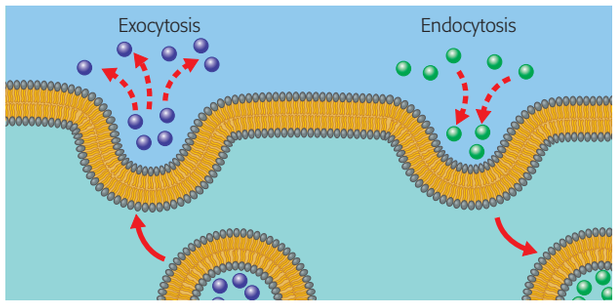
Passive Transport, Osmosis, and Active Transport



Most animal cells require an isotonic environment, with equal concentrations of water within and outside the cell. Plant cells need a hypotonic environment, which causes water to flow inward, keeping walled cells turgid.

Exocytosis and Endocytosis: Traffic of Large Molecules

Exocytosis is the secretion of large molecules within vesicles. Endocytosis is the import of large substances via vesicles into the cell.



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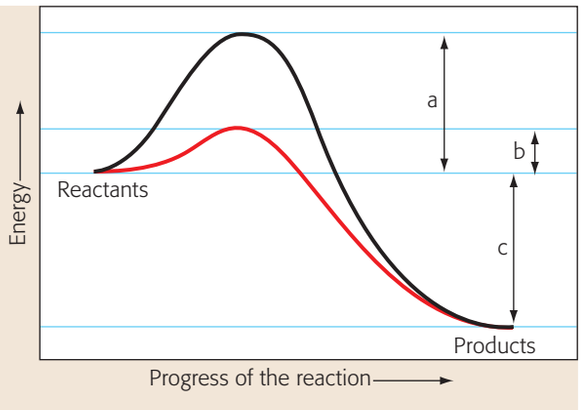
SELF-QUIZ

1. Describe the energy transformations that occur when you climb to the top of a stairway.
2. An object at rest has no _____ energy, but it may have _____ energy resulting from its location or structure.
3. The label on a tin of tomato soup says that it contains 56 Calories. If you could convert all of that energy to heat, you could raise the temperature of how much water by 15°C?
4. Why does removing a phosphate group from the triphosphate tail in a molecule of ATP release energy?
5. Your digestive system uses a variety of enzymes to break down large food molecules into smaller ones that your cells can assimilate. A generic name for a digestive enzyme is hydrolase. What is the chemical basis for that name? (*Hint*: Review Figure 3.4.)
6. Explain how an inhibitor can disrupt an enzyme's action without binding to the active site.
7. If someone at the other end of a room smokes a cigarette, you may breathe in some smoke. The movement of smoke is similar to what type of transport?
   a. osmosis
   b. diffusion
   c. facilitated diffusion
   d. active transport
8. Explain why it is not enough just to say that a solution is "hypertonic."
9. What is the primary difference between passive and active transport in terms of concentration gradients?
10. When a person cries, tears are exported from cells through the process of
    a. facilitated diffusion.
    b. active transport.
    c. endocytosis.
    d. exocytosis.

Answers to these questions can be found in Appendix: Self-Quiz Answers.

THE PROCESS OF SCIENCE

11. Cyclic guanosine monophosphate (cGMP) is a signaling molecule that relaxes smooth muscle tissues, thereby allowing blood flow into the corpus cavernosum and triggering penile erection. It is degraded by the enzyme cGMP-specific phosphodiesterase type 5 (PDE5). Sildenafil (Viagra), used to treat erectile dysfunction, has a molecular structure similar to that of cGMP. Propose a model for how sildenafil prevents erectile dysfunction.
12. Gaining and losing weight are matters of caloric accounting: Calories in the food you eat minus Calories that you spend in activity. One pound of human body fat contains approximately 3,500 Calories. Using Figure 5.3, compare ways you could burn off those Calories. How far would you have to run, swim, or walk to burn the equivalent of 1 pound of fat, and how long would it take? Which method of burning Calories appeals the most to you? The least? How much of each food would you have to consume in order to gain a single pound? How does one pound's worth of food compare with one pound's worth of exercise? Does it seem like an even trade-off?
13. **Interpreting Data** The graph below illustrates the course of a chemical reaction with and without an enzyme. Which curve represents the reaction in the presence of the enzyme? What energy changes are represented by the lines labeled a, b, and c?



BIOLOGY AND SOCIETY

14. Obesity is a serious health problem for many Americans. Several popular diet plans advocate low-carbohydrate diets. Most low-carb dieters compensate by eating more protein and fat. What are the advantages and disadvantages of such a diet? Should the government regulate the claims of diet books? How should the claims be tested? Should diet proponents be required to obtain and publish data before making claims?
15. Many detox and diet plans recommend drinking large amounts of water—sometimes even up to 5 liters per day. Which properties of water may have led to this recommendation? Is a massively high intake of water always recommendable? How can drinking large amounts of water have a negative physiological effect? Compare recommendations for water intake found on different detoxing/dieting websites, and determine whether they are scientifically sound.
16. Lead acts as an enzyme inhibitor, and it can interfere with the development of the nervous system. One manufacturer of lead-acid batteries instituted a "fetal protection policy" that banned female employees of childbearing age from working in areas where they might be exposed to high levels of lead. These women were transferred to lower-paying jobs in lower-risk areas. Some employees challenged the policy in court, claiming that it deprived women of job opportunities available to men. The U.S. Supreme Court ruled the policy illegal. But many people are uncomfortable about the "right" to work in an unsafe environment. What rights and responsibilities of employers, employees, and government agencies are in conflict? What criteria should be used to decide who can work in a particular environment?